When Trouble Travels: Peer Effects of Criminal Exposure in Schools

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Abstract

A child's peers are an important determinant of their performance at school. While a large literature documents the adverse effects of exposure to disruptive peers on educational outcomes, little is known about the impact of being exposed to a peer with a criminal background. Using rich administrative data linking criminal justice and education records for nearly the entire population of school students in England, I study whether and to what extent criminal behavior spreads within schools through exposure to juvenile offenders. I exploit the plausibly exogenous timing of school transfers by students with a criminal record and find that exposure increases an exposed student's probability of starting a criminal career by 12.9% within four months of the transfer, with effects persisting for at least three years. These peer effects generate approximately 20 million pounds in social costs, equivalent to 5% of the total social cost of youth crime in England. The effects vary substantially across school characteristics, which means that reassigning transferring offenders across schools could reduce the volume of crime induced by peer exposure by 36% under reasonable assumptions. Exposure also reduces educational attainment. Approximately one-third of this decline reflects a direct effect of exposure, while the remaining two-thirds operate through the exposed student's subsequent engagement in criminal behavior.

Keywords: human capital, youth crime, peer effects

JEL Classification: I21, K14, K42

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1 Introduction

Early adolescence marks the peak period of first contact with the criminal justice system in many countries. 12.5% of boys already have an offense record by age 17 in England (Cattan et al., 2025). Youth crime imposes substantial costs on both individuals and society. A large body of theoretical and empirical work shows that involvement in crime can hinder offenders' human capital accumulation and ultimately worsen their labor market outcomes.[1] Youth crime also generates substantial social and economic costs through policing, prevention, victim compensation, and other channels. In England and Wales, the annual cost of serious youth violence alone is estimated to be as high as £1.3 billion ($1.8 billion) (Irwin-Rogers, Muthoo, and Billingham, 2020).

Preventing youth crime, however, is challenging if policy focuses only on individual offenders. A growing literature shows that criminal behavior can propagate through social networks, including in jails (Bayer, Hjalmarsson, and Pozen, 2009; Damm and Gorinas, 2020; Johnsen and Khoury, 2024; Ouss and Philippe, 2025), neighborhoods (Dustmann and Landersø, 2021; Billings and Schnepel, 2022; Domínguez, 2026), and families (Bhuller et al., 2018*a*,*b*; Norris, Pecenco, and Weaver, 2021; Byström, 2025; Ahrsjö et al., forthcoming). Schools and youth clubs are the most important places for adolescents to interact socially; hence, their essential role in peer effects is well documented (Billings, Deming, and Ross, 2019; Díaz et al., 2021; Lee et al., 2021; Billings and Hoekstra, 2023; Villa, 2024; Lindquist et al., 2024).

Despite growing evidence of peer effects on crime in schools, little is known about how exposure to offending peers affects the initiation of criminal behavior among younger adolescents. Existing studies tend to focus on older youth and intensive margins, even though understanding the onset of offending is crucial for assessing its long-run consequences for human capital. Moreover, much of the existing literature captures the combined influence of schools and neighborhoods, leaving the relative importance of these two settings unclear. Distinguishing between them is important for policy because interventions targeted at schools differ substantially from those targeted at neighborhoods.

This paper advances this literature by studying the peer effects of exposure to offenders in secondary schools in England. I examine whether exposure to an incoming offender causes previously nonoffending students to begin committing crimes and whether it also affects overall criminal activity, including among students who have already begun offending. I study heterogeneity in these effects by the characteristics of offenders, peers, and destination schools and investigate potential mechanisms. I also isolate school-based peer effects from those associated with neighborhoods. I estimate heterogeneous treatment effects across crime types with different social costs and use

[1] An extensive literature investigates the effects of crime on human capital. See Becker (1962); Pager (2003); Holzer, Raphael, and Stoll (2006); Holzer (2007); Hjalmarsson (2008); Aizer and Doyle Jr. (2015); Mueller-Smith (2015); Agan and Starr (2017); Doleac and Hansen (2020); Eren and Mocan (2021); Rose (2021); Cullen, Dobbie, and Hoffman (2023); Burton and Wasser (2025); Garin et al. (2025); Lavy and Yancu (2025).

them to calculate the total social cost of crime attributable to peer effects. Beyond criminal behavior, I examine effects on academic achievement and conduct a mediation analysis to distinguish between two channels: a direct effect of exposure to a disruptive offender and an indirect effect operating through the exposed student's own entry into criminal activity. Finally, I conduct a counterfactual assignment exercise in which a social planner changes the destination schools of transferring offenders to assess how alternative school assignments could mitigate crime spillovers given the substantial heterogeneity in peer effects.

Identification relies on variation in the timing of offenders' school transfers. Conditional on a rich set of observable characteristics, the remaining variation in when students are first exposed to an offender transferring into their school and grade is plausibly as good as random. Because the timing of offenders' transfers differs across individuals, I use a staggered event-study design to estimate average treatment effects on the treated, comparing students who have already been exposed to an incoming offender with students who have not yet been exposed.

Using rich administrative data sets in England, the National Pupil Database (NPD) on demographics and educational attainments and the Police National Computer (PNC) on crime records, I estimate the peer effects of criminal activities. When an offender transfers into a student's school and cohort, the probability that the student commits a first offense within the following four months increases by 12.9%. This magnitude corresponds to approximately one additional new offender for every three offender transfers. This suggests that the peer effects are intense enough to drag non-offenders into a criminal life at a very early stage of life. The cumulative effect grows over the following two years and remains persistent, suggesting potentially long-lasting consequences for human capital. An offender's transfer also increases the number of crimes committed by students in the recipient school over the following four months by 24.3%. Together, these findings indicate that incoming offenders increase both the initiation of criminal activity and overall offending.

I then disentangle school effects from neighborhood effects. By comparing transfers into the same school but different neighborhoods with transfers into a different school within the same neighborhood, I show that school-based effects are substantially larger than neighborhood effects. Consistent with this interpretation, I also show that most estimated peer effects are concentrated in crimes committed during school terms rather than during holidays or weekends.

To investigate the underlying mechanisms, I organize several potential channels using the human capital framework of Lochner (2004), in which individuals weigh the returns to and costs of criminal activity. Factors that may increase the returns to crime include the accumulation of criminal capital—through spillovers in skills or knowledge—and changes in idiosyncratic tastes or beliefs, including role-model effects. On the cost side, relevant factors include marginal expected loss when caught—the opportunity cost of crime—and expected penalty. Combining the predictions of this framework with heterogeneity analyses, I find evidence consistent with an important role for role-model effects on the return side and for opportunity costs on the cost side.

I also show that treatment effects vary substantially with the characteristics of offenders, peers, and destination schools. Most notably, the existing concentration of offenders at the destination school matters. In "offender-dense" schools, where the fraction of incumbent offenders in a school-grade is high, the effect on offense initiation disappears. One interpretation is that peers in these schools are already exposed to offenders, so an additional offender provides little new information and is less salient as a role model. In contrast, offender-dense schools amplify the effect on the number of crimes committed. Consistent with evidence from prison settings (Bayer, Hjalmarsson, and Pozen, 2009), this pattern suggests that concentrating offenders may strengthen peer interactions among students who are already engaged in criminal activity.

To quantify the economic magnitude of these effects, I calculate the total cost of youth crime attributable to school-based peer effects using estimates of the social cost of different crime types from Heeks et al. (2018). I estimate that approximately £20.0 million ($26.4 million), equivalent to 5% of the total cost of youth crime during my sample period, can be attributed to peer effects in schools.

Next, I examine academic achievement as an additional negative externality of exposure to offending peers. I focus on the age-14 assessment and compare students exposed to an incoming offender up to one year before the exam with students exposed only after the exam. Exposure before the exam reduces age-14 test scores by 0.016 standard deviations. A mediation analysis suggests that approximately one-third of this effect operates directly through exposure to an offending peer, for example through classroom disruption, while the remaining two-thirds operates indirectly through induced offending: exposure causes some students to become offenders before the exam, which further worsens their educational achievement.

Finally, to assess how school assignment can mitigate crime spillovers, I conduct a simulation exercise in which a social planner reassigns transferring offenders across destination schools, exploiting the substantial heterogeneity in treatment effects. Focusing on variation in the existing density of offenders across schools, I find that an assignment rule designed to minimize total crime could reduce the number of crimes generated by peer effects by 36% under reasonable assumptions. However, the choice of objective is crucial: a policy that instead minimizes the number of newly induced offenders may substantially increase the total number of crimes.

This paper builds on a growing literature on peer effects in crime. The closest study is Billings, Deming, and Ross (2019), which documents social spillovers in crime when disadvantaged students attending the same school are also concentrated within small neighborhoods. Their setting captures the combined influence of schools and neighborhoods, whereas I separately identify each contribution. This distinction matters for policy because interventions aimed at schools and neighborhoods differ in both design and implementation. Their analysis also focuses on joint offending, a mechanism that does not necessarily operate when students attend the same school but live in different neighborhoods. Finally, their offense records begin at age 16, whereas first contact with the criminal justice system in England peaks at age 15 (Cattan et al., 2023*a*,*b*, 2025). Studying

younger adolescents therefore provides new evidence on how criminal behavior begins.

Another closely related study is Lee et al. (2021), which identifies key players in offenders' networks and shows through simulation that removing a small fraction of these individuals can substantially reduce crime. While that exercise identifies which offenders are most influential, this paper focuses on who begins offending because of exposure to such individuals, providing a complementary measure of criminal spillovers. Billings and Hoekstra (2023) studies the effects of schoolmates whose parents have been arrested on students' subsequent criminal behavior. Their treatment captures exposure to potentially disruptive peers more broadly, while I focus directly on the transmission of criminal behavior following exposure to an offender.

More broadly, this paper contributes to the literature on how schools and education affect crime. A large body of evidence shows that additional schooling and education reduce criminal activity (Lochner and Moretti, 2004; Machin, Marie, and Vujić, 2011; Cook and Kang, 2016; Landersø, Nielsen, and Simonsen, 2017; Bacher-Hicks, Billings, and Deming, 2024; Bell, Costa, and Machin, 2022), as does access to early childhood education (Depew and Eren, 2016; Ando, Mori, and Yamaguchi, 2025). Several studies examine these relationships through the lens of human capital accumulation (Aizer and Doyle Jr., 2015; Mueller-Smith, 2015; Eren and Mocan, 2021), while others model youth behavior as a choice between schooling and criminal activity (Lochner, 2004).[2] I contribute to this literature by identifying a factor within schools that can increase criminal activity, particularly crime onset. Such endogenous peer effects are difficult to identify because students' peer groups are generally not randomly assigned. I address this challenge by exploiting detailed information on student characteristics and variation in the timing of school transfers that is plausibly exogenous to students already enrolled in the destination school.[3]

The paper also adds to evidence on the transmission of criminal behavior across peers. Existing studies have documented peer effects within prisons (Bayer, Hjalmarsson, and Pozen, 2009; Stevenson, 2017; Ouss and Philippe, 2025; Damm and Gorinas, 2020; Johnsen and Khoury, 2024), neighborhoods (Kling, Ludwig, and Katz, 2005; Sviatschi, 2022; Dustmann and Landersø, 2021; Billings and Schnepel, 2022; Domínguez, 2026), and families (Bhuller et al., 2018*a*,*b*; Norris, Pecenco, and Weaver, 2021; Byström, 2025; Ahrsjö et al., forthcoming).[4] I provide evidence on the transmission of criminal behavior within schools, a setting in which such effects are particularly

[2]Related work in criminology and psychology also studies youth crime (Sabates, 2008; McAra and McVie, 2010; Welch and Payne, 2012; McGloin and Thomas, 2019).

[3]The paper also relates to the broader literature on peer effects, much of which focuses on academic achievement in schools. See, for example, Hoxby (2000); Sacerdote (2001); Zimmerman (2003); Kang (2007); Figlio (2007); Ding and Lehrer (2007); Carrell, Fullerton, and West (2009); Ammermueller and Pischke (2009); Carrell and Hoekstra (2010); Duflo, Dupas, and Kremer (2011); Arcidiacono et al. (2012); Lavy, Silva, and Weinhardt (2012); Burke and Sass (2013); Angrist (2014); Lu and Anderson (2015); Feld and Zölitz (2017); Murphy and Weinhardt (2020); Carneiro et al. (forthcoming). Related literature studies school segregation through the lens of peer effects (Billings, Deming, and Rockoff, 2014).

[4]Related studies in school settings examine spillovers in problematic behaviors such as drinking and smoking and generally find negative peer effects (Powell, Tauras, and Ross, 2005; Nakajima, 2007; Carrell, Malmstrom, and West, 2008; Balsa et al., 2011; Fletcher, 2012; Eisenberg, Golberstein, and Whitlock, 2014).

difficult to measure. The school environment is important because it allows me to observe the transition from a non-offender to an offender at a young age, a margin that cannot be studied in prison settings. In addition, compulsory schooling requires adolescents to interact with schoolmates regularly, making schools a potentially important setting for the transmission of criminal behavior.

Finally, this paper contributes to the literature on school assignment. A large literature develops mechanisms for allocating students to schools based on their preferences (Abdulkadiroğlu and Sönmez, 2003; Abdulkadiroğlu, Pathak, and Roth, 2005; Abdulkadiroğlu et al., 2005; Abdulkadiroğlu, Pathak, and Roth, 2009; Pathak and Sönmez, 2013; Kamada and Kojima, 2015, 2024). A separate literature examines the consequences of excluding disadvantaged or disruptive students from school and how their allocation across schools can mitigate negative externalities (McAra and McVie, 2010; Balestra, Eugster, and Liebert, 2022). This paper connects these two strands by applying ideas from school assignment to a counterfactual exercise aimed at reducing crime spillovers across students.

The remainder of the paper is organized as follows. Section 2 describes the institutional background, and Section 3 presents the data. Section 4 describes the empirical strategy and identification assumptions, and Section 5 presents the main results. Section 6 reports the counterfactual assignment exercise. Section 7 concludes. The appendix provides additional details on the data, robustness checks, and heterogeneity analyses.

2 Institutional Background

2.1 Education System in England

In the UK, about 94% of children are enrolled in state schools (Kynaston and Green, 2019), which are equivalent to "public" or non-private schools in other countries. Compulsory education in England begins at age 6 in primary school and continues through age 16 in secondary school.[5][6] Pupils take four national assessments known as Key Stages (KS), typically at ages 7 for KS1, 11 for KS2, 14 for KS3, and 16 for KS4. KS4, also known as the General Certificate of Secondary Education (GCSE), is particularly important for progression to further education. Contrary to the US, grade retention is extremely rare in the UK. After the minimum school-leaving age of 16, some children continue their studies in upper secondary education until age 18, while others pursue alternative pathways, such as vocational training. At the end of upper secondary education, students take A-Levels (KS5), which are an important qualification for entry into higher education,

[5]Prior to elementary school, some go to nursery schools or use an area-based holistic support called Sure Start. See Carneiro et al. (2025) for the effects of Sure Start on juvenile criminal behaviors.

[6]In 2013, the minimum leaving age was lifted to 17 in England, and further to 18 in 2015, through the Education and Skills Act 2008. Currently, after age 16, children must be in some form of education or training, although they do not have to attend a state school.

including colleges and universities.

In this paper, I focus on secondary schools for students ages 11 to 16 for two reasons. First, this period falls entirely within compulsory schooling, meaning that children are required to attend school regularly, which limits selection into education. Second, under English law, individuals are criminally responsible from age 10. Since most offenders begin engaging in criminal activity considerably later (Cattan et al., 2023*a*,*b*, 2025), focusing on ages 11 to 16 allows me to capture nearly the full range of juvenile crime trajectories.

Under the compulsory education system, children can transfer to another state school when necessary. To do so, their parents must apply either directly to the school in which they want their child to enroll or to the local authority responsible for the preferred school. In principle, if a place is available, a child should be admitted regardless of their background, including their prior offense history.[7]

The application process takes time. Parents must apply two months before the transfer, and decisions are typically announced within two to three weeks. Although school principals should be informed about the backgrounds of incoming students, teachers responsible for their classrooms are not informed in advance that they will receive additional students. As a result, they cannot adjust their teaching practices in anticipation of the characteristics of incoming students, including whether those students have a history of offending. Similarly, parents of children in the receiving class are not informed about the transfer in advance. Therefore, students at the destination school have little opportunity to respond to an incoming student's transfer before it occurs. Most importantly, because the transfer application process takes time, students at the receiving school cannot move to another school in anticipation of the arrival of an offending student. Transfers typically occur at the beginning of a school term: September for the autumn term, January for the spring term, and April for the summer term.[8]

In some schools, students are assigned to classrooms primarily according to academic ability, particularly in math, while behavioral issues play little role in classroom assignment. Because offenders tend to exhibit lower educational achievement (Cattan et al., 2023*a*,*b*, 2025), classrooms containing students with lower academic achievement may have a higher concentration of offenders than classrooms containing higher-achieving students.

[7]Under Section 86(5)18 of the School Standards and Framework Act 1998, a school can refuse a child's school transfer if the child is eligible for special education needs (SEN) or has a physical disability that is not compatible with the school's existing resources (Department for Children, Schools and Families, 2009). However, in principle, schools cannot refuse them on the basis of their behavioral issues. If more transfer requests exceed the school's available slots, the school must follow strict admission criteria. The criteria typically include whether a child is on a look-after status, whether they have a sibling at the same school, and their commuting distance. However, the offense history should be irrelevant to the criteria.

[8]Another way to relocate students to another school is through exclusion. If a student misbehaves severely at school, school principals have the right to exclude the student. However, on only a very few occasions is the student permanently excluded from the school (Cattan et al., 2023*a*,*b*, 2025). During compulsory education, students need to move to another school. I do not distinguish the effects induced by regular school transfers from those by permanent exclusions, as the number of latter cases is too small to affect the results.

2.2 Criminal Justice System in England

In England, individuals are criminally responsible for their offenses from age 10. If a child aged 10 or older commits an offense, they are subject to police intervention. For relatively minor offenses, the police issue a caution and release the child without further court proceedings. Even in such cases, the offense is formally recorded. More serious offenses may result in the child being accused, charged, and referred to court for further proceedings. If there is sufficient evidence to prosecute, the child may be convicted and receive a sanction. Punishments range from fines to community orders, probation, or incarceration. In most cases, children receive only a caution. The likelihood of imprisonment or a youth rehabilitation order following a juvenile offense is less than 1% (Cattan et al., 2023*a*,*b*, 2025). Juvenile offenders may also be excluded from school, but such exclusions are generally temporary. As a result, most juvenile offenders return to regular classroom instruction shortly after committing an offense.

Once a person commits an offense, is caught by the police, and is found guilty, the record is stored in the Police National Computer (PNC) and retained until the person's 100th birthday. Employers may ask job applicants to provide a criminal record certificate through the government service known as the Disclosure and Barring Service (DBS). During my sample period, these certificates included all youth convictions for offenses committed within the previous 5.5 years and all youth cautions for offenses committed within the previous 2 years. Convictions for particularly serious offenses, including murder, rape, kidnapping, assault on a child, and many sexual offenses and offenses against children, were disclosed regardless of when they were committed. Therefore, a criminal record can have lasting consequences for children's human capital.[9]

3 Data

This study uses linked administrative records on educational outcomes and criminal justice involvement for nearly the entire population of children in England. The linked dataset is provided through the Data First initiative, a data-linking program led by the Ministry of Justice (MoJ).

3.1 Offense Records

The criminal justice outcomes are drawn from the Home Office Court Appearance System (HOCAS) and the National Offender Management Information System (NOMIS), which collectively form a combined sequence of databases such as the Police National Computer (PNC). The PNC

[9]Since November 2020, youth cautions are not automatically disclosed by DBS, although they are still stored in PNC, and some jobs and activities require the thorough checks directly with PNC, such as national security, the police, judicial appointments, and firearms certificates. See the UK government's website for the recent updates (https://www.gov.uk/government/publications/filtering-rules-for-criminal-record-check-certificates/filtering-rules-for-dbs-certificates-criminal-record-checks, last accessed: October 12th, 2026).

contains individual-level panel records of offenses from age 10 onward, the age of criminal responsibility in England. My PNC extract covers recordable offenses committed between 2000 and 2017. Because the cohorts I focus on in this paper were born between 1995/96 and 1998/99, all individuals have complete records of recordable offenses between ages 10 and 17, covering their entire secondary-school period. My NPD extract contains 1.1 million individuals and 164.6 million panel records. Throughout the paper, I define offenses as those recorded in the PNC and offenders as individuals with at least one offense recorded in the PNC, committed by the end of secondary school at the age of 16.

The PNC contains highly detailed information on offense types, with as many as 2,820 distinct categories. Following Cattan et al. (2025), I aggregate these into ten categories: theft/robbery, battery, drug-related, violent, fraud, weapons, arson/damage, sexual, summary motoring, and a residual category comprising other miscellaneous or unknown offenses, loosely following the MoJ's classification.[10][11]

3.2 Education Records

The education records come from the National Pupil Database (NPD), provided by the Department for Education (DfE). These data cover all students attending state schools, who account for approximately 94% of all children in England (Kynaston and Green, 2019).[12] Part of the NPD is drawn from the school census, which contains information on students' demographic characteristics and school enrollment. My extract contains three school censuses per academic year from 2007/08 onward, compared with only one or two censuses per year in earlier periods.[13] Therefore, in the main analysis, I focus on cohorts for which three census observations are available in every year of secondary school, namely the 1995/96 to 1998/99 cohorts.

The NPD contains information on children's demographic characteristics, family backgrounds, and educational outcomes. Demographic characteristics include gender, self-reported ethnicity, year and month of birth, whether English is their first language, and whether they are registered as having special educational needs (SEN). In most analyses, I aggregate ethnicity into 12 broader categories to ensure sufficient observations within each group.[14] Measures of family background include children's area of residence and socioeconomic status. Residential location is measured

[10] See Table A5 for the most common offenses within each crime type.

[11] In analyses by offense type, I further aggregate these into four broader categories to ensure sufficient statistical power: property crimes (theft/robbery, fraud, and arson/damage), violent crimes (battery, violent, and sexual), drug-related crimes, and others (weapons, summary motoring, and a residual group).

[12] Strictly speaking, students in private schools are assigned a Pupil Matching Reference in NPD if they take a GCSE, but basically they do not have information on enrollment.

[13] The autumn Census in my extract starts in 2007, the spring one in 2002, and the summer one in 2006.

[14] The 12 categories are a summary of the original 99 detailed categories and loosely based on the DfE's classification. I use the same categories in my previous papers (Cattan et al., 2023*a*,*b*, 2025), which are "White British," "Other White," "African Black," "Caribbean Black," "Other Black," "Indian," "Pakistani," "Bangladeshi," "Other Asian," "Mixed," "Others," and "Unknown."

using lower layer super output areas (LSOAs), which are comparable to US census tracts and contain approximately 1,000 to 3,000 residents each. Socioeconomic status is proxied by eligibility for free school meals (FSMs), which roughly identifies the bottom quartile of children in the household income distribution. SEN and FSM status vary across school censuses, while the other demographic characteristics are effectively time-invariant. For SEN and FSM status, I construct indicators for whether a child was ever eligible by age 11, prior to entering secondary school. For the time-invariant characteristics, I use the latest available record for each individual.

Educational outcomes include school enrollment histories and measures of academic achievement at each Key Stage. School enrollment is observed at the school-census level, with censuses conducted on the first Thursday in October for the autumn census, the third Thursday in January for the spring census, and the third Thursday in May for the summer census. Because most school transfers occur at the beginning of each term, these census records are likely to capture nearly all such transfers.

Coverage of Key Stage assessments differs across cohorts, as shown in Table A1. In the main analysis, I focus on 2.4 million children born between 1995/96 and 1998/99, most of whom have test-score records at ages 7 and 11. However, none of these cohorts have test-score records at age 14. This creates an important limitation for studying the effects on educational attainment. To compare students who have already been exposed to an offender's transfer with those who have not yet been exposed during secondary school, I need an educational outcome that is observed for both already-treated and not-yet-treated students during secondary school. The age-14 assessment is the only educational measure that satisfies this requirement. Therefore, I use the earlier 1990/91–1993/94 cohorts, for which age-14 test scores are available, when analyzing educational outcomes in Subsection 5.6.[15] One limitation of these earlier cohorts is that they have only one school record per year, so school transfers can be identified only at the annual level.

3.3 Data Linkage

The PNC and NPD are linked using anonymized individual identifiers provided by the MoJ. These identifiers are constructed through probabilistic matching based on an individual's name, gender, date of birth, and postal code of residence. Table A2 reports the linkage rate and offense rate in my sample. As shown in Column (1), each cohort contains approximately 600,000 individuals with complete information on the full set of controls, yielding a total sample of 2.4 million individuals across my target cohorts, 1995/96–1998/99.[16]

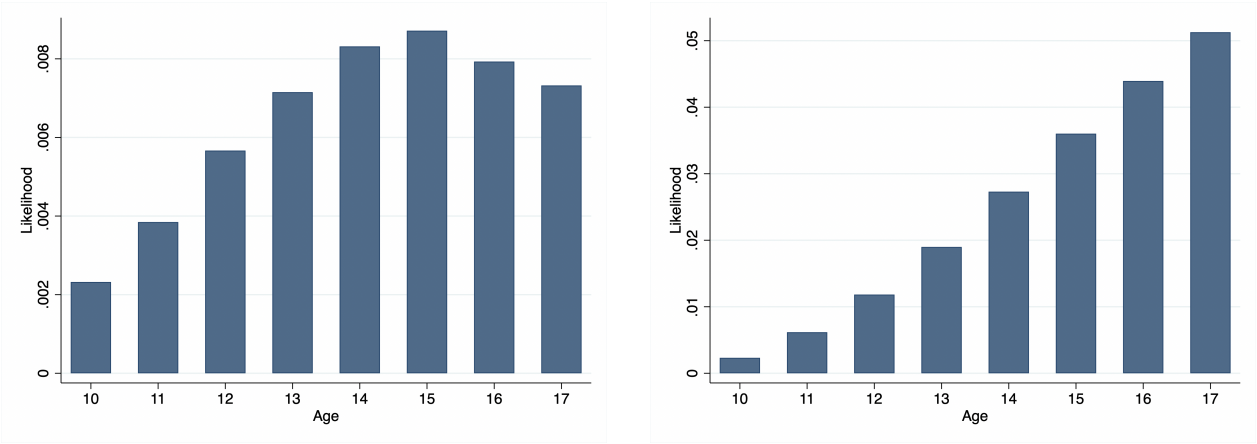
Column (2) reports the fraction of individuals with an offense ID, indicating that they had contact with the criminal justice system at some point, including outside my data period, which

[15] I show the descriptives and the replication of the main results using these cohorts in Appendix G. Despite the aforementioned differences, the main results still hold.

[16] See Appendix A for the data details. Table A1 shows the original sample size, and about 95% of the observations have full information on the controls.

varies from 4.5% to 12% for the target cohorts. Columns (3) and (4) report, among individuals with an offense ID, the fraction with at least one offense recorded during my data period, that is, those successfully matched to an offense record between 2000 and 2017. Column (3) includes offenses committed at any age, while Column (4) restricts attention to offenses committed by the end of secondary school. Roughly half of individuals with an offense ID have at least one offense recorded by the end of secondary school.

Column (5) reports the overall offense rate for each cohort, calculated as Column (2) × Column (4). The offense rate ranges from 2.8% to 5.9% and exhibits a clear downward trend, consistent with the patterns documented in Youth Justice Board and Ministry of Justice (2023); Cattan et al. (2025). The final column reports the precision of the linkage. Approximately 80% of linked records are based on an exact match on an individual's name, gender, date of birth, and postal code of residence across the NPD and PNC.[17]



(a) Age of Onset of First Crime

(b) Cumulative Offense Rate

Figure 1. Age of the First Entry into the Criminal Justice System

Notes: Offenders in this figure are defined as individuals in the 1995/96–1998/99 cohorts in the NPD with at least one recordable offense in the PNC committed between ages 10 and 17 that resulted in a caution or conviction. I exclude individuals with an offense record before age 10. Data sources: Ministry of Justice and Department for Education.

3.4 Data Summary

In this subsection, I describe the summary statistics for the resulting dataset. I begin by describing the characteristics of offenders. Figure 1a shows the distribution of age at first offense. The incidence of first offending peaks at age 15, when 0.83% of all children in England commit their first offense.[18] Figure 1b shows the cumulative offense rate. By age 16, when secondary school

[17]The remaining 20% is based on a fuzzier match, which does not match at least one of these four identifiers. However, even the fuzziest matching is based on the exact forename, surname, and date of birth, ensuring high linkage quality.

[18]These figures and some of the subsequent tables follow the style of those in Cattan et al. (2025). Numbers differ from them because I focus on a subset of their cohorts and include both boys and girls. Notably, the offense rate in my sample is significantly lower because younger cohorts and girls have lower offense rates.

ends, 4.4% of children have committed at least one offense. Thus, the secondary school period covers both the early stages of criminal onset and the age at which initiation peaks, underscoring its importance for studying juvenile crime.

In the main analysis, I restrict the sample to individuals with complete school enrollment records throughout secondary school in England, from the autumn census at age 11 through the summer census at age 16. I exclude students who ever attended a very small secondary school, defined as one in which the number of pupils in a school-grade is fewer than 20. Table A3 describes this sample-selection process in detail. Together, these two restrictions remove 11.8% of observations with complete control variables, and the students remaining in the sample are less disadvantaged.[19][20]

I further restrict the sample to students who are exposed to an offender's school transfer at some point by age 16, excluding the never-treated group. This restriction makes the overlap assumption more plausible, as it requires the probability that a school receives an offender to be bounded in $(0, 1)$. Excluding never-treated students ensures that the estimation sample consists only of individuals with a positive probability of exposure. The first half of Table 1 describes this final step of the sample-selection process.[21] Columns (1) to (4) compare students who are ever exposed to an offender's school transfer with those who are never exposed. Students who are ever exposed are slightly more disadvantaged. However, these differences are small after conditioning on individual demographic characteristics, suggesting that this restriction is unlikely to substantially reduce external validity relative to the broader population. Within the resulting estimation sample, the treatment group consists of students who have already been exposed to an offender's transfer, while the control group consists of students who have not yet been exposed but will be later during secondary school. The final sample contains 1,050,567 individuals. and 15,758,505 panel records.[22]

The second half of the table compares students who ever transfer schools ("movers") with those who do not ("non-movers"). Movers are less likely to be male or belong to an ethnic minority, more

[19]Columns (1) to (4) of Table A3 highlight the differences in observable characteristics, among those in the educational dataset (NPD) with a full set of control variables, between those with a complete record of school censuses during secondary school and those without it. Those who do not have a complete record are more disadvantaged in many ways than those with a complete record. This could be for many reasons: one major possibility is that those without a complete record might be immigrants joining the English school system mid-secondary school, reflected in high minority status and low native-English speaker status. I omit those without a complete record, who account for about 10% of the original sample. Columns (5) to (8) describe the differences between those who had ever been in a very small school where the number of pupils in a school-grade is less than 20 and those who had not, among those with a complete record of school censuses during secondary school. The former group is more disadvantaged, although its sample size is small. Omitting them reduces the sample size further by 1.6%.

[20]In a robustness check, I further limit the sample to individuals who have never transferred to another school to fully control for school fixed effects in the staggered event study design, at the risk of ignoring many pupils who move schools.

[21]Columns (9) to (12) of Table A3 show the same differences but in more detail.

[22]I conduct a robustness check including the "never-treated" group, and the results do not change.

Table 1. Summary Statistics and Transfer Experience

	Ever Exposed to An Offender's School Transfer				Among Those in Column (1) Transfer by Age 16			
	Exposed	Never Exposed	Uncond. Diff. [(1) = (2)]	Cond. Diff. [(1) = (2)]	Non-movers	Movers	Uncond. Diff. [(5) = (6)]	Cond. Diff. [(5) = (6)]
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Demographics								
Male	.518	.494	.024***	.004***	.522	.507	.015***	.040***
			(.001)	(.001)			(.001)	(.002)
Minority	.185	.231	-.046***	.000	.193	.160	.033***	-.013***
			(.001)	(.000)			(.001)	(.001)
English as the first language	.902	.882	.019***	.001	.897	.918	-.021***	-.012***
			(.000)	(.000)			(.001)	(.001)
Ever with SEN by age 11	.365	.316	.049***	.003***	.352	.409	-.057***	-.061***
			(.001)	(.001)			(.001)	(.002)
Ever eligible for FSM by age 11	.296	.243	.054***	.004***	.273	.368	-.095***	-.131***
			(.001)	(.001)			(.001)	(.002)
Offender by age 16	.038	.021	.016***	.003***	.031	.060	-.029***	-.048***
			(.000)	(.000)			(.001)	(.001)
Any school transfer by age 16	.243	.176	.067***	-.006***	0	1	-	-
			(.001)	(.001)				
N of school transfers by age 16 | any transfer	1.18	1.14	.036***	-.012***	-	1.18	-	-
	[.445]	[.410]	(.001)	(.002)		[.445]		
B. Education								
Age 5 overall score	-.011	.059	-.070***	.016	.019	-.107	.126***	.032
	[.967]	[.985]	(.007)	(.008)	[.959]	[.985]	(.012)	(.019)
Age 7 overall score	-.030	.109	-.139***	-.002	.001	-.126	.128***	.058***
	[.961]	[.979]	(.001)	(.001)	[.953]	[.976]	(.002)	(.003)
Age 11 overall score	-.066	.140	-.206***	-.007***	-.016	-.225	.209***	.092***
	[.983]	[.981]	(.002)	(.002)	[.972]	[1.00]	(.003)	(.004)
Age 16 score: taken	1.00	1.00	.000	.000	1.00	1.00	.000***	.001***
			(.000)	(.000)			(.000)	(.000)
Age 16 score: 5 A* – G | taken	.857	.912	-.056***	-.006***	.878	.790	.088***	.082***
			(.000)	(.001)			(.001)	(.002)
Age 16 score: 5 A* – C | taken	.557	.676	-.119***	-.010***	.591	.450	.140***	.108***
			(.001)	(.001)			(.001)	(.002)
Age 16 score: point score | taken	.018	.178	-.160***	.001	.083	-.182	.265***	.320***
	[.912]	[.913]	(.001)	(.001)	[.887]	[.959]	(.002)	(.003)
Sample size	1,050,567	1,038,164	2,088,731	2,088,731	795,011	255,556	1,050,567	1,050,567

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard deviations are reported in brackets, and heteroskedasticity-robust standard errors clustered at the secondary-school level are reported in parentheses. I use the earliest available secondary-school record. Conditional differences control for gender, minority status, an indicator for English as the first language, and SEN and FSM status at age 11, unless the variable is itself the outcome. All test scores are standardized within each cohort, except for the indicator variables for the age 16 tests. Early-years test scores are available only for a quasi-random 10% of the sample comprising children born on the 5th, 15th, or 25th of each month (Department for Education, 2010). The age 7 overall score is calculated as a weighted average of teachers' assessments of Reading, Writing, Speaking and Listening, Math, and Overall Science. Within English, Reading, Writing, and Speaking and Listening are weighted 1:1:2 so that the four underlying abilities contribute equally. English, Math, and Science are then weighted 1:1:1 in constructing the overall score. The age 11 overall score is calculated as a weighted average of English, Math, and Science, with equal weight assigned to each subject. Some observations are missing information for certain variables. The reported sample size is the number of observations with demographic information. Data sources: Ministry of Justice and Department for Education.

likely to be native English speakers, and slightly more disadvantaged along several dimensions.[23] On average, movers transfer schools only once during secondary school.

Table A4 compares non-offenders and offenders. The patterns are consistent with those documented in Cattan et al. (2023*a*,*b*, 2025): offenders are substantially more likely to be male and White British, to have English as their first language, to come from disadvantaged backgrounds, and to have lower educational achievement. Offenders are also more likely to transfer to another school and, conditional on ever transferring, tend to make slightly more transfers.

Table 2 compares two types of offenders: (1) those who have transferred to another school *after* committing an offense and (2) those who have not, that is, offenders who never transferred schools or who transferred *only before* their first offense. I refer to the former as treaters, because their transfers generate treatment for other students, and to the latter as non-treaters.[24] Panel A shows that treaters are more likely to be male, to belong to an ethnic minority, to have English as an additional language, and to be less disadvantaged than non-treaters. They commit fewer offenses overall (Panel B) and commit their first offense at an older age (Panel C). There are only small differences in the types of offenses they commit, both for their first offense and overall, but treaters are more likely to receive a caution, the least severe sanction, rather than a more serious penalty (Panel D).[25] These patterns suggest that the peer effects identified from offenders' school transfers may provide a lower bound on the peer effects generated by offenders more generally.

Column (5) of Panel D reports descriptive statistics for treaters based only on offenses committed before their first transfer, providing a measure of the offending history that could plausibly be transmitted to peers. By construction, these offenses occur at younger ages and are more likely to result in a caution. Otherwise, the patterns are broadly similar to those in Columns (1) and (2). This comparison further suggests that the estimated peer effects might be a lower bound on the effects of exposure to offenders more generally.

Table 3 presents the age at school transfer separately for offenders and non-offenders. Offenders here are defined as students who had committed an offense before a given school transfer, while non-offenders are those who had not committed an offense before that transfer, including students who never offend. Columns (1) to (3) of Panel A report school transfers by offenders, while Columns (4) to (6) report transfers by non-offenders. The first term at age 11 marks the start of secondary school and is therefore excluded from my definition of a school transfer.[26] The modal

[23]Since movers are more likely to be offenders and more disruptive, a valid concern is that estimates of peer effects of offenders' transfers might be contaminated by the effects of disruptive students' transfers, not necessarily those of offenders (Billings and Hoekstra, 2023). Therefore, I conduct a robustness check by examining the effects of transfers of non-offenders with disruptive/disadvantaged characteristics, and it confirms that most peer effects are from offenders, not from disruptive non-offenders.

[24]This sample includes all offenders, not trimmed by the sample selection above, to minimize potential attrition and measurement error in transfer students' characteristics.

[25]Table A5 lists common youth crimes across different crime types.

[26]This timing could be used as another type of exogenous exposure to offenders (Lavy and Yancu, 2025). In NPD-PNC, we see few offenders at age 11 (Figure 1b), so I do not adopt this identification strategy in this paper.

Table 2. Offenders

	Offending Status by Age 16				
	Treaters	Non-treaters	Uncond. Diff. [(1) = (2)]	Cond. Diff. [(1) = (2)]	Treaters, Before First Transfers
	(1)	(2)	(3)	(4)	(5)
A. Demographics					
Male	.723	.706	.017***	.013***	-
			(.004)	(.004)	
Minority	.226	.185	.041***	-.003	-
			(.003)	(.003)	
English as the first language	.907	.936	-.030***	-.007***	-
			(.002)	(.002)	
Ever with SEN by age 11	.629	.682	-.054***	-.033***	-
			(.004)	(.004)	
Ever eligible for FSM by age 11	.607	.688	-.081***	-.056***	-
			(.004)	(.004)	
B. Crime Onset and Offense Numbers					
Number of offenses	2.65	3.66	–1.02***	–1.03***	-
	[3.76]	[4.99]	(.037)	(.039)	
Number of offenses per offending day	1.22	1.20	.019***	.012**	-
	[.658]	[.564]	(.005)	(.005)	
C. First Offense					
Age at first offense	13.8	12.5	1.28***	1.30***	-
	[1.64]	[1.46]	(.012)	(.012)	
Caution	.820	.882	-.062***	-.051***	
			(.003)	(.003)	
Theft/Robbery	.327	.332	-.006	-.015***	
			(.004)	(.004)	
Battery	.181	.199	-.017***	-.010***	
			(.003)	(.003)	
Drug Related	.061	.027	.035***	.031***	
			(.001)	(.002)	
Violent	.137	.133	.005	.008***	
			(.003)	(.003)	
Fraud	.020	.015	.005***	.004***	
			(.001)	(.001)	
Weapons	.037	.039	-.002	-.006***	
			(.002)	(.002)	
Arson/Damage	.158	.195	-.038***	-.033***	
			(.003)	(.003)	
Sexual	.026	.024	.002	.001	
			(.001)	(.001)	
Summary Motoring	.019	.010	.008***	.008***	
			(.001)	(.001)	
Others/Unknown	.035	.026	.009***	.012***	
			(.001)	(.001)	
Sample size	72,859	20,953	93,812	93,812	
D. All Offenses					
Offense age	14.1	13.6	.530***	.551***	12.8
	[1.51]	[1.62]	(.008)	(.008)	[1.46]
Caution	.526	.466	.061***	.064***	.690
			(.002)	(.002)	
Theft/Robbery	.317	.317	-.000	-.010***	.336
			(.002)	(.002)	
Battery	.169	.175	-.006***	-.003	.190
			(.002)	(.002)	
Drug Related	.064	.045	.019***	.015***	.029
			(.001)	(.001)	
Violent	.131	.132	-.001	.003	.129
			(.002)	(.002)	
Fraud	.022	.020	.002***	.002***	.017
			(.001)	(.001)	
Weapons	.030	.027	.002***	-.000	.031
			(.001)	(.001)	
Arson/Damage	.151	.166	-.015***	-.009***	.191
			(.002)	(.002)	
Sexual	.019	.016	.004***	.004***	.021
			(.001)	(.001)	
Summary Motoring	.024	.022	.002***	.002***	.015
			(.001)	(.001)	
Others/Unknown	.073	.080	-.007***	-.004***	.042
			(.001)	(.001)	
Sample size	155,327	62,143	217,470	217,470	35,109

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard deviations are reported in brackets, and heteroskedasticity-robust standard errors clustered at the secondary-school level are reported in parentheses. I use the earliest available secondary-school record. Conditional differences control for gender, minority status, an indicator for English as the first language, and SEN and FSM status at age 11, unless the variable is itself the outcome. Offenders in this table are defined as individuals in the NPD with at least one recordable offense in the PNC committed between the September ages of 11 and 16 that resulted in a caution or conviction. Treaters are offenders who transferred to another school after committing their first offense. Non-treaters are offenders who either never transferred during secondary school or committed their first offense only after their final transfer to another secondary school, so that no subsequent transfer could induce treatment. Column (5) reports summary statistics for offenses committed by treaters before their first school transfer. Offense types are based loosely on the offense groups defined by the MoJ. The order of offenses is determined as follows. First, offenses are sorted chronologically for each offender based on the offense date. An offense committed on the earliest date is classified as the first offense, and subsequent offenses are ordered analogously. If multiple offenses occur on the same date, they are ordered according to the severity of the penalty, the length of any custodial sentence, and the amount of any fine, in that order. If these are all identical, one offense is selected at random. Thus, only one offense is retained for each offense date in this table. Panel D includes only offenses committed during the secondary-school period. Data sources: Ministry of Justice and Department for Education.

Table 3. Age of School Transfers

Age in Sept.	Offenders' Transfers			Non-Offenders' Transfers		
	Autumn (1)	Spring (2)	Summer (3)	Autumn (4)	Spring (5)	Summer (6)
A. Transfer						
11	-	423	722	-	18,856	30,998
12	2,059	1,131	1,388	87,990	22,859	31,001
13	4,308	2,074	1,733	146,986	24,126	19,517
14	4,428	2,510	1,838	88,152	20,340	14,235
15	3,104	2,944	936	40,334	16,364	10,217
B. First Contact to Others' Transfers						
11	-	31,518	59,783	-	1,269,477	434,327
12	98,326	75,242	102,565	270,010	44,740	19,738
13	122,645	106,842	112,271	20,164	8,026	2,775
14	115,099	70,539	77,019	3,798	1,704	682
15	55,483	14,881	8,354	2,255	10*1	

Notes: Offenders are those who have committed a crime by the time of their school transfer, while non-offenders are those who have never committed a crime or have not yet committed a crime by the time of their school transfer. Panel A counts school transfers rather than individuals; if an individual transfers schools multiple times, each transfer is counted separately, while Panel B counts individuals only once.
∗1: There are 10 transfers for Spring and Summer at age 15 combined. Showing them separately is disclosive.

age of transfer is 13 for non-offenders and 14 for offenders. This difference partly reflects the fact that many children have not yet begun offending by age 13 (Figure 1b).

Panel B reports the age at which students are first exposed to a peer's school transfer. Columns (1) to (3) show the timing of first exposure to an offender's transfer, that is, the timing of treatment, while Columns (4) to (6) show first exposure to a non-offender's transfer. Most students are first exposed to a non-offender's transfer at the beginning of secondary school, whereas first exposure to an offender's transfer is more evenly distributed across ages. The distribution of the latter peaks at age 13, and the main peer-effect estimates are therefore identified primarily from students first exposed around this age.

Finally, I describe the characteristics of the origin and destination schools of transfer students. Columns (1) to (4) of Table 4 show the origin and the destination schools of offenders' transfers. Offenders tend to transfer to schools with a higher concentration of boys, white British students, children whose first language is English, students eligible for FSM and SEN by age 11, students with lower educational achievement, students with a prior offense history, and fewer students. If I control for other variables, all magnitudes become smaller. Although the disparity is not large, offenders tend to move to a slightly more disadvantaged school.

Columns (5) to (8) report the corresponding statistics for non-offenders' transfers. Differences between origin and destination schools are generally small, except for school size: non-offenders tend to move to larger schools. Columns (9) and (10) compare the transfer patterns of offenders

Table 4. Summary Statistics on Schools and School Transfers

		Transfer (school-term) level									
		Offenders' transfers				Non-offenders' transfers				Difference-in-differences	
	All school	Origin school	Destination school	Uncond. Diff. [(2)=(3)]	Cond. Diff. [(2)=(3)]	Origin school	Destination school	Uncond. Diff. [(6)=(7)]	Cond. Diff. [(6)=(7)]	Uncond. Diff. [(4)=(8)]	Cond. Diff. [(4)=(8)]
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Male	.531	.536	.544	-.008***	-.002**	.515	.516	-.000*	-.001***	-.008***	-.002
	[.174]	[.133]	[.122]	(.001)	(.001)	[.126]	[.125]	(.000)	(.000)	(.001)	(.001)
Minority	.216	.193	.185	.008***	-.002**	.192	.184	.008***	.000**	.000	-.003***
	[.256]	[.235]	[.222]	(.002)	(.001)	[.242]	[.231]	(.000)	(.000)	(.002)	(.001)
English as the first language	.885	.890	.899	-.009***	-.004***	.891	.896	-.005***	-.001***	-.004**	-.003***
	[.174]	[.167]	[.153]	(.002)	(.001)	[.174]	[.165]	(.000)	(.000)	(.002)	(.001)
Ever with SEN by age 11	.416	.422	.444	-.022***	-.014***	.387	.383	.003***	-.009***	-.026***	-.004***
	[.199]	[.143]	[.158]	(.001)	(.001)	[.122]	[.123]	(.000)	(.000)	(.001)	(.001)
Ever eligible for FSM by age 11	.331	.378	.397	-.019***	-.006***	.325	.319	.006***	-.005***	-.025***	.000
	[.197]	[.188]	[.200]	(.002)	(.001)	[.178]	[.173]	(.000)	(.000)	(.002)	(.001)
Early years overall score	-.152	-.013	-.015	.003	-.004	-.006	-.005	-.000	-.000	.003	-.002
	[.686]	[.060]	[.049]	(.003)	(.003)	[.046]	[.045]	(.000)	(.000)	(.003)	(.003)
Age 7 overall score	-.210	-.132	-.168	.036***	.002	-.048	-.042	-.005***	.004***	.041***	-.007***
	[.744]	[.307]	[.338]	(.003)	(.002)	[.306]	[.316]	(.001)	(.000)	(.003)	(.002)
Age 11 overall score	-.180	-.201	-.227	.026***	.008***	-.138	-.127	-.010***	.001***	.036***	.003
	[.590]	[.283]	[.283]	(.003)	(.002)	[.280]	[.281]	(.001)	(.000)	(.003)	(.002)
Offender	.042	.054	.091	-.036***	-.028***	.022	.027	-.005***	-.006***	-.032***	-.027***
	[.101]	[.103]	[.150]	(.001)	(.001)	[.035]	[.047]	(.000)	(.000)	(.001)	(.001)
N of students	154	169	179	-9.28***	-16.2***	164	210	-46.3***	-45.4***	37.0***	31.1***
	[75.6]	[73.1]	[92.8]	(.800)	(.669)	[65.0]	[94.3]	(.153)	(.143)	(.815)	(.695)
Sample size	4,559	21,803	21,803	43,606	43,606	559,419	559,419	1,118,838	1,118,838	1,162,444	1,162,444

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard deviations are reported in brackets, and heteroskedasticity-robust standard errors clustered at the secondary-school level are reported in parentheses. I use the earliest available secondary-school record. Offenders in this table are defined as individuals in the NPD who, by the time of their school transfer, had committed at least one recordable offense in the PNC between the September ages of 11 and 16 that resulted in a caution or conviction. Non-offenders are individuals who had not committed such an offense by the time of their transfer, including those who never committed an offense. Column (1) reports averages of school-level characteristics, calculated by first taking the average of each outcome within each school. Columns (2), (3), (6), and (7) report leave-one-out averages of school-level outcomes. The unit of observation is a school transfer. Conditional differences control for school-level measures of gender, minority status, an indicator for English as the first language, and SEN and FSM status at age 11, unless the variable is itself the outcome. All test scores are standardized within each cohort. Early-years test scores are available only for a quasi-random 10% of the sample comprising children born on the 5th, 15th, or 25th of each month (Department for Education, 2010). The age 7 overall score is calculated as a weighted average of teachers' assessments of Reading, Writing, Speaking and Listening, Math, and Overall Science. Within English, Reading, Writing, and Speaking and Listening are weighted 1:1:2 so that the four underlying abilities contribute equally. English, Math, and Science are then weighted 1:1:1 in constructing the overall score. The age 11 overall score is calculated as a weighted average of English, Math, and Science, with equal weight assigned to each subject. Some observations are missing information for certain variables. The reported sample size is the number of observations with demographic information. Data sources: Ministry of Justice and Department for Education.

and non-offenders. Most differences are small, with the notable exception of school size: offenders tend to move to smaller schools than non-offenders. Overall, transfer students tend to move to slightly more disadvantaged schools. This pattern is reasonable because schools accepting new students must have available places. Importantly, as discussed in the next section, my main identification strategy exploits variation in the timing of school transfers by comparing treated students with not-yet-treated students, allowing me to control for school characteristics.

4 Identification

This section outlines the empirical strategy used to identify the causal peer effects of exposure to others' criminal behavior at school. Exposure to offenders is generally endogenous. For example, a student may be bullied and subsequently commit an offense against the students who bullied them in retaliation. In such a case, exposure to the retaliatory offense is clearly not exogenous. Identifying causal peer effects therefore requires plausibly exogenous variation in exposure to offending.

This paper exploits the timing of offenders' school transfers as a plausibly exogenous shock. As discussed in Subsection 2.1, these transfers cannot be anticipated by other students, their parents, or even the teachers responsible for the receiving classrooms. Therefore, conditional on an offender transferring into a school-cohort, the timing of peers' exposure to that offender is plausibly exogenous to their behavior prior to the transfer at the school-term level.[27] In other words, after conditioning on a rich set of observable characteristics, the remaining variation in exposure generated primarily by the timing of offenders' school transfers is plausibly random.[28] Because the timing of offenders' school transfers varies across individuals, I adopt a staggered event-study design:

$$y_{ijst} = \sum_{k \neq -1} \beta_k \mathbb{1}\{D_{it+k} = 1\} + \lambda_{js} + \gamma_t + \boldsymbol{x}_{ijst}^{\mathrm{T}} \boldsymbol{\theta} + \varepsilon_{ijst}$$

where y_{ijst} denotes an outcome for pupil i in grade j of school s at time t, such as an indicator for whether individual i has committed an offense *by* time t, an indicator for whether individual i has committed an offense *at* time t, and the number of offenses committed.[29] The first one captures criminal onset, while the latter two variables capture overall criminal activity. D_{it} denotes the treatment status of individual i at time t, $\boldsymbol{x}_{ijst}$ is a vector of predetermined individual-level

[27] As explained in Subsection 2.1, the standard timing of school transfers is at the beginning of school terms. The randomness I exploit here is across the timing of school transfers at the start of terms over secondary school.

[28] Balsa et al. (2023) uses the same strategy in their analysis of the peer effects of students' mental disorders. It is close to the identification in Bayer, Hjalmarsson, and Pozen (2009), which views the timing of prison inmates' entry into jail as a random shock to incumbent prisoners. It is also close to Fadlon and Nielsen (2019, 2021), which view the timing of a health shock as exogenous variation.

[29] I do not control for the individual fixed effects, as most children have never committed a crime, whose behaviors are totally absorbed in the fixed effects and the remaining variation would be very small.

characteristics, and ε_{ijst} is an error term. λ_{js} and γ_t denote school-cohort and time fixed effects, respectively.

I estimate the coefficients β's following Callaway and Sant'Anna (2021).[30] In the main specification, I condition on a linear age term; individual characteristics including gender and FSM and SEN status by age 11; an indicator for whether English is the individual's first language; a minority indicator; and a linear month-of-birth term. Standard errors are clustered at the level of exposure, defined by the school-cohort-timing cell. I drop records at the time of transfer to avoid conflating the effects of exposure to offenders with those of entering a new school environment.

To identify the average treatment effect on the treated, I require several assumptions. First, I assume that post-treatment outcomes would have followed parallel trends in the absence of treatment conditional on observed covariates. Following standard practice, I examine pre-treatment trends and use their similarity as supporting evidence for the plausibility of parallel counterfactual trends after treatment (Roth, 2026).

Second, I assume no treatment anticipation. That is, children and their parents cannot predict whether an offender will transfer into their school and grade and therefore cannot adjust their behavior in advance. As discussed in Subsection 2.1, children, their parents, and even the teachers responsible for the receiving classroom do not know whether they will receive a transfer student before the transfer occurs. This institutional setting makes the no-anticipation assumption plausible. Moreover, any anticipatory behavioral response would be reflected in the pre-treatment trend.[31]

Third, I assume overlap: the probability of each school receiving an offender is bounded in $(0, 1)$. I exclude the never-treated group from the sample so that the estimation sample consists only of students who are exposed at some point during secondary school, making the overlap assumption more plausible.

Fourth, I assume treatment irreversibility, meaning that once students are exposed to an offender, their treatment status cannot revert to untreated. In this setting, this assumption is natural because exposure cannot be undone once contact with an offender has occurred.[32]

Finally, I assume random sampling, such that the outcome, pre-treatment characteristics, and treatment status are independently and identically distributed across observational units. Under these assumptions, the average treatment effect on the treated can be identified and estimated.

[30] See Appendix B for the formal setup of the estimation.

[31] I check their behaviors in offending and school transfers to avoid treatment anticipation, which would show up as a pre-trend, and there is no evidence of anticipation, supporting plausibility of this assumption.

[32] As a robustness check, I repeat the analysis censoring the panel records if pupils are treated a second time, and the results do not change. This robustness suggests the validity of the irreversibility assumption.

5 Results

5.1 Peer effects of crime

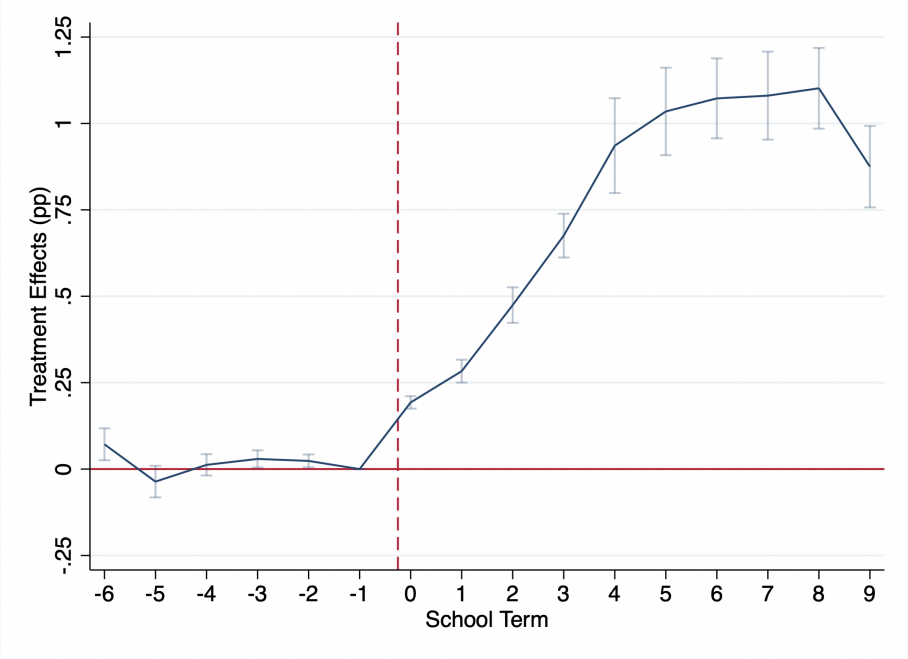
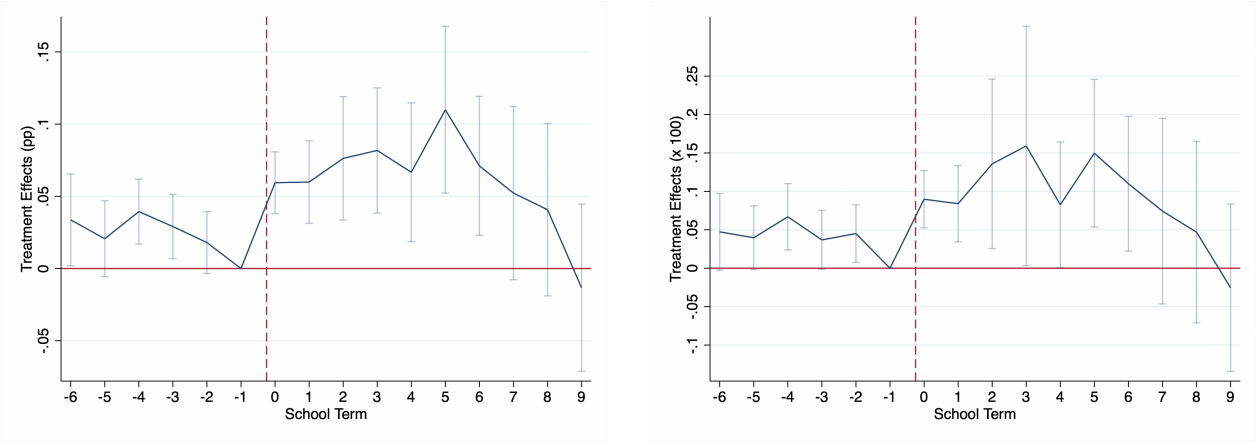


Figure 2. Effects of Receiving an Offender's Transfer on Offense Initiation

Note: This figure reports the treatment effects of exposure to offenders transferring into students' school-grades on their peers' criminal behavior. The treatment effect is normalized at $t = -1$ and multiplied by 100 for visibility. The treatment group consists of students who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021). The bars show 95% confidence intervals based on standard errors clustered at the treatment level. Data sources: Ministry of Justice and Department for Education.

Having discussed the richness of the data and the plausibility of the identification strategy, I now turn to the main findings on the peer effects of crime in schools. I begin with the visual evidence. Figure 2 plots the dynamic treatment effects on offense initiation over six pre-treatment and nine post-treatment school terms, spanning approximately two years before and three years after an offender transfers into a peer's school and grade. I use a two-year pre-treatment window and a three-year post-treatment window because treatment is most common around the Autumn Census at age 13 (Table 3), and these periods approximately cover the full duration of secondary school for this population. I normalize the treatment effect at -1 and multiply the estimates by 100, so the y-axis in Figure 2 can be interpreted as the percentage-point change in the probability of offense initiation. First, there is no evidence of a systematic pre-trend difference between the treatment and control groups. This finding supports the plausibility of both the parallel-trends assumption in the absence of treatment and the no-anticipation assumption. Turning to the main result, exposure to an offender's school transfer increases the probability that peers begin offending. At the time of the transfer, the probability of becoming an offender increases by 0.193

percentage points. This magnitude implies that approximately one new offender is induced for every three offender transfers.[33] The effect continues to grow for approximately two years, reaching 1.1 percentage points. Thereafter, it levels off but does not disappear, indicating a persistent gap in cumulative offense initiation between treated and not-yet-treated students.[34]



(a) Any Crime at Each Period

(b) Number of Crimes at Each Period

Figure 3. Effects of Receiving an Offender's Transfer on Overall Criminal Activities

Note: This figure reports the treatment effects of exposure to offenders transferring into students' school-grades on their peers' criminal behavior. The treatment effect is normalized at $t = -1$ and multiplied by 100 for visibility. The treatment group consists of students who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. I omit observations when they move to another school. The estimation follows Callaway and Sant'Anna (2021). The bars show 95% confidence intervals based on standard errors clustered at the treatment level. Data sources: Ministry of Justice and Department for Education.

Figure 3 shows the effects on overall criminal activity: whether pupils commit an offense in each term and the number of offenses they commit in each term, including among students who had already begun offending. Although the pre-trends are slightly noisier, there is a clear increase at time 0.[35] Once pupils are exposed to an offender's school transfer at time 0, the treatment effects on both the probability of committing an offense and the number of offenses increase sharply. These effects continue to rise for approximately one year, remain positive for around two years, and fade toward zero within three years. This pattern is consistent with the slowdown in the growth of the extensive-margin effect after approximately two years in Figure 2.

There are two possible explanations for why the treatment effect on overall criminal activity fades over time. One is that the increase in offending among treated pupils is temporary. The

[33]This number comes from a back-of-the-envelope calculation: (the effect size) $\times$ (the number of students in the destination school in Table 4) $= 0.00193 * 178 = 0.345$, meaning that there is a new offender at the destination school upon the arrival of an offender with 34.5% probability.

[34]If we examine the effects separately on caution and conviction experiences in Figure F1, we see the same pattern.

[35]According to Table 4, the destination school is slightly more disadvantaged. This small pre-trend might reflect this fact. To assess how much this pre-trend matters, I conduct a robustness check using the method proposed by Rambachan and Roth (2023), which bounds the treatment effects based on the pre-trend patterns. Figures C7 and C8 show the result. Even under a very conservative assumption and regardless of the method, the results are robust and we see clear treatment effects.

Table 5. Effects of Offenders' Transfers

Effects on:	Crime Onset (1)	Any Crime (2)	Number of Crimes (3)
Treatment Effects at $t = 0$	.193***	.059***	.090***
	(.009)	(.011)	(.019)
Romano-Wolf Step-down p-value	[.000]	[.000]	[.000]
Pre-Treatment Mean	1.50	.273	.369
Relative Effect Size	12.9%	21.8%	24.3%
Sample Size (Panel Count)	15,458,144	15,458,144	15,458,144
Number of Individuals	1,050,567	1,050,567	1,050,567

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. p-values based on the Romano-Wolf step-down procedure are reported in brackets (Romano and Wolf, 2005; Clarke, Romano, and Wolf, 2020). Standard errors reported in parentheses are clustered at the treatment level. Control variables include FSM and SEN status before secondary school, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, a linear month-of-birth term, and a linear age term. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. Relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Data sources: Ministry of Justice and Department for Education.

other is that pupils in the control group begin offending before they themselves are treated, so that treatment primarily shifts the timing of criminal activity forward. Taken together with the offense-initiation results in Figure 2, the evidence is more consistent with the former interpretation. During the first two years after treatment, the effect on offense initiation continues to grow while the effect on contemporaneous offending remains positive. Once the latter approaches zero, the initiation effect also stops increasing. This pattern suggests that exposure to an offender produces a relatively short-lived increase in contemporaneous criminal activity but a persistent difference in the probability of acquiring an offense record. Such an effect can have longer-run consequences for human capital and potentially for labor market outcomes.[36]

Table 5 summarizes the treatment effects at $t = 0$, together with the pre-treatment mean, calculated as the mean of each outcome one school term before treatment.[37] All coefficients are multiplied by 100 for visibility. Column (1) reports the effect on whether pupils have begun offending by each school term. The contemporaneous treatment effect within one school term, approximately four months, is 0.193 percentage points, corresponding to a 12.9% increase relative to the pre-treatment mean of 1.50%. Columns (2) and (3) report the effects on whether pupils

[36] As discussed in Subsection 2.2, during my data period, crime records could be shared with employers at the job applications upon their request.

[37] Since the offending activities are monotonically increasing almost always during secondary school, according to Figure 1, these means could be slightly underestimated. However, this is still the best possible reference point for understanding how large the treatment effects are, so I show them despite this caveat.

commit an offense in a given term and on the number of offenses committed in that term, respectively. The treatment effect on the former is 0.00059, representing a 21.8% increase relative to the pre-treatment mean of 0.00273, while the treatment effect on the latter is 0.00090, corresponding to a 24.3% increase.

Because the three outcome variables are likely to be highly correlated, testing them separately raises the probability of making a Type I error. Therefore, I apply the Romano-Wolf stepdown procedure to control the family-wise error rate (Romano and Wolf, 2005; Clarke, Romano, and Wolf, 2020). I report the adjusted p-values, and the main results remain statistically significant. Overall, the findings show that exposure to an offender's school transfer produces a sizable immediate increase in criminal activity and a persistent increase in the probability of entering the criminal justice system. In most of the analyses that follow, I focus on the effect on crime onset because it most directly captures whether exposure induces a student to acquire an offense record.

5.2 Robustness Checks

In the previous subsection, we see that an offender's school transfer accelerates peers' criminal activity, both in offense initiation and overall criminal activity. Here, I present robustness checks showing that these effects are attributable to offenders' school transfers.

First, I repeat the main analysis focusing on not-yet offenders: students who had not offended by the time of their transfer but did so by the end of secondary school. They are more similar to offenders than non-offenders in characteristics such as disruptiveness, except that they have no prior offense history.[38] Figure C1 shows no significant effects before or after treatment, suggesting that prior offending experience is important for crime transmission.

Not-yet offenders may nevertheless be less disruptive than offenders. Thus, I examine transfers of non-offenders with disadvantaged or disruptive characteristics. Disadvantage is proxied by eligibility for free school meals and special education needs before transfer, while disruptiveness is proxied by prior school suspensions. Panel A of Table C2 shows positive and significant effects for some of these groups, consistent with the literature on disruptive peers (Billings, Deming, and Ross, 2019; Billings and Hoekstra, 2023).[39] However, the magnitudes are much smaller than the main treatment effects. Moreover, conditioning on offenders' disadvantaged or disruptive characteristics changes the estimates little. Together, these results suggest that prior offending is the key characteristic driving crime transmission.

I further separate school effects from neighborhood effects. Previous studies document peer effects within neighborhoods (Dustmann and Landersø, 2021; Billings and Schnepel, 2022), while

[38] Also, according to Tables 3 and C1, the distribution of the timing of exposure to offenders' school transfers is closer to that of not-yet offenders' school transfers than that of non-offenders. Even with this similarity, I use the weight to adjust for differences in exposure timing across grades and seasons of school censuses.

[39] Column (1) of Panel A in Table C2 shows the effects of not-yet offenders' transfer, which is a replication of Figure C1.

Table 6. Locality

Locality Measure:	School	MSOA			LSOA		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Effects on Crime Onset							
Treatment Effects at $t = 0$	.193***	.138***	.158***	.122***	.168***	.177***	.135***
	(.009)	(.009)	(.013)	(.011)	(.018)	(.012)	(.023)
Pre-Treatment Mean	1.50	1.51	1.16	1.17	2.16	1.33	1.59
Relative Effect Size	12.9%	9.12%	13.6%	10.5%	7.78%	13.3%	8.50%
Panel B: Effects on Number of Crimes							
Treatment Effects at $t = 0$	.090***	.032*	.098***	.052**	.030	.089***	.012
	(.019)	(.019)	(.021)	(.022)	(.037)	(.020)	(.040)
Pre-Treatment Mean	.369	.374	.289	.303	.505	.329	.410
Relative Effect Size	24.3%	8.47%	33.9%	17.1%	5.99%	27.2%	2.89%
School	Same	Any	Same	Different	Any	Same	Different
Locality	Any	Same	Different	Same	Same	Different	Same
Sample Size (Panel Count)	15,458,144	16,617,101	9,883,495	10,952,721	5,969,939	13,489,046	3,504,026
Number of Individuals	1,050,567	1,127,226	668,104	738,093	406,544	912,593	236,932

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. Control variables include FSM and SEN status before secondary school, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, a linear month-of-birth term, and a linear age term. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. MSOA stands for Middle Layer Super Output Area, a geographic unit containing approximately 5,000 to 15,000 residents. LSOA stands for Lower Layer Super Output Area, a geographic unit containing approximately 1,000 to 3,000 residents. In Columns (3) and (6), I exclude students who are treated at the neighborhood level before or at the same time as they are treated at the school level. Similarly, in Columns (4) and (7), I exclude students who are treated at the school level before or at the same time as they are treated at the neighborhood level. Data sources: Ministry of Justice and Department for Education.

others emphasize the combined role of schools and neighborhoods (Billings, Deming, and Ross, 2019). I distinguish these channels by comparing offenders who transfer into the same school and/or neighborhood. Table 6 summarizes the contemporaneous treatment effects of these transfers.

I use two geographic units for this analysis. The first is the Middle-layer Super Output Area (MSOA), which contains roughly 2,000 to 6,000 households and 5,000 to 15,000 residents. Given that the average secondary school size is about 150 (Table 4), MSOAs are broadly comparable in size to school catchment areas. The second is the Lower-layer Super Output Area (LSOA), a finer geographic unit containing 400 to 1,200 households and 1,000 to 3,000 residents. LSOAs are similar in size to US census tracts.

Column (1) replicates Table 5, while Columns (2) and (5) report the effects of offenders' transfers into the same locality.[40] Comparing Columns (1), (2), and (5), I find evidence of neighborhood peer effects, consistent with prior studies (Dustmann and Landersø, 2021; Billings and Schnepel, 2022), but the effects of exposure through school are substantially larger.

I next distinguish the two channels more directly by comparing offenders who move into the

[40]The move into the neighborhood is measured by the change in the LSOA registered in the school census. Therefore, I can measure only up to three times a year. If they move more than once during the school census period, this will introduce measurement error, although moving to another place within 4 months is very unlikely.

same school but a different neighborhood with those who move into the same neighborhood but a different school. The LSOA analysis has relatively few observations because children living very close to one another are also likely to attend the same school, making the estimates slightly noisier. Despite this limitation, the estimated school effects in Columns (3) and (6) are substantially larger than the neighborhood effects in Columns (4) and (7) in both the MSOA and LSOA comparisons. One possible explanation is that students attending the same school interact much more frequently than children who simply live in the same area. Overall, these results suggest that a large fraction of the peer effects of crime operate through offenders' entry into peers' schools rather than through residential proximity.

If crime transmission occurs mainly through school interactions, spillovers should be stronger during school terms and on weekdays. Table C3 confirms this pattern: treatment effects are substantially larger during school terms than holidays and larger on weekdays than weekends. This provides further evidence that transmission occurs primarily through school interactions.[41]

In the main analysis, I include students who change schools in order to capture the overall effects of exposure. However, if a student in the control group transfers into a school-grade that has already been treated, that student may be surrounded by treated peers despite not yet having been directly exposed to an offender's transfer. Although this does not necessarily violate SUTVA, it may create contamination between treated and control groups. To address this concern, I censor each student's panel at their first school transfer, at the risk of losing the panel structure of the data. Figure C2 shows that the results are unchanged.

As Table 3 shows, some pupils are exposed to multiple offender transfers. Because the irreversibility assumption treats exposure as absorbing, subsequent treatments should not alter treatment status. Therefore, I censor each student's panel at their second treatment. Figure C3 shows that the results are unchanged, supporting the plausibility of the irreversibility assumption.

Although my identification relies on the conditional parallel-trends assumption and the plausibility of the exogenous shock, it is also informative to report unconditional estimates. Importantly, conditional parallel trends do not imply unconditional parallel trends. Figure C4 presents the results without controls. The effect on crime onset in Figure C4a exhibits a pre-treatment trend difference. However, this difference is approximately linear, with a clear jump at time zero, providing additional suggestive evidence of peer effects. To assess the sensitivity of the estimates to this pre-trend, I apply the method of Rambachan and Roth (2023), which allows for deviations from parallel trends based on the observed pre-treatment pattern. Figures C5a and C6a show that the estimated effects remain robust after accounting for the pre-trend.[42] The effects on overall criminal activities with the unconditional specification in Figures C4b and C4c do not show any

[41] Children might commit a different type of crime in and out of school. See Boshoff, Machin, and Sandi (2026) for a discussion of this.

[42] As a further robustness check, I use outcome variables residualized with respect to demographic characteristics. The pattern of results remains unchanged in Figure C9.

pre-trend difference, and the main result still holds.[43]

Next, I conduct additional checks of the identifying assumptions and sample restrictions. Because the main analysis excludes never-treated students, I re-estimate the model using the never-treated group as the control to assess external validity.[44] Figure C10 shows similar immediate treatment effects. Although the effect on overall activity remains positive even after three years, the main result is unchanged. This suggests that excluding the never-treated group is unlikely to drive the findings and supports the external validity of the main finding to the rest of the population.

If peers transfer to another school in response to an offender's arrival, the treatment effects of exposure might be underestimated. To assess this possibility, I examine whether offenders' school transfers increase the probability that incumbent students transfer out. Figure C11 shows no evidence of such a response.

I also examine what happens when offenders themselves leave a school. Because transfer-outs may be endogenous, these estimates should be interpreted as descriptive rather than causal. Table C4 shows that a peer's departure increases offense initiation *regardless of* whether the departing student is an offender, not-yet offender, or non-offender, while there is no effect on overall criminal activity when offenders or non-offenders leave. Thus, there is no evidence that removing offenders reduces crime transmission. Consistent with this result, Figure C12 shows that the main estimates are unchanged when I restrict the sample to schools from which no offender leaves.

A further concern is that offender transfers may be endogenous because offenders or their parents choose to move.[45] This may threaten identification if peers can anticipate which offender will enter their school-grade before the transfer occurs. To examine this possibility, I study heterogeneity by whether, and how far, the offender changes residence around the time of the school transfer. Transfers accompanied by longer-distance residential moves are more likely to be unexpected by peers at the destination school. Such moves may also be more likely to reflect parental circumstances, such as job changes, rather than factors directly related to the offender. Table C5 shows slightly larger treatment effects when offenders move farther away, consistent with this interpretation. However, sizable effects remain even when there is no residential move. This provides little evidence that anticipatory information spillovers within neighborhoods drive the main results.

In relatively rare cases, an offender who transfers out later returns to the original school, in which case exposure is less plausibly a new shock. Excluding these cases leaves the results unchanged (Figure C13).

[43] Nevertheless, I run the honest difference-in-differences. Figures C5b, C6b, C5c, and C6c confirm the results are robust to the specification.

[44] Since the sample size is too large for the memory that I work with, I draw a random 50% of individuals for this analysis.

[45] Another case of an offender's school transfer is due to permanent exclusion. However, permanent exclusion is a very rare punishment and does not affect my main analysis (Cattan et al., 2023*a*,*b*).

Finally, because the main outcomes are binary and crime is relatively rare, I examine whether the linear specification drives the findings. I estimate Poisson and logit models following Wooldridge (2023, 2025). Table C6 shows qualitatively the same pattern, confirming that the main findings are robust to alternative functional forms.

5.3 Heterogeneity and Potential Mechanisms

So far, I have documented peer effects of crime arising from offenders who transfer into their peers' school and grade. I now investigate the mechanisms underlying these effects. I first summarize the theoretical predictions from a human capital framework in Lochner (2004), and then examine heterogeneity in the estimated treatment effects in light of these predictions.

Lochner (2004) models crime as a dynamic choice over the allocation of time among education, work, and criminal activity. In this framework, an individual commits a crime when the expected net return from doing so exceeds the expected net cost.

For my setting, the net return depends primarily on (i) criminal capital and (ii) idiosyncratic shocks, while the net cost depends on (a) the expected loss from being caught and acquiring a criminal record and (b) the expected penalty.[46]

(i) Criminal capital: Individuals accumulate criminal capital alongside human capital. Exposure to an offender may provide peers with information about criminal opportunities, techniques, or behaviors, thereby increasing their criminal capital. The strength of this channel may depend on the offender's criminal experience and on the intensity of interaction between the offender and peers. It may also be stronger when peers have more opportunities to interact with the offender or share characteristics that facilitate communication. If criminal-capital transmission is important, peers should become more likely to commit the same types of crimes as the transferring offender, as in Bayer, Hjalmarsson, and Pozen (2009).

(ii) Idiosyncratic shock: Individual characteristics and preferences may amplify the treatment effect. For example, students from disadvantaged backgrounds may receive relatively greater returns from crimes with monetary benefits, such as theft or fraud. Some characteristics may also make an offender more influential as a role model, increasing peers' propensity to offend, which can be viewed as an i.i.d. taste. Further, offenders who commit riskier crimes associated with more severe punishment or who have more extensive criminal histories may attract greater attention from peers and become more salient role models.

[46] In the original discussion of Lochner (2004), there are two more elements for the net return and another element for the net cost exist: human capital and time spent to commit a crime for net return and return to work in addition to an i.i.d. shock for wage for net cost. I focus on secondary schools, and most of them are light crimes, so there is no variation in time spent to commit a crime and required skill level, and I drop these factors.

(a) Marginal expected loss when being caught: Being caught can lower an individual's future value because a criminal record may reduce subsequent opportunities. As discussed in Subsection 2.2, criminal records can damage human capital by limiting access to the labor market. The resulting opportunity cost is likely to vary with expected earnings, which depend on educational achievement, demographic and socioeconomic characteristics, and local deprivation. Students with better educational outcomes or more advantaged backgrounds may therefore face a greater cost of acquiring a criminal record and respond less strongly to exposure.

(b) Expected penalty: The expected severity of punishment also affects the decision to offend. Exposure to an offender may change peers' perceptions of these penalties. In particular, if observing an offender leads peers to revise downward their beliefs about the likelihood or severity of punishment, their propensity to commit a crime may increase.

I investigate these channels through heterogeneity in the treatment effects, summarized in Table 7.

Table 7. Summary of Heterogeneities and Mechanisms Tested

Heterogeneity	Category	Potential Mechanisms	Table and Figure
Return side			
Treatment intensity	(i), (ii)	(i) More opportunity to develop criminal capital and (ii) to have a new role model	Table D1
Demographics	(i), (ii)	(i) Sharing characteristics enables (i) smooth skill transfers and (ii) being a new role model	Table D2
Offender-dense schools	(i), (ii)	(i) New information to commit crime and (ii) a new role model	Table 8
Offenders' experience	(i), (ii)	(i) Accumulated skill transmission and (ii) influential role model	Table D3
Crime types	(i)	Accumulated skill transmission	Table 9
Joint crime	(i)	Organized Crime	Figure D1
Behaviors of transferring offenders	(i)	Organized Crime	Figure D2
Seniority of transferring offenders	(ii)	Role model effect	Table 10
Treatment age	(ii)	Role model effect	Table D4
Cost side			
Educational achievement	(a)	Opportunity cost	Table 11
Early-year skills	(a)	Opportunity cost	Table D5
FSM and SEN	(a)	Opportunity cost	Table D6
Local deprivation	(a)	Opportunity cost and returns to crime	Table 12
School Types	(a)	Opportunity cost through school discipline and suspension	Table D7
Punishment	(b), (ii)	(b) Updates in expected penalty and (ii) role model	Table 13

(i) Criminal capital; (ii) Idiosyncratic shock; (a) Marginal expected loss when being caught; (b) Expected penalty.

5.3.1 Channels of the return side

First, I examine whether greater opportunities to interact with transfer offenders amplify the spillovers. I define treatment intensity as the fraction of offenders transferring into a school-grade relative to the number of pupils in that grade and divide the sample into quartiles. Table D1 reports the results. Panel A shows the mean treatment intensity in each quartile. In the lowest-intensity group, transfer offenders account for 0.833% of students in the school-grade, compared with 5.83% in the highest-intensity group. Panel B reports the contemporaneous treatment effects on offense initiation. The effects increase monotonically with treatment intensity, consistent with both the criminal-capital and role-model channels.[47]

Treatment effects may also vary with children's characteristics, particularly if homophily strengthens peer interactions (Billings, Deming, and Ross, 2019; Lee et al., 2021). Shared characteristics may facilitate communication and therefore accelerate crime transmission. Also, those who share the characteristics are more likely to be role models (Beaman et al., 2009; Bhalotra, Clots-Figueras, and Iyer, 2018). I therefore examine heterogeneity by gender, first-language status, and ethnicity.[48]

The first part of Table D2 focuses on gender. Unsurprisingly, the treatment effects are larger for boys than for girls. However, the proportional increases relative to their pre-treatment means are similar. Effects are generally larger when the transferred offender and the exposed student share the same gender. A notable asymmetry emerges for opposite-gender exposure: male offenders increase offense initiation among girls, whereas female offenders *decrease* offense initiation among boys, although the estimate is a little imprecise. One possible explanation is that female offenders are relatively rare and may therefore be less likely to generate the same type of peer interaction or role-model effect among boys.

The second part of Table D2 examines heterogeneity by first-language status. Panel A shows the effects of exposure to offenders with the same first-language status on offense initiation. As with gender, the effects are larger when offenders and exposed students share the same characteristic. The relative effect is particularly large for students whose first language is not English. Because they account for only 10% of the sample (Table 1), exposure to an offender with the same language background may generate stronger interactions. In contrast, Panel B shows little evidence of

[47] On the other hand, the main results may be driven by unusually large numbers of transfer students entering a school-grade at the same time. To address this concern, I exclude observations in which the number of transfer students is above the 95th percentile of the distribution and repeat the main analysis. Figure C14 confirms that the results are unchanged.

[48] Although I do not have information on friendship in the data, the richness of demographic information allows me to investigate how important homophily in terms of observable characteristics is in peer effects. Based on a nationwide UK household longitudinal study, Understanding Society, 83.3% of those between 11 and 16 years old have their best friends in the same ethnic group, and 50% have ones in the same ethnic-gender-age groups. This suggests that homophily is crucial for friendship. So, it is reasonable to think that if a transfer student already has the same observable characteristics as children in the destination school, they are more likely to interact with the transfer student.

spillovers across first-language groups, which is because communication among them is not smooth enough to transmit crimes, nor can they serve as role models for their peers.

The final part examines heterogeneity by ethnicity. Panel A reports the effect of exposure to offenders from the same ethnic group on offense initiation, while Panel B reports the effect of exposure to offenders from a different ethnic group. In contrast to previous findings on ethnic homophily in Germany and the US (Leszczensky and Pink, 2015; Patacchini and Zenou, 2016), I find little evidence that ethnic matching plays an important role in the peer effects of crime. This might be because racial segregation is less pronounced in the UK (Jenkins, Micklewright, and Schnepf, 2008; Iceland, Mateos, and Sharp, 2011).

Table 8. Offender-Dense School

Quartile in offender density:	Lowest (1)	Middle (2)	Highest (3)
Panel A: Effects on Crime Onset			
Treatment Effects at $t = 0$	.023***	.243***	.034
	(.009)	(.055)	(.128)
Pre-Treatment Mean	.324	1.06	3.11
Relative Effect Size	7.15%	22.9%	1.11%
Panel B: Effects on Number of Crimes			
Treatment Effects at $t = 0$	-.012	.041	.284***
	(.015)	(.029)	(.097)
Pre-Treatment Mean	.100	.274	.738
Relative Effect Size	–12.0%	14.8%	38.5%
Sample Size (Panel Count)	5,101,909	5,084,637	5,049,704
Number of Individuals	344,858	344,858	344,858

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. Control variables include FSM and SEN status before secondary school, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, a linear month-of-birth term, and a linear age term. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Offender density is calculated as the ratio of the number of offenders in a school-grade to the total number of students in that school-grade, measured one school term before treatment. Data sources: Ministry of Justice and Department for Education.

Next, I examine whether new information and role-model effects matter for offense initiation. Schools differ substantially in their existing offender density, which I define as the fraction of

existing offenders to the number of children in the school-grade in the term before treatment. Table 8 reports effects on the results. Panel A examines offense initiation, while Panel B reports effects on the number of crimes. The pre-treatment means in Panel A illustrate the differences across terciles: in the least offender-dense schools, only 0.324% of pupils have an offense record, compared with 3.11% in the most offender-dense schools.

The effects on offense initiation are large and statistically significant except in the most offender-dense schools. One interpretation is that an incoming offender provides more new information or a more salient role model in schools where few offenders are already present. In the most offender-dense schools, by contrast, similar information and role models may already be available, leaving less scope for an additional offender to affect initiation. The non-monotonicity may also reflect weaker role-model effects in the least offender-dense schools, where an incoming offender may differ too much from other students. In contrast, the effect on the number of crimes is larger in more offender-dense schools. This pattern is consistent with evidence from prison settings that offenders can increase other offenders' criminal activity (Bayer, Hjalmarsson, and Pozen, 2009; Damm and Gorinas, 2020). In less offender-dense schools, the increase in offense initiation is not accompanied by an increase in the total number of crimes, suggesting offsetting changes in offending among existing offenders.[49]

More experienced offenders may also be more influential. Cattan et al. (2023*a*,*b*, 2025) document substantial heterogeneity in the number of crimes committed during the juvenile years. Table D3 divides transfer offenders into three groups according to the number of crimes committed before transfer: one crime, two to five crimes, or more than five crimes. The treatment effects increase with prior offending experience, suggesting that more experienced offenders generate stronger peer effects, potentially through criminal-capital transmission or role modeling.

If exposure to offenders helps peers accumulate criminal capital through skill acquisition, exposed students should be more likely to commit the same types of crimes that the transfer offenders committed before their transfers. Table 9 examines this possibility. To ensure sufficient statistical power, I aggregate the 10 crime categories into four broader groups: violent crimes (battery, violence, and sexual offenses), property crimes (theft, arson/damage, and fraud), drug-related crimes, and others (weapons, motoring, and other offenses). Panel A reports the effects of exposure to an offender on offense initiation by crime type, distinguishing between the same and different types of crime, while Panel B reports effects on the number of crimes of each type. Because baseline offense rates differ substantially across crime types, I also compare the estimates relative to their pre-treatment means. I find no systematic evidence that peers disproportionately adopt the same crime types as transfer offenders, especially for crimes that are more likely to require some skills,

[49]Some may be concerned that differences in demographics in these schools make the difference in the treatment effects. To address this concern, I partial out the outcome variables first, then use the residualized outcome variables for the analysis. As noted in Footnote 42, the interpretation of the coefficient is not straightforward, but the result pattern is the same, which means that the difference in demographics does not drive the results.

Table 9. Crime Types

Crime Types:	All Crimes		Violent Crimes		Property Crimes		Drug Crimes		Other Crimes	
	Same	Different	Same	Different	Same	Different	Same	Different	Same	Different
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Effects on Crime Onset										
Treatment Effects at $t = 0$	.089***	.082***	.068***	.104***	.106***	.082***	.019	.179**	-.027	.054
	(.007)	(.008)	(.011)	(.014)	(.010)	(.009)	(.034)	(.072)	(.018)	(.046)
Pre-Treatment Mean	.714	.806	.661	1.30	1.08	.714	.121	2.21	.155	1.96
Relative Effect Size	12.4%	10.2%	10.2%	7.98%	9.82%	11.5%	15.8%	8.08%	–17.7%	2.77%
Panel B: Effects on Number of Crimes										
Treatment Effects at $t = 0$	.018**	.039***	.020	.051**	.030**	.061***	-.024	.264**	.010	-.052
	(.009)	(.015)	(.012)	(.025)	(.013)	(.016)	(.029)	(.131)	(.019)	(.053)
Pre-Treatment Mean	.121	.184	.114	.273	.167	.166	.043	.451	.044	.376
Relative Effect Size	14.6%	21.2%	17.2%	18.8%	17.9%	36.6%	–56.1%	58.5%	21.9%	–13.9%
Sample Size (Panel Count)	12,184,487	12,184,487	6,182,281	6,182,281	8,351,676	8,351,676	699,820	699,820	1,628,495	1,628,495
Number of Individuals	827,507	827,507	420,119	420,119	567,629	567,629	47,586	47,586	110,664	110,664

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. I exclude observations that were simultaneously exposed to an offender with a different type of offense history at the time of treatment. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Data sources: Ministry of Justice and Department for Education.

such as property crimes. This result suggests that crime-specific skill transmission is unlikely to be a major mechanism.

Even in the absence of crime-specific skill transmission, transfer offenders may encourage peers to commit crimes together. Previous work shows that criminal networks can be sustained by a small number of "key players" (Lee et al., 2021), and that gang recruitment in school is an important channel for offense participation (Byström, 2025). Because I do not observe co-offending directly, I proxy for joint crime using offenses committed on dates when a transfer offender in the same school-cohort also commits an offense. I do not require the offenses to be of the same type, since offenders in organized crime may take on different roles and therefore commit different types of offenses. Figure D1 shows that this measure accounts for only a tiny share of the overall treatment effect, providing little evidence that coordinated offending with transfer offenders is a major mechanism.

Moreover, I examine how school transfers are associated with the subsequent offending behavior of the transfer offenders themselves. Because the decision to move is endogenous, these estimates are descriptive rather than causal. Figure D2 shows that school transfers are associated with a *decrease* in transferring offenders' criminal activity, which is also consistent with the finding that joint offending with transfer offenders is unlikely to explain much of the overall peer effect.

Given the limited evidence for skill transmission, a remaining mechanism consistent with the results is a role-model effect. To investigate this channel further, I examine heterogeneity by offenders' seniority. If an offender transfers to the same school but a different grade, peers have fewer opportunities to interact with the offender. At the same time, older offenders may be more salient role models than younger offenders.

Table 10. Treatment Seniority Heterogeneity

Treatment Cohort:	Two Cohorts Younger (1)	One Cohort Younger (2)	Same Cohort (3)	One Cohort Older (4)	Two Cohorts Older (5)
Treatment Effects at $t = 0$	.015	.051***	.183***	.204***	.073***
	(.011)	(.011)	(.034)	(.015)	(.026)
Pre-Treatment Mean	.400	.697	1.39	1.93	2.78
Relative Effect Size	3.65%	7.38%	13.1%	10.6%	2.61%
Sample Size (Panel Count)	6,704,483	9,790,276	7,321,414	4,737,409	1,398,616
Number of Individuals	453,765	663,904	497,302	322,733	95,296

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. Control variables include FSM and SEN status before secondary school, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, a linear month-of-birth term, and a linear age term. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into the school and grade specified in each column, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. In Columns (1), (2), (4), and (5), I exclude students who are exposed to an offender's transfer into their cohort before or at the same time as the treatment. In Column (3), I exclude students who are exposed to an offender's transfer into a different cohort before or at the same time as the treatment. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. Data sources: Ministry of Justice and Department for Education.

Table 10 reports treatment effects when offenders transfer into peers' own grade (Column (3)), one or two cohorts below (Columns (1) and (2)), and one or two cohorts above (Columns (4) and (5)).[50] The effects are stronger for older offenders than for younger offenders, consistent with the role-model channel. As the grade gap widens, the effects generally weaken, likely because opportunities for interaction become more limited. Notably, however, the point estimate is larger when the incoming offender is one cohort older than when the offender is in the same grade. This pattern provides further evidence consistent with an important role-model mechanism.

Relatedly, I examine whether the age at which peers are exposed to an offender matters. Offenders who begin offending at very young ages may be more disruptive (Cattan et al., 2025) and therefore more influential, making them better role models. At the same time, as children grow older, offending becomes more common, which may lower the psychological barrier to initiating crime.

Table D4 reports the results. Because the control mean increases monotonically with age, as shown in Figure 1b, the relative effect size is particularly informative. The treatment effects follow a U-shaped pattern: relative effects are larger at ages 11 and 14 than at ages 12 and 13. This pattern is consistent with two offsetting mechanisms. The influence of particularly disruptive early offenders may decline with age, while the psychological barrier to initiating crime may also fall as offending becomes more prevalent. Together, these forces can generate the observed U-shaped relationship, consistent with the role-model explanation.

[50] To ensure the SUTVA assumptions, I omit those who have received an offender transfer in their cohorts prior to the targeted cohorts in each column.

5.3.2 Channels of the cost side

Table 11. Heterogeneity in Educational Achievement

Quartile in Educational Achievement:	Highest (1)	2nd Highest (2)	3rd Highest (3)	Lowest (4)
Panel A: Measured by Key Stage 1 at Age 7				
Treatment Effects at $t = 0$	.074***	.141***	.192***	.354***
	(.014)	(.017)	(.020)	(.026)
Pre-Treatment Mean	.769	1.23	1.51	2.57
Relative Effect Size	9.65%	11.5%	12.7%	13.8%
Sample Size (Panel Count)	3,687,155	3,679,891	3,674,242	3,666,407
Number of Individuals	249,863	249,861	249,863	249,859
Panel B: Measured by Key Stage 2 at Age 11				
Treatment Effects at $t = 0$	.028***	.083***	.139***	.198***
	(.011)	(.015)	(.020)	(.024)
Pre-Treatment Mean	.582	1.12	1.78	2.82
Relative Effect Size	4.85%	7.39%	7.84%	7.03%
Sample Size (Panel Count)	2,982,887	2,973,715	2,966,242	2,955,983
Number of Individuals	201,761	201,760	201,761	201,758

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. Control variables include FSM and SEN status before secondary school, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, a linear month-of-birth term, and a linear age term. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. All test scores are standardized within each cohort. The age 7 overall score is calculated as a weighted average of teachers' assessments of Reading, Writing, Speaking and Listening, Math, and Overall Science. Within English, Reading, Writing, and Speaking and Listening are weighted 1:1:2 so that the four underlying abilities contribute equally. English, Math, and Science are then weighted 1:1:1 when constructing the overall score. The age 11 overall score is calculated as a weighted average of English, Math, and Science, with equal weight assigned to each subject. Data sources: Ministry of Justice and Department for Education.

So far, the evidence on the return side suggests that role-model effects are the main mechanism. I now turn to mechanisms operating through the cost of crime, beginning with expected loss from being caught through opportunity costs. If peers face a higher opportunity cost of offending, they should be less responsive to exposure to offenders.

I first examine heterogeneity by educational achievement before secondary school, which serves as a proxy for the opportunity cost. Table 11 reports the effects on offense initiation. Panel A shows heterogeneity by age-7 test scores, while Panel B uses age-11 test scores. I divide students into quartiles based on standardized educational attainment at each age. Both panels show a clear pattern: treatment effects decline monotonically with educational achievement. This is consistent with higher-achieving students facing a greater opportunity cost of acquiring a criminal record.[51]

[51] As discussed in Subsection 2.1, in many schools, classroom allocation is based on pupils' educational level. Since offenders tend to have lower achievement (Cattan et al., 2023*a*,*b*, 2025), those in the lower quartile might have more intense exposure to offenders' transfers. However, if I compare the treatment effects among those with similar treatment intensity, we still observe this monotonicity by educational level.

Another explanation for the monotonicity by educational achievement is that students with higher test scores may also have better self-control and therefore be less likely to offend. To investigate this channel, I use measures of non-cognitive skills at age 5.[52] Because the sample is relatively small, I divide students into only two groups based on whether their scores are above or below the median.[53]

Table D5 shows that treatment effects are generally smaller among students with stronger non-cognitive skills. Across the detailed domains, the clearest difference appears in "Personal social and emotional development." This component measures children's skills to manage emotions and self-control, and to persist and wait for what they want and direct attention as necessary, all of which are consistent with placing greater weight on future consequences (lower time discount factor).[54][55] Overall, these patterns suggest that self-control may reduce susceptibility to crime spillovers by increasing the perceived future cost of offending.

I further examine heterogeneity by peers' individual and family backgrounds. I first focus on FSM and SEN eligibility by age 11. As shown in Table D6, the treatment effects are concentrated among disadvantaged students, particularly those eligible for FSM. This pattern suggests that students from more disadvantaged backgrounds may be more susceptible to exposure to offenders, potentially because they face a lower opportunity cost of offending. This interpretation is also consistent with evidence on intergenerational mobility showing that children growing up in lower-income households have lower predicted adult income (Krutikova et al., 2023).

Next, related to FSM, I examine heterogeneity by local deprivation. Although the peer effects operate primarily through schools rather than neighborhoods, the opportunity cost of offending may still vary with local economic conditions. I measure local deprivation using the 2007 Income Deprivation Affecting Children Index (IDACI). IDACI measures the proportion of children aged 0 to 15 living in income-deprived households receiving certain means-tested benefits (Noble et al., 2008). I divide the sample into quartiles based on the IDACI value of peers' LSOA of residence at the time of treatment.[56][57]

[52] Although non-cognitive abilities are measured at age 5, they may remain relatively stable over time (Cornelissen and Dustmann, 2019), making them a useful proxy for non-cognitive abilities during secondary school.

[53] I have a record of non-cognitive measures at the age of 5 only from the 1997/8 cohort onwards. See Table A1.). Also, it is only for a 10% quasi-random sample of children, as the outcomes are recorded for those born on the 5th, 15th, and 25th of each month (Department for Education, Schools Analysis and Research Division, 2010).

[54] See the UK government's website for the details (https://help-for-early-years-providers.education.gov.uk/areas-of-learning/personal-social-and-emotional-development, last accessed: October 12th, 2026).

[55] Another important point is that those with high "Knowledge and understanding of the world" are *more* likely to start committing crime. This measure evaluates how children make sense of their physical world and community, reflecting their curiosity. Therefore, we can interpret this result as those with higher curiosity about others being more likely to be affected.

[56] See Figure A1 for the distribution of deprived areas across England and London.

[57] Relatedly, some might be interested in the geographical difference in the treatment effects. First, Table F4 shows the heterogeneity with respect to urban and rural areas. The treatment effects tend to be higher in urban areas. This is consistent with the fact that the crime rate is higher in urban areas in England (Hall et al., 2026).

Table 12. Local Deprivation

Quartile in Deprivation Index:	Least Deprived (1)	2nd Least Deprived (2)	3rd Least Deprived (3)	Most Deprived (4)
All Crimes	.073***	.119***	.209***	.309***
	(.016)	(.017)	(.023)	(.021)
Pre-Treatment Mean	.784	1.10	1.59	2.22
Relative Effect Size	9.35%	10.8%	13.1%	13.9%
Theft	.014	.048***	.082***	.131***
	(.010)	(.010)	(.015)	(.014)
Pre-Treatment Mean	.326	.462	.655	.934
Relative Effect Size	4.31%	10.4%	12.6%	14.0%
Fraud	.002	.001	.005	.008*
	(.002)	(.003)	(.006)	(.004)
Pre-Treatment Mean	.011	.026	.028	.049
Relative Effect Size	21.1%	1.96%	18.2%	15.7%
Sample Size (Panel Count)	3,299,643	3,503,179	3,970,134	4,685,188
Number of Individuals	223,594	237,727	270,064	319,182

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Deprivation is measured using the 2007 Income Deprivation Affecting Children Index (IDACI), defined as the proportion of children aged 0 to 15 living in income-deprived families at the LSOA level. LSOA stands for Lower Layer Super Output Area, a geographic unit containing approximately 1,000 to 3,000 residents. Data sources: Ministry of Justice and Department for Education.

The first rows of Table 12 report treatment effects on all crime types. Both the absolute and relative effects increase monotonically with local deprivation, consistent with the previous results for FSM. The second and third rows focus on theft and fraud. Because receiving stolen goods is the most common fraud offense in my data (Table A5), these two crime types have more direct monetary returns. The result shows the same broad pattern, but strikingly, the treatment effects on theft and fraud are zero in the least deprived areas, and the effect on fraud appears only in the most deprived areas. These results are consistent with monetary returns making offending relatively more attractive by lowering the *net* opportunity cost.[58]

Some schools impose disciplinary sanctions when students commit offenses, with exclusion being a common punishment. Descriptive evidence shows that the timing of offending and exclusion is closely correlated (Cattan et al., 2023*a*,*b*). In England, some state schools, known as "academies," have greater autonomy and tend to enforce stricter discipline; students in these schools also commit fewer crimes (Duchini, Lavy, and Machin, 2020). In Table D7, I examine heterogeneity by school type. It shows no significant treatment effects in academies, consistent with stricter disciplinary environments increasing the opportunity cost of offending.[59]

[58] I show the results on the crime types that directly address the potential returns to crime with the measure in 2007. However, the overall results are the same with other crimes and with the measure in 2004. See Tables F1 and F2.

[59] Not only strict school discipline, but also police intervention in schools is common in the US. See Owens (2017) for the analysis on this aspect.

Table 13. Heterogeneity in Punishment that Transfer Offenders Have Experienced

	Caution Only		At least One Conviction	
	Caution Expected (1)	Conviction Expected (2)	Caution Expected (3)	Conviction Expected (4)
Treatment Effects at $t = 0$	.212***	.361***	.460***	.488**
	(.015)	(.088)	(.088)	(.196)
Pre-Treatment Mean	1.57	1.63	1.72	1.82
Relative Effect Size	13.5%	22.1%	26.7%	26.7%
Sample Size (Panel Count)	8,058,932	1,676,311	1,054,148	563,559
Number of Individuals	547,754	113,902	71,760	38,342

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. This table focuses on transfers of offenders who committed crimes with an estimated probability of resulting in a caution between 0.45 and 0.55 in the final data. Caution-expected crimes are those with an estimated probability of caution between 0.50 and 0.55, while conviction-expected crimes are those with an estimated probability of caution between 0.45 and 0.50. Data sources: Ministry of Justice and Department for Education.

The final piece of evidence on mechanisms concerns the expected penalty. If individuals perceive the punishment for an offense to be more severe, they should be less likely to offend. Although actual penalties do not vary exogenously, peers' *perceptions* of their severity may change after observing the punishment received by a transfer offender. In particular, exposure to an offender who received a lighter-than-expected punishment may lead peers to revise their beliefs about punishment downward, while exposure to a heavier-than-expected punishment may lead them to revise their beliefs upward. On the other hand, if the role model channel is still strong, those who commit a more serious crime and receive a severer punishment might attract more attention. Such risky behavior can lead to popularity among peers during adolescence (Balsa et al., 2011).

Table 13 examines this mechanism. To control for offense severity, I first restrict the sample to offense types for which the probability of receiving a caution lies between 0.45 and 0.55.[60] Within this range, I distinguish two types of offenses: caution-expected offenses, for which the probability of receiving a caution is between 0.50 and 0.55, and conviction-expected offenses, for which the probability of receiving a caution is between 0.45 and 0.50.[61] I then distinguish four groups according to the punishment received by the transfer offender and the punishment typically associated with that offense: (i) caution for a caution-expected offense (Column (1));

[60] See Table A7 for the list of such offenses.

[61] More specifically, I select these offenses by calculating the likelihood of caution and conviction in PNC, and limit the offenses that result in caution to those with a frequency between 0.45 and 0.55.

(ii) caution for a conviction-expected offense (Column (2)); (iii) conviction for a caution-expected offense (Column (3)); and (iv) conviction for a conviction-expected offense (Column (4)).[62] Thus, offenders in group (ii) receive a lighter punishment than would typically be expected for their offense, whereas those in group (iii) receive a heavier punishment than expected.

If belief updating is an important mechanism, I expect the following inequalities in the relative treatment effects: (1) < (2), (3) < (4), (1) > (3), and (2) > (4). In contrast, if the role-model effect dominates belief updating, I expect (1) < (2), (3) < (4), (1) < (3), and (2) < (4).

Because baseline offense rates differ across groups, I focus on relative effect sizes. Table 13 shows that (1) < (2), (1) < (3), and (2) < (4). In particular, the latter two inequalities are opposite to those predicted by the belief-updating mechanism and are instead consistent with the role-model channel. This interpretation is also consistent with the heterogeneity in non-cognitive abilities. Students who discount the future more heavily, and who are therefore more likely to initiate offending, may place less weight on the possibility of more severe punishment in the future and instead place greater weight on the immediate utility from risk-taking influenced by role models. Overall, these results provide little evidence that updating beliefs about expected punishment is an important mechanism and are more consistent with a strong role-model effect.

5.4 Social Cost of Peer Effects of Crime

To assess the economic importance of these peer effects, I estimate their social cost. First, I calculate the social cost of youth crime in my data using monetary cost estimates from Heeks et al. (2018). I classify all 2,820 crime types into 13 categories with non-negative costs, following Cattan et al. (2025).[63] Table 14 summarizes the calculation. For each crime category, Column (3) reports the total social cost, calculated as the number of youth offenses in Column (1) multiplied by the social cost per offense from Heeks et al. (2018) in Column (2).[64]

Then, I re-estimate the peer effects separately for each crime category (Table F3) and use these estimates to calculate the number of additional crimes attributable to peer effects, reported in Column (4).[65] Column (5) converts these additional offenses into social costs by multiplying Column (2) by Column (4). Finally, I divide Column (5) by Column (3) to calculate the share of the total social cost attributable to peer effects. Overall, approximately 5% of the social cost

[62] This analysis does not mean anything about how (un)fair a certain sentence is. It is just the deviation from the unweighted likelihood of cautions for the offenses in the sample, without accounting for offenders' characteristics, such as how many crimes they have committed prior to.

[63] I omit the categories of homicides and cybercrimes because of potentially disclosive low counts. Homicides have the highest estimated cost among the crimes considered in (Heeks et al., 2018), at £3,217,740 per offense. Excluding them may lead me to underestimate the total social cost of youth crime and may partly explain the difference between my estimate and that in Irwin-Rogers, Muthoo, and Billingham (2020).

[64] The social cost per crime for all crimes is not reported directly in Heeks et al. (2018). I calculate it by dividing the total cost, obtained as the sum across crime types in Column (3), by the total number of crimes in Column (1).

[65] I replace the point estimate with zero if it is not statistically significantly different from zero at a 10% significance level.

of youth crime in my data period, equivalent to about 20 million pounds (26.4 million dollars), is attributable to these peer effects.

Table 14. Social Costs

	Number of Youth Crimes (1)	Social Cost per Crime (2)	Social Cost of the Crime Type (3)	Increase in the Number Crimes due to Peer Effects (4)	Increase in Social Cost due to Peer Effects (5)	Fraction of Increase (6)
All Crimes	61,538	6,456	397,267,948	2,922	20,000,725	5.03%
Cost of Crime for Each Crime Type						
Violence with Injury	16,233	14,050	228,073,648	355	4,990,295	2.19%
Violence without Injury	4,125	5,930	24,461,250	940	5,571,984	22.8%
Rape	426	39,360	16,767,360	78.9	3,103,717	18.5%
Other Sexual Offence	2,595	6,520	16,919,400	123	799,296	4.72%
Robbery	2,964	11,320	33,552,480	-42.0	-475,684	-1.42%
Domestic Burglary	4,025	5,930	23,868,250	513	3,041,085	12.7%
Theft of Vehicle	711	10,290	7,316,190	247	2,542,533	34.8%
Theft from Vehicle	653	870	568,110	-119	-103,151	-18.2%
Theft from Person	17,080	1,380	23,570,400	449	618,974	2.63%
Criminal Damage – Arson	720	8,420	6,062,400	-83.7	-704,632	-11.6%
Criminal Damage – Other	10,612	1,350	14,326,200	444	598,877	4.18%
Fraud	1,344	1,290	1,733,760	0	0	0%
Commercial Theft	50	970	48,500	18.0	17,430	35.9%

Notes: Column (1) reports the number of crimes committed by individuals in the final sample described in Sections 3 and A. The crimes listed in the second part of the table constitute a subset of all crimes in the data. To calculate the social cost of crime, I map the 2800 crime types recorded in the PNC to the 13 categories used by Heeks et al. (2018). Some crimes recorded in the PNC, such as weapon offenses, do not have a corresponding category. I assign a cost of zero to these crimes and exclude them from the list. This approach therefore provides a lower-bound estimate of the social cost, since these offenses are likely to impose some social cost. The cost for each crime is calculated by multiplying the cost per crime by the number of crimes induced by peer effects, as shown in Table F3. If the treatment effect is not statistically significant at the 10% level, I set the effect equal to zero. Data sources: Ministry of Justice and Department for Education.

5.5 Effects on Other Behavioral Responses

So far, I have focused on criminal behavior. Because the NPD also contains rich information on behavioral problems, I next examine whether exposure to offenders affects milder forms of misconduct. Even if children do not commit a crime, they may still respond to exposure through other problematic behaviors. I focus on suspension (fixed-term exclusion) and unauthorized absence. Offenders are more likely to experience longer suspensions and more unauthorized absences (Cattan et al., 2023*a*,*b*, 2025), making these outcomes useful proxies for less severe behavioral problems.

Figure F2 presents the results. Figures F2a and F2b show that offenders' transfers increase both the probability and duration of exclusion in the short run.[66] Figure F2c also shows an increase in the share of unauthorized absence sessions. These patterns are broadly consistent with those observed for offenders. Overall, the results suggest that exposure to offenders increases problematic behavior even among students whose behavior does not escalate to criminal offending.

[66]The effects become *negative* in the long-run. This may be because of substitution from disruptive behavior at school toward absence from school. Indeed, the unauthorized absences and fixed exclusions do not decrease (Figures F2d and F2e), which suggests that a decrease in exclusion does *not* mean that exposure to offenders improves peers' behavior in the long run.

5.6 Effects on Educational Achievement

Table 15. Effects on Educational Outcomes at Age 14

Transfer Student:	Offender		Non-offender	
	(1)	(2)	(3)	(4)
Any Offender's Transfer	-.041***	-.016***	-.129***	.004
	(.010)	(.005)	(.010)	(.009)
Controls		x		x
Sample Size	862,553	862,553	69,297	69,297

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors clustered at the treatment level are reported in parentheses. Control variables include FSM and SEN status before the age 14 exam, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, and month-of-birth fixed effects. The age 11 and 14 overall scores are calculated as a weighted average of English, Math, and Science, with equal weight assigned to each subject. Data sources: Ministry of Justice and Department for Education.

In the school context, educational achievement is a key outcome. However, the cohorts used in the main analysis, born between 1995/96 and 98/99, do not have measures of educational achievement during secondary school in my data extract. Therefore, I cannot compare the outcomes of students who have already been exposed to an offender with those of students who have not yet been exposed.[67] I instead focus on earlier cohorts born between 1990/91 and 1993/94, for whom educational achievement at age 14 is observed.[68] A major limitation is that these cohorts have only one school census per year, in the spring, so the exact timing of school transfers cannot be observed. This introduces potential measurement error in treatment status. Despite this limitation, these cohorts allow me to examine an important educational outcome.[69]

Because I use the age-14 standardized test score (KS3), I define the treatment group as students exposed to an offender transferring into their school during the academic year beginning in September at age 13; that is, up to one year before the age-14 exam. The control group consists of students exposed after the exam, up to one year later. I estimate:

$$y_{ics} = \alpha + \beta T_{cs} + \boldsymbol{x}_i^{\mathrm{T}}\boldsymbol{\delta} + \eta_c + \nu_s + \varepsilon_{ics}, \tag{1}$$

where y_{ics} is the age-14 test score for individual i in cohort c and school s, T_{cs} is the treatment

[67] I do not use the GCSE exam at age 16 because people may leave state school at 16, and the choice is likely correlated with educational achievement and offense history, which may cause selection bias if I adopt a difference-in-differences approach.

[68] The data linkage information is in Table A2. The linkage rate is similar to the cohorts born between 1995/96–98/99, although the offense rate is higher, which is consistent with the fact that the offense rate is decreasing (Youth Justice Board and Ministry of Justice, 2023; Cattan et al., 2025).

[69] Appendix G replicates the summary statistics and main results using these cohorts. The results are robust to the choice of the cohorts.

indicator, $\boldsymbol{x}_i$ is a vector of individual characteristics, and η_c and ν_s are cohort and secondary-school fixed effects, respectively. Standard errors are clustered at the level of exposure, defined by the school-cohort-timing cell.[70]

Table 15 presents the results. Column (1) reports the unconditional difference, while Column (2) reports the conditional estimate. Exposure to an offender reduces educational attainment by 0.016 standard deviations. As a robustness check, I instead focus on non-offenders' transfers on educational achievement. Column (4) shows that there is no adverse impact from a non-offender's school transfer.

Understanding the source of this negative effect is important. Therefore, I decompose the decline in educational achievement into an indirect effect operating through offense initiation and a direct effect of exposure itself. To do so, I compare children who differ in both the timing of exposure to an offender's transfer and the timing of their first offense.

I define four groups. Groups A and B are exposed to an offender at age 13 and form the treatment group, while Groups C and D are exposed at age 14 and form the control group. Within each exposure group, I further distinguish children by their own offending behavior. Children in Groups A and C commit their first offense in the same year as exposure, at ages 13 and 14, respectively, while those in Groups B and D do not commit an offense by age 14.[71]

These comparisons allow me to separate the association between educational attainment and becoming an offender following exposure from the direct effect of earlier exposure itself. Panel I of Table 16 summarizes the four groups and reports their educational achievement at age 14. Students who begin offending following exposure have lower educational attainment than those who do not, and earlier exposure is also associated with lower test scores, consistent with my prior paper (Cattan et al., 2025).

I now formalize these comparisons using the following regression model. For simplicity, I omit controls and the school and cohort subscripts, s and c, as well as the corresponding fixed effects:

$$Y_i = \alpha + \beta T_i + \gamma C_i + \delta T_i * C_i + \varepsilon_i, \tag{2}$$

where Y_i is individual i's educational achievement at age 14, T_i is the treatment indicator, C_i is a "complier" indicator identifying students who are vulnerable to the shock of exposure to an offender, and ε_i is an error term. I include T_i to capture the direct effect of exposure, which

[70] Because there is only one school census per year for these cohorts, clustering at the school-cohort level is equivalent to clustering at the school-cohort level.

[71] One caveat of this specification is that I compare students who are exposed to an offender at age 13 but do not commit a crime for the following year (Group B) with those who are exposed at age 14 but do not commit a crime only during that same year (Group D). This difference arises from the limited frequency of the school information and cannot be fully avoided in the mediation analysis. To address this concern partially, as a robustness check, I exclude from Group D students who commit a crime at age 15, so that Groups B and D are required to remain nonoffending for the same length of time following exposure. This alternative definition conditions group membership on a future outcome and should therefore be interpreted only as a robustness exercise. The results are unchanged under either definition of the groups.

Table 16. Mediation Analysis

Panel I: Age 14 Educational Achievements

Group	Criteria			Mean	SD	Sample Size
	Age at which they are exposed to an offender	First crime at the age 13 or 14	Age of the first crime (13 or 14)			
A:	13	Yes	13	-.570	[.905]	8,623
B:	13	No	-	-.011	[.914]	421,568
C:	14	Yes	14	-.469	[.896]	9,663
D:	14	No	-	.028	[.912]	387,368

Panel II: Mediation

Parameter:	$\gamma + \delta$ (1)	$\beta + \delta$ (2)	γ (3)	β (4)	δ (5)
Conditional Difference	-.225***	-.066***	-.190***	-.016***	-.034***
	(.007)	(.018)	(.007)	(.002)	(.010)
Controls	x	x	x	x	x
Sample Size (Panel Count)	430,191	18,286	397,031	808,936	827,222

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors clustered at the treatment level are reported in parentheses. Control variables include FSM and SEN status before the age 14 exam, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, and month-of-birth fixed effects. The age 14 overall score is calculated as a weighted average of English, Math, and Science, with equal weight assigned to each subject. Some observations are missing information for certain variables. Data sources: Ministry of Justice and Department for Education.

may arise, for example, through psychological distress; previous studies document adverse effects of exposure to violent crime on educational achievement (Brown and Velásquez, 2017; Koppensteiner and Menezes, 2021; Cabral et al., 2026). I include C_i because compliers may differ systematically from never-takers, given that their offending behavior responds to the relatively small shock of an offender's school transfer. Finally, the interaction $T_i * C_i$ identifies students who become offenders following treatment. Thus, δ captures the indirect effect of exposure operating through offense initiation, while β captures the direct effect of exposure on educational achievement.

For each group, the following applies.

Group A	Group B	Group C	Group D
$\alpha + \beta + \gamma + \delta$	$\alpha + \beta$	$\alpha + \gamma$	α

To estimate β, I compare Groups B and D, while I estimate δ using the difference-in-differences $(A - B) - (C - D)$. In all specifications, I include school and cohort fixed effects, as well as observed individual characteristics that may be correlated with treatment and complier status.

Panel II of Table 16 presents the results. Exposure to an offender directly reduces educational achievement by 0.016 standard deviations. This suggests that exposure itself adversely affects learning, even without inducing the student to offend. One possible mechanism is that disruptive peers interfere with the learning environment or require teachers to redirect resources toward them, consistent with the literature on disruptive students (Carrell, Hoekstra, and Kuka, 2018; Carneiro et al., forthcoming). In addition to this direct effect, becoming an offender following exposure reduces test scores by a further 0.034 standard deviations, approximately twice the magnitude of the direct exposure effect.[72] Thus, the indirect effect operating through offense initiation appears larger than the direct effect of exposure. This pattern is consistent with my previous findings in Cattan et al. (2025) – students who offend before the exam have substantially lower educational achievement. One possible explanation is that acquiring an offense record reduces the expected return to further human-capital investment, thereby weakening incentives to exert effort in school.

6 Simulation

So far, I have shown that peer effects of crime are substantial and contribute meaningfully to youth offending. Now I consider a potential policy aimed at reducing crime through the peer-composition channel. Given the substantial heterogeneity in treatment effects documented above, reallocating transfer students across schools may reduce criminal spillovers. Because my identification strategy relies on offenders' school transfers, a natural counterfactual is to consider how alternative school assignments could mitigate these negative peer effects. Any such policy would raise important

[72] In this regression, the sample is slightly different from Table 15. I also add fixed effects and controls in all regressions. Therefore, the point estimates in Table 16 do not sum up to those in Table 15.

ethical and practical concerns, but quantifying the potential reduction in crime provides a useful benchmark for policy discussion.

Suppose there were a centralized school-transfer market in which all pupils wishing to move schools registered in advance, and a social planner assigned them to schools based on their characteristics, preferences, and available capacity.[73] Using a matching algorithm from mechanism design, the planner can construct a stable allocation that reduces predicted criminal spillovers.[74]

For simplicity, I focus only on contemporaneous treatment effects and abstract from the dynamic effects documented earlier. Incorporating the full dynamics would require a substantially more complex model in which students jointly make decisions about school transfers, friendship formation, and criminal activity in every single school term. Such a complex model would be difficult to implement in practice, and a simpler framework remains informative for quantifying the potential gains from reassignment.

I also focus on a single source of treatment-effect heterogeneity. Although the previous analysis documents heterogeneity along many dimensions, there is no natural way to summarize them in a single index without imposing additional assumptions. Moreover, a simpler rule is more transparent and potentially more relevant for policy implementation. Accordingly, I treat offenders and non-offenders as homogeneous and consider a policy that swaps the destination school of an offender with that of a non-offender.

6.1 Simulation Algorithm

I now describe the simulation algorithm step by step and discuss the desirable properties of the resulting allocation.

Step 1: Interpolate the treatment effect. The heterogeneity by offender density reported in Table 8 is estimated by tercile. So, I interpolate the treatment effects over the distribution of offender density. Specifically, I linearly interpolate both the point estimates and the 95% confidence intervals, and set the treatment effect to zero whenever the confidence interval includes zero. Figure E1 provides a visual illustration. I restrict the analysis to the range of offender density between the top and bottom terciles to avoid extrapolating treatment effects.

Step 2: Calculate the potential treatment effect of those who move. I identify students who transfer to another school at the beginning of each school term. I assume that

[73] There are many other possible alternative policies as a thought experiment, ethically right or not, such as denying offenders' school transfer for example. However, in this analysis, I assume all pupils who want to move can move to another school. Another alternative would be assigning offenders and victims (if any) to different schools if there were linked data on crimes and victims, as theoretically discussed in Kasuya (2016).

[74] See Appendix E for the detail.

each departure creates one available slot in the school-grade the student leaves, so the number of available slots equals the number of departing students. For every student in the transfer market, I calculate the predicted peer effect associated with assigning that student to each school-grade with an available slot, based on the student's characteristics and the characteristics of the potential destination school. Because Table C4 shows no effect of offenders leaving a school, I exclude any effect at the origin school. Because non-offenders' transfers do not increase crime at any tercile (Figure E2), I assign them a predicted treatment effect of zero.

Step 3: Stack the peer effects at each school into one vector for each student. For each transfer student, I collect the predicted peer effects associated with all feasible destination school-grades into a column vector. If a school has multiple available slots, I duplicate the corresponding row. I assign a value of ∞ when a school is too far from the student's original destination school. Distance is measured between the student's original destination school and each school with an available slot.[75] I consider distance thresholds of 1, 3, 5, 10, and 25 miles.

Table 17 provides an example. Suppose a transfer market contains one offender (A) and two non-offenders (B and C), and there are three schools. Originally, A is assigned to School 1, B to School 2, and C to School 3, as shown in the gray-shaded cells. If Offender A is assigned to Schools 1, 2, and 3, the predicted increases in crime are 3, 1, and 2, respectively. Non-offenders B and C do not increase crime. Assume that Schools 1 and 3 lie beyond the allowed distance from B's original destination, so I assign ∞ to these options.

Table 17. Example of Vectors of Preferences for Simulation

	Offender A	Non-offender B	Non-offender C
School 1	3	∞	0
School 2	1	0	0
School 3	2	∞	0

Step 4: Interpret the vector as an individual's preference. I interpret each individual's vector as their *preference* over school choices, with lower predicted peer effects corresponding to higher preferences. For example, for Offender A, School 2 is the most preferred, School 3 is the second most preferred, and School 1 is the least preferred. For Non-offender B, the outside option is preferred to Schools 1 and 3, which are equally preferred. The outside option is defined as transferring to the individual's original destination school. Non-offender

[75] The location is identified as the mode of the LSOA of students' residences.

C is indifferent among all schools. I assume that schools have no preferences over students.[76][77]

Step 5: Apply an algorithm from mechanism design theory. Given the set of preferences, I apply a random priority mechanism. I assign a random number to each individual and assign the most preferred school from the pool of available slots in ascending order of that random number.

The resulting assignment has several attractive properties (Bogomolnaia and Moulin, 2001). First, the mechanism is strategy-proof, so no student has an incentive to misreport their preferences. It also satisfies equal treatment of equals. Finally, it is ex post efficient: no reassignment can make one student better off without making another worse off.

In the example above, the status quo generates three additional crimes. Suppose the random-priority order is B, A, and C. The social planner then assigns B to School 2, A to School 3, and C to School 1, as highlighted in red, reducing the predicted number of additional crimes by one. This example illustrates how reassignment can reduce crime by reallocating transfer students across schools.

6.2 Simulation Results

Table 18 presents the simulation results. Panel A reports outcomes when the planner minimizes the number of crimes, while Panel B reports outcomes when the planner minimizes the number of offenders. Each panel contains three sets of two rows. The first reports the number of offenders reassigned away from their original destination school and the corresponding share of all offender transfers. The second and third report the number of crimes and offenders generated by peer effects under the alternative assignment, together with the percentage change relative to the status quo.

Column (1) reports the status quo: the number of crimes and offenders generated by contemporaneous peer effects and the number of offender transfers. Columns (2) to (6) report simulations in which alternative schools must lie within an m-mile radius of the original destination, where $m \in \{1, 3, 5, 10, 25\}$. Column (7) removes the distance restriction and allows reassignment across England. The statutory walking distance for secondary-school students under the Education Act 1996 is 3 miles, making this a useful benchmark (Department for Education, 2026).

Under the status quo, peer effects generate 379 additional crimes and 887 additional offenders. When the planner minimizes the number of crimes, allowing reassignments within 3 miles reduces peer-effect-induced crimes by 36%. Achieving this requires swapping the destination schools of

[76] In the empirical exercise, I impose strict preferences by randomly ordering schools that are otherwise equally preferred.

[77] I make the technical assumption that, even if a student enters the reassignment market and thereby creates an available slot at their original school, they may still choose the outside option if the resulting assignment is worse. In that case, they are allowed to remain at their original destination school even if the slot they created has already been assigned to another incoming student; they are not forced to move to another school.

Table 18. Simulation Results

		Maximum Distance between the Originally-assigned and the Re-assigned Schools (miles)					
	Original	1	3	5	10	25	Any
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Minimizing N of Crimes							
N of offenders swapped	3661	310	1649	2283	3071	3543	3657
Fraction of offenders swapped		8%	45%	62%	84%	97%	100%
N of crimes	379	350	244	193	137	113	91
Δ crimes		-8%	-36%	-49%	-64%	-70%	-76%
N of offenders	887	893	917	932	924	918	826
Fraction of offenders		1%	3%	5%	4%	3%	-7%
Panel B: Minimizing N of Offenders							
N of offenders swapped	3661	309	1579	2224	2961	3470	3653
Δ offenders swapped		8%	43%	61%	81%	95%	100%
N of crimes	379	392	498	609	805	1089	1244
Δ crimes		3%	31%	61%	113%	187%	228%
N of offenders	887	848	686	617	574	603	648
Δ offenders		-4%	-23%	-30%	-35%	-32%	-27%

Notes: This table shows the simulation results based on the random priority mechanism (Bogomolnaia and Moulin, 2001). I drop the school outside the interpolation window in Figure E1. I repeat the simulation 10 times and take the average.

1,649 offenders with those of 1,649 non-offenders. If reassignment is allowed across England, the reduction reaches 76%. The number of additional offenders, however, changes relatively little.

Panel B highlights an important trade-off. When the planner instead minimizes the number of offenders, a 3-mile reassignment reduces the number of additional offenders by 23%, but increases the number of crimes by 31%. When reassignment is allowed across England, the number of crimes more than doubles relative to the status quo. This occurs because assigning offenders to offender-dense schools reduces offense initiation but increases criminal activity among existing offenders (Table 8). Thus, reassignment can substantially reduce crime spillovers, but the choice of objective function is crucial: a policy that improves one margin may worsen another.

7 Conclusion

This paper investigates peer effects in offending among secondary-school students in England by examining whether exposure to a peer with a prior offense affects students' subsequent criminal behavior. I examine both offense initiation and contemporaneous overall criminal activity, as well as heterogeneity in these effects and the mechanisms underlying them. To identify causal effects, I exploit the plausibly exogenous timing of school transfers by students who had committed an offense before transferring. Using linked administrative data from the National Pupil Database (NPD) and the Police National Computer (PNC), I estimate the effects in a staggered event-study design comparing already-treated students with those not yet exposed.

Exposure to an offender's transfer increases the probability of offense initiation by 12.9% within one school term, with the effect persisting for several years. The mechanism analysis, guided by the human capital framework of Lochner (2004), points to several channels. On the return side, the evidence is most consistent with a role-model effect rather than crime-specific skill transmission. On the cost side, the opportunity cost of offending appears important. The effects extend beyond criminal behavior: exposure also increases other forms of problematic behavior and reduces educational attainment.

The economic magnitude of these effects is substantial. Using crime-specific social cost estimates from Heeks et al. (2018), I estimate that peer effects account for approximately 5% of the total social cost of youth crime in my data, equivalent to about 20 million pounds. A counterfactual reassignment exercise further suggests that reallocating transfer students across schools could reduce crimes generated by peer effects by 36% under a 3-mile reassignment constraint. However, the choice of the planner's objective is crucial: minimizing the number of offenders can increase the total number of crimes, highlighting an important trade-off in the design of such policies.

These findings also have implications for school exclusion policies following student offending (Cattan et al., 2023*a*,*b*). In the extreme case of permanent exclusion, a student must transfer to another school because of compulsory education requirements. Previous studies show that

exclusion can increase subsequent offending among excluded students (McAra and McVie, 2010; Bacher-Hicks, Billings, and Deming, 2024). My results add another concern: when an offender transfers to a new school, students at the destination school become more likely to offend. At the same time, I find no evidence that removing offenders reduces criminal behavior among peers at the origin school. Taken together, these findings suggest that exclusion policies based on criminal behavior may shift negative externalities across schools rather than eliminate them. Policies addressing youth offending should therefore account not only for the consequences for excluded students, but also for the spillovers their reassignment may generate for other students.

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A Data Details

The NPD extract I received contains 164.6 million panel records for 1.1 million individuals. To make the dataset manageable and to facilitate tracking the analysis, I restrict the sample to the 1995/96–1998/99 cohorts and exclude records outside England.[78] I then restrict the sample to individuals observed at ages 10 and 15 in September so that I can track them throughout secondary school. Table A1 in the Appendix summarizes the resulting data structure, drawing on the Appendix tables in Cattan et al. (2023*a*,*b*, 2025).

My raw PNC extract contains 8,844,526 recorded offenses committed by 1,431,443 individuals, together with 15,805,405 penalty records. I first exclude offenses with missing dates or dates before 2000 or after 2017, leaving 8,749,087 offenses, 15,647,338 penalty records, and 1,423,472 individuals.[79] I then exclude offenses that did not result in a finding of guilt, leaving 6,836,862 offenses, 13,683,785 penalty records, and 1,422,793 individuals.

Using these cleaned datasets, I merge the NPD and PNC through the unique offender identifiers provided by the MoJ. I then drop 558 individuals with offense records before age 10.[80] The resulting analysis sample contains 1,957,990 individuals and 17,519,984 panel records, of whom 281,005 have an offense record. Table A2 in the Appendix reports the linkage rates.

Column (1) reports the number of individuals with a complete set of controls, including year and month of birth, secondary-school ID, SEN status, FSM status, an indicator for whether English is their first language, and ethnicity. Most individuals have complete information on these controls, although completeness rates are slightly lower among minority groups (i.e., those who are not White British), replicating the pattern in Column (2) of Table A1.

Among individuals in Column (1), Column (2) reports the fraction with a unique offender identifier. Among those in Column (2), Column (3) reports the fraction with an actual offense record in my PNC extract, while Column (4) restricts this measure to offenses committed by the end of secondary school. Column (5) calculates the juvenile offense rate as Column (2) times Column (4). Finally, Column (6) reports the fraction of linked observations matched at the most precise level, using first and last names, gender, postcode, and date of birth.

[78] Also, before 1995 cohorts, the school ID was collected annually, not for each census. This would cause the analysis based on school transfers to be much more imprecise.

[79] Some cases have multiple offense start dates for the same case and offense. In this cleaning process, I keep cases and offenses if they have at least one record between 2000 and 2017.

[80] I dropped these observations after merging the two data sets because I calculated the offense age using the year and month of birth in NPD but not PNC.

Table A1. Data Structure

Year of Birth Cohort	Available test scores/exams: Early Yrs Age 5	KS1 Age 7	KS2 Age 11	KS3 Age 14	KS4 Age 16	KS5 Age 18	Sample size (1) NPD Records	(2) NPD Records with control	(2)/(1)
1990/91			X	X	X	X	659,546	637,949	96.7%
1991/92			X	X	X	X	661,511	635,280	96.0%
1992/93			X	X	X	X	645,005	613,926	95.2%
1993/94			X	X	X	X	649,414	612,748	94.4%
1994/95		X	X		X	X	641,308	599,006	93.4%
1995/96		X	X		X	X	638,645	592,244	92.7%
1996/97		X	X		X	X	654,837	602,703	92.0%
1997/98	X	X	X		X	X	641,619	589,475	91.9%
1998/99	X	X			X	X	632,221	583,545	92.3%

Notes: We report details of the data linkage between the National Pupil Database (NPD) and the Police National Computer (PNC), where we refer to the PNC as shorthand for linked data from the Police National Computer, the Home Office Court Appearance System, and the National Offender Management Information System. Each row shows information by year of birth cohort. The controls referred to in Column (2) are whether the individual is ever classified as having special educational needs, is ever eligible for free school meals, whether English is a first language, their minority status (as captured through 12 categories: White British, Other White, Black African, Black Caribbean, Other Black, Indian, Pakistani, Bangladeshi, Other Asian, Mixed, Others and Unknown), cohort, month of birth, and the secondary school identifiers.

Table A2. Data Linkage

Birth Cohort	NPD sample with controls	Share of (1) with offense ID	Share of (2) with		Share of Individuals with offenses by end of sec. school	Share of (5) with the perfect match
			any offense	offenses by end of sec. school		
	(1)	(2)	(3)	(4)	(5)	(6)
1990/91	637,949	25.0%	81.9%	45.5%	11.4%	80.7%
1991/92	635,280	23.1%	82.6%	48.5%	11.2%	80.8%
1992/93	613,926	20.7%	82.8%	50.5%	10.4%	81.2%
1993/94	612,748	17.9%	83.1%	51.2%	9.15%	81.0%
1994/95	599,006	15.0%	83.2%	50.9%	7.65%	80.2%
1995/96	592,244	11.9%	83.1%	49.5%	5.90%	79.3%
1996/97	602,703	9.15%	83.8%	49.3%	4.51%	79.4%
1997/98	589,475	6.66%	85.9%	51.8%	3.45%	77.5%
1998/99	583,545	4.53%	91.3%	62.3%	2.82%	76.3%

Notes: I report details of the data linkage between the National Pupil Database (NPD) and the Police National Computer (PNC) where I refer to the PNC as shorthand for linked data from the Police National Computer, the Home Office Court Appearance System, and the National Offender Management Information System. Each row shows information by year of birth cohort. Column (6) indicates the quality of data linkages: we show the share of links that are perfect matches using exact information on names, date of birth, zip code of residence, and gender.

Table A3. Sample Selection

	Among NPD sample with controls Complete Records during Secondary School				Among Those in Column (1) Experienced a School-grade less than 20 Students				Among Those in Column (5) Ever Exposed to An Offender's School Transfer			
	Complete	Not Complete	Uncond. Diff. [(1) = (2)]	Cond. Diff. [(1) = (2)]	Never Experienced	Experienced	Uncond. Diff. [(5) = (6)]	Cond. Diff. [(5) = (6)]	Exposed	Never Exposed	Uncond. Diff. [(9) = (10)]	Cond. Diff. [(9) = (10)]
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
A. Demographics												
Male	.510	.540	-.031***	-.019***	.506	.710	-.203***	-.124***	.518	.494	.024***	.004***
			(.001)	(.001)			(.002)	(.004)			(.001)	(.001)
Minority	.207	.500	-.293***	-.084***	.208	.191	.016***	.009***	.185	.231	-.046***	.000
			(.001)	(.001)			(.002)	(.003)			(.001)	(.000)
English as the first language	.892	.650	.242***	.058***	.892	.908	-.016***	-.018***	.902	.882	.019***	.001
			(.001)	(.001)			(.002)	(.002)			(.000)	(.000)
Ever with SEN by age 11	.351	.353	-.002**	.006***	.341	.941	-.600***	-.296***	.365	.316	.049***	.003***
			(.001)	(.001)			(.001)	(.004)			(.001)	(.001)
Ever eligible for FSM by age 11	.274	.322	-.049***	-.010***	.270	.508	-.238***	-.175***	.296	.243	.054***	.004***
			(.001)	(.001)			(.003)	(.005)			(.001)	(.001)
Offender by age 16	.214	.237	-.024***	-.025***	.210	.451	-.241***	-.555***	.243	.176	.067***	-.006***
			(.001)	(.001)			(.003)	(.004)			(.001)	(.001)
Any school transfer by age 16	1.17	1.26	-.091***	-.070***	1.16	1.30	-.144***	-.220***	1.18	1.14	.036***	-.012***
	[.439]	[.559]	(.002)	(.003)	[.431]	[.607]	(.005)	(.008)	[.445]	[.410]	(.001)	(.002)
N of school transfers by age 16 | any transfer	.031	.135	-.104***	-.110***	.030	.124	-.094***	-.204***	.038	.021	.016***	.003***
			(.001)	(.001)			(.002)	(.004)			(.000)	(.000)
B. Education												
Age 5 overall score	.017	-.268	.285***	.080***	.029	–1.51	1.54***	.425***	-.011	.059	-.070***	.016
	[.992]	[1.08]	(.015)	(.013)	[.978]	[1.49]	(.057)	(.054)	[.967]	[.985]	(.007)	(.008)
Age 5 score: personal, social and emotional development	.018	-.282	.300***	.132***	.030	–1.47	1.50***	.480***	-.005	.055	-.061***	.018**
	[.989]	[1.12]	(.015)	(.014)	[.975]	[1.53]	(.059)	(.057)	[.970]	[.977]	(.007)	(.009)
Age 5 score: communication, language, and literacy	.019	-.290	.309***	.075***	.030	–1.41	1.44***	.383***	-.025	.070	-.095***	.023***
	[.992]	[1.07]	(.015)	(.012)	[.982]	[1.27]	(.049)	(.048)	[.978]	[.983]	(.007)	(.008)
Age 5 score: mathematical development	.016	-.257	.274***	.065***	.028	–1.48	1.51***	.416***	-.018	.062	-.079***	.020**
	[.990]	[1.12]	(.015)	(.013)	[.975]	[1.58]	(.061)	(.058)	[.977]	[.972]	(.007)	(.009)
Age 5 score: knowledge and understanding the world	.012	-.191	.204***	.042***	.022	–1.27	1.29***	.393***	-.006	.042	-.048***	.031***
	[.992]	[1.11]	(.015)	(.014)	[.980]	[1.56]	(.060)	(.060)	[.983]	[.977]	(.007)	(.009)
Age 5 score: physical development	.011	-.178	.189***	.051***	.022	–1.41	1.43***	.443***	-.001	.039	-.040***	.033***
	[.992]	[1.11]	(.015)	(.014)	[.975]	[1.78]	(.068)	(.068)	[.973]	[.976]	(.007)	(.009)
Age 5 score: creative development	.012	-.190	.202***	.040***	.022	–1.31	1.33***	.330***	-.009	.045	-.054***	.022**
	[.993]	[1.09]	(.015)	(.014)	[.980]	[1.61]	(.062)	(.059)	[.982]	[.977]	(.007)	(.009)
Age 7 overall score	.026	-.375	.401***	.099***	.039	–1.85	1.89***	.504***	-.030	.109	-.139***	-.002
	[.988]	[1.09]	(.003)	(.002)	[.972]	[1.42]	(.012)	(.012)	[.961]	[.979]	(.001)	(.001)
Age 7 score (English, math, science)	.025	-.357	.382***	.094***	.037	–1.84	1.88***	.503***	-.028	.105	-.133***	-.003
	[.990]	[1.08]	(.003)	(.002)	[.973]	[1.49]	(.013)	(.012)	[.959]	[.983]	(.001)	(.001)
Age 7 score: English reading	.025	-.367	.392***	.098***	.037	–1.62	1.66***	.428***	-.030	.105	-.135***	-.000
	[.988]	[1.10]	(.003)	(.002)	[.976]	[1.26]	(.011)	(.011)	[.974]	[.973]	(.001)	(.001)
Age 7 score: English writing	.025	-.365	.391***	.103***	.037	–1.68	1.71***	.472***	-.028	.103	-.131***	-.002
	[.987]	[1.11]	(.003)	(.002)	[.975]	[1.23]	(.011)	(.011)	[.970]	[.975]	(.001)	(.001)
Age 7 score: English listening and speaking	.020	-.293	.314***	.065***	.031	–1.60	1.64***	.418***	-.025	.090	-.115***	-.003
	[.992]	[1.06]	(.003)	(.003)	[.980]	[1.41]	(.012)	(.012)	[.967]	[.989]	(.001)	(.002)
Age 7 score: overall English	.025	-.364	.389***	.092***	.038	–1.78	1.82***	.476***	-.030	.107	-.137***	-.002
	[.989]	[1.09]	(.003)	(.002)	[.974]	[1.36]	(.012)	(.012)	[.964]	[.979]	(.001)	(.001)
Age 7 score: math	.022	-.318	.340***	.089***	.033	–1.65	1.68***	.471***	-.025	.093	-.118***	-.002
	[.991]	[1.07]	(.003)	(.003)	[.978]	[1.45]	(.013)	(.013)	[.969]	[.984]	(.001)	(.001)
Age 7 score: science	.020	-.292	.312***	.076***	.031	–1.58	1.61***	.424***	-.022	.085	-.107***	-.003
	[.993]	[1.06]	(.003)	(.003)	[.980]	[1.49]	(.013)	(.013)	[.966]	[.991]	(.001)	(.002)
Age 11 overall score	.026	-.387	.412***	.148***	.029	–1.14	1.17***	.418***	-.066	.140	-.206***	-.007***
	[.989]	[1.08]	(.004)	(.003)	[.987]	[1.15]	(.019)	(.019)	[.983]	[.981]	(.002)	(.002)
Age 11 score: English	.025	-.373	.398***	.128***	.028	–1.21	1.23***	.412***	-.067	.142	-.209***	-.008***
	[.990]	[1.07]	(.003)	(.003)	[.987]	[1.11]	(.018)	(.017)	[.983]	[.980]	(.002)	(.002)
Age 11 score: math	.024	-.351	.375***	.138***	.027	–1.06	1.08***	.404***	-.058	.128	-.186***	-.005***
	[.992]	[1.05]	(.003)	(.003)	[.990]	[1.08]	(.016)	(.017)	[.987]	[.984]	(.002)	(.002)
Age 11 score: science	.026	-.380	.407***	.161***	.031	–1.18	1.21***	.464***	-.047	.123	-.170***	-.004**
	[.986]	[1.12]	(.004)	(.003)	[.982]	[1.26]	(.017)	(.018)	[.990]	[.965]	(.002)	(.002)
Age 16 score: taken	1.00	.805	.195***	.198***	1.00	.997	.002***	.003***	1.00	1.00	.000	.000
			(.001)	(.001)			(.000)	(.001)			(.000)	(.000)
Age 16 score: 5 A* – G	.872	.536	.336***	.330***	.884	.150	.734***	.432***	.857	.912	-.056***	-.006***
			(.001)	(.001)			(.002)	(.004)			(.000)	(.001)
Age 16 score: 5 A* – C	.607	.324	.283***	.249***	.616	.051	.565***	.209***	.557	.676	-.119***	-.010***
			(.001)	(.001)			(.001)	(.003)			(.001)	(.001)
Age 16 score: point score | taken	.083	-.932	1.01***	.994***	.098	–2.00	2.09***	1.17***	.018	.178	-.160***	.001
	[.932]	[1.24]	(.003)	(.003)	[.916]	[.713]	(.006)	(.009)	[.912]	[.913]	(.001)	(.001)
Sample size	2,123,028	244,939	2,367,967	2,367,967	2,088,731	34,297	2,123,028	2,123,028	1,050,567	1,038,164	2,088,731	2,088,731

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard deviations are reported in brackets, and heteroskedasticity-robust standard errors clustered at the secondary-school level are reported in parentheses. I use the earliest available secondary-school record. The sample in Columns (1)–(4) consists of observations in the NPD with the full set of control variables. The sample size corresponds to the sum of the 1995/96–1998/99 cohorts reported in Column (2) of Table A1 and Column (1) of Table A2. Conditional differences control for gender, minority status, an indicator for English as the first language, and SEN and FSM status at age 11, unless the variable is itself the outcome. All test scores are standardized within each cohort, except for the indicator variables for the age 16 tests. Early-years test scores are available only for a quasi-random 10% of the sample comprising children born on the 5th, 15th, or 25th of each month (Department for Education, 2010). The age 7 overall score is calculated as a weighted average of teachers' assessments of Reading, Writing, Speaking and Listening, Math, and Overall Science. Within English, Reading, Writing, and Speaking and Listening are weighted 1:1:2 so that the four underlying abilities contribute equally. English, Math, and Science are then weighted 1:1:1 in constructing the overall score. The age 11 overall score is calculated as a weighted average of English, Math, and Science, with equal weight assigned to each subject. Some observations are missing information for certain variables. The reported sample size is the number of observations with demographic information. Data sources: Ministry of Justice and Department for Education.

Table A4. Summary Statistics on Non-offenders and Offenders

	Offending Status by Age 16			
	Non-offenders	Offenders	Uncond. Diff. [(1) = (2)]	Cond. Diff. [(1) = (2)]
	(1)	(2)	(3)	(4)
A. Demographics				
Male	.511	.697	-.186***	-.157***
			(.002)	(.002)
Minority	.185	.171	.014***	-.007***
			(.002)	(.001)
English as the first language	.901	.930	-.029***	-.007***
			(.001)	(.001)
Ever with SEN by age 11	.356	.604	-.248***	-.148***
			(.003)	(.002)
Ever eligible for FSM by age 11	.285	.591	-.306***	-.235***
			(.003)	(.002)
Any school transfer by age 16	.238	.387	-.149***	-.091***
			(.002)	(.002)
N of school transfers by age 16 | any transfer	1.17	1.30	-.130***	-.093***
	[.434]	[.587]	(.005)	(.005)
Sample size	1,010,977	39,590	1,050,567	1,050,567
B. Age 5 Education				
Age 5 overall score	.018	-.328	.346***	.033
	[.939]	[.964]	(.031)	(.028)
Age 5 score: personal, social and emotional development	.024	-.325	.349***	.066**
	[.941]	[1.02]	(.032)	(.032)
Age 5 score: communication, language, and literacy	.004	-.400	.404***	.030
	[.957]	[.977]	(.031)	(.028)
Age 5 score: mathematical development	.012	-.344	.356***	.059
	[.945]	[1.05]	(.033)	(.031)
Age 5 score: knowledge and understanding the world	.017	-.178	.194***	-.012
	[.959]	[1.02]	(.032)	(.033)
Age 5 score: physical development	.022	-.098	.120***	-.088***
	[.941]	[.983]	(.031)	(.032)
Age 5 score: creative development	.016	-.245	.261***	-.001
	[.955]	[1.05]	(.033)	(.034)
Sample size	35,736	1,013	36,749	36,749
C. Age 7 Education				
Age 7 overall score	.010	-.444	.454***	.117***
	[.927]	[.953]	(.005)	(.004)
Age 7 score (English, math, science)	.010	-.400	.410***	.104***
	[.925]	[.942]	(.005)	(.004)
Age 7 score: English reading	.007	-.476	.483***	.135***
	[.948]	[.994]	(.005)	(.004)
Age 7 score: English writing	.009	-.460	.469***	.119***
	[.944]	[1.01]	(.005)	(.005)
Age 7 score: English listening and speaking	.009	-.365	.374***	.091***
	[.941]	[.949]	(.005)	(.004)
Age 7 score: overall English	.009	-.458	.467***	.120***
	[.934]	[.962]	(.005)	(.004)
Age 7 score: math	.008	-.327	.336***	.081***
	[.940]	[.966]	(.005)	(.005)
Age 7 score: science	.011	-.310	.321***	.083***
	[.938]	[.944]	(.005)	(.005)
Sample size*2	961,786	37,914	999,700	999,700
D. Age 11 Education				
Age 11 overall score	-.042	-.577	.535***	.226***
	[.976]	[.971]	(.006)	(.005)
Age 11 score: English	-.042	-.616	.574***	.217***
	[.975]	[.972]	(.006)	(.005)
Age 11 score: math	-.038	-.462	.424***	.168***
	[.982]	[.971]	(.005)	(.005)
Age 11 score: science	-.022	-.529	.507***	.240***
	[.979]	[1.03]	(.006)	(.005)
Sample size*3	783,184	31,952	815,136	815,136
E. Age 16 Education				
Age 16 score: taken	1.00	.998	.001***	.001***
			(.000)	(.000)
Age 16 score: 5 A* – G | taken	.871	.491	.380***	.275***
			(.003)	(.002)
Age 16 score: 5 A* – C | taken	.573	.143	.430***	.238***
			(.002)	(.002)
Age 16 score: point score | taken	.078	-.924	1.00***	.720***
	[.868]	[.995]	(.005)	(.005)
Sample size	1,010,795	39,526	1,050,321	1,050,321

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard deviations are reported in brackets, and heteroskedasticity-robust standard errors clustered at the secondary-school level are reported in parentheses. I use the earliest available secondary-school record. Conditional differences control for gender, minority status, an indicator for English as the first language, and SEN and FSM status at age 11, unless the variable is itself the outcome. Offenders in this table are defined as individuals in the NPD with at least one recordable offense in the PNC committed between the September ages of 11 and 16 that resulted in a caution or conviction. All test scores are standardized within each cohort, except for the indicator variables for the age 16 tests. Early-years test scores are available only for a quasi-random 10% of the sample comprising children born on the 5th, 15th, or 25th of each month (Department for Education, 2010). The age 7 overall score is calculated as a weighted average of teachers' assessments of Reading, Writing, Speaking and Listening, Math, and Overall Science. Within English, Reading, Writing, and Speaking and Listening are weighted 1:1:2 so that the four underlying abilities contribute equally. English, Math, and Science are then weighted 1:1:1 in constructing the overall score. The age 11 overall score is calculated as a weighted average of English, Math, and Science, with equal weight assigned to each subject. Some observations are missing information for certain variables. The reported sample size is the number of observations with demographic information. Data sources: Ministry of Justice and Department for Education.

Table A5. Detailed Common Offense Descriptions

Rank	Count	Detailed offense description
Theft Offenses		
1	23640	Theft from shops
2	6554	Other stealing and unauthorised taking: Offences under the Theft Act 1968 S.1 not classified elsewhere
3	6169	Robbery
4	4735	Burglary in a building other than a dwelling with intent to steal/inflict grievous bodily harm/commit damage – triable either way
5	4370	Other burglary in a dwelling
6	2572	Theft of pedal cycle
7	2392	Stealing from the person of another
8	1589	Theft in a dwelling other than from automatic machine or meter
9	1442	Theft from a motor vehicle
10	2143	Others
Battery Offenses		
1	30631	Common assault and battery
2	101	Assaulting a designated or accredited person, or person assisting him or her, in the execution of his or her duty
3	16	Resisting or wilfully obstructing a designated or accredited person, or person assisting, in the execution of his or her duty
4	15	Others
Drug-related Offenses		
1	6231	Having possession of a controlled drug – class B (cannabis, including cannabis resin, cannabinol and cannabinol derivatives)
2	264	Having possession of a controlled drug – class B (other)
3	192	Possession of a controlled drug with intent to supply – class B (cannabis, including cannabis resin, cannabinol and cannabinol derivatives)
4	116	Supplying or offering to supply a controlled drug – class B (cannabis, including cannabis resin, cannabinol and cannabinol derivatives)
5	105	Having possession of a controlled drug – class C (cannabis, including cannabis resin, cannabinol and cannabinol derivatives) – historic
6	66	Having possession of a controlled drug – class A (cocaine)
7	57	Production of or being concerned in production of a controlled drug – class B (cannabis, including cannabis resin, cannabinol and cannabinol derivatives)
8	51	Possession of a controlled drug with intent to supply – class A (heroin)
9	48	Possession of a controlled drug with intent to supply – class A (crack cocaine)
10	525	Others
Violent Offenses		
1	7401	Assaults occasioning actual bodily harm
2	5242	Causing harassment, alarm or distress
3	1952	Causing fear or provocation of violence
4	1386	Assault on a constable
5	1252	Breach of Anti-Social Behaviour Order and Interim ASBO (order made to protect from alarm, distress or harassment)
6	1148	Affray
7	1121	Causing intentional harassment, alarm or distress
8	596	Wound / inflict grievous bodily harm without intent
9	309	Racially or religiously aggravated common assault or beating
10	1822	Others
Fraud Offenses		
1	1294	Receiving stolen goods
2	1079	Undertaking or assisting in the retention, removal, disposal or realisation of stolen goods, or arranging to do so
3	567	Dishonestly making a false representation to make a gain for oneself or another or to cause loss to another or to expose another to a risk
4	178	Travelling by railway without paying correct fare, failing to show ticket, failing to give name and address, etc.
5	81	Attempting to pervert the course of public justice (fabrication of false evidence, causing person to be wrongly convicted, interference with witness) – indictable only
6	35	Communicating false information alleging the presence of bombs
7	31	Possession etc. of articles for use in frauds
8	23	Passing etc. counterfeit coin or note as genuine
9	23	Obtaining pecuniary advantage by deception (except railway frauds)
10	74	Others
Weapon Offenses		
1	1774	Having an article with a blade or point in a public place
2	1323	Possession of offensive weapons without lawful authority or reasonable excuse
3	588	Having an article with a blade or point on school premises
4	232	Possession of offensive weapons without lawful authority or reasonable excuse on school premises
5	158	Possess imitation firearm or loaded/unloaded air weapon in public place
6	152	Carrying a loaded or unloaded or imitation firearm or air weapon in public place
7	98	Possession of a firearm or imitation firearm, with intent to cause fear of violence (Group I)
8	67	Possessing anything with intent to destroy or damage property
9	52	Possessing or distributing prohibited weapons designed for discharge of noxious liquid etc. (Group I)
10	148	Others
Damage/Arson Offenses		
1	21708	Causing damage to an allotment through negligence or any unlawful act
2	3410	Other criminal damage
3	1503	Arson not endangering life
4	114	Threat to commit criminal damage
5	105	Arson endangering life
6	70	Unlawfully and maliciously causing or conspiring to cause explosion likely to endanger life or property
7	60	Racially or religiously aggravated criminal damage
8	43	Racially aggravated other criminal damage
9	17	Others
Sexual Offenses		
1	635	Sexual assault on a female aged 13 or over – no penetration
2	365	Sexual assault on a female child aged under 13 – touching
3	245	Taking, permitting to be taken or making, distributing or publishing indecent photographs or pseudo photographs of children
4	185	Rape of a female child aged under 13 by a male
5	177	Exposure
6	142	Sexual assault on a male child aged under 13 – touching
7	107	Rape of a male child aged under 13 by a male
8	103	Sexual assault on a female child aged under 13 – penetration
9	100	Sexual activity with a female child aged 13 to 15 – offender aged under 18 – penetration
10	1070	Others
Motoring Offenses		
1	1161	Unauthorised taking or driving or attempt taking or driving of a motor vehicle
2	457	Being carried knowing vehicle to have been taken or driven away
3	457	Interference with motor vehicle/trailer
4	344	Using motor vehicle uninsured against third party risks
5	247	Taking or riding a pedal cycle without consent etc.
6	195	Driving, causing or permitting a person to drive other than in accordance with a licence (full or provisional) (except HGV)
7	124	Unauthorised taking or driving of conveyance other than motor vehicle or pedal cycle
8	53	Aiding, abetting, causing or permitting reckless driving
9	50	Driving whilst disqualified from holding or obtaining a licence
10	196	Others
Other Offenses		
1	3692	[description missing in source]
2	1385	Absconding by person released on bail
3	1222	Being guilty, while drunk, of disorderly behaviour
4	708	Resisting or obstructing a constable in execution of duty – offences under the Police Act 1996
5	495	Racially or religiously aggravated intentional harassment or alarm or distress – words or writing
6	485	Offence of harassment
7	311	Improper use of public electronic communications network
8	300	Racially or religiously aggravated harassment or alarm or distress – words or writing
9	277	Being on enclosed premises for an unlawful purpose
10	1671	Others

Table A6. Summary Statistics on Schools and School Transfers, Not-yet-offenders

		Transfer (school-term) level									
		Offenders' transfers				Not-yet-offenders' transfers				Difference-in-differences	
	All school (1)	Origin school (2)	Destination school (3)	Uncond. Diff. [(2)=(3)] (4)	Cond. Diff. [(2)=(3)] (5)	Origin school (6)	Destination school (7)	Uncond. Diff. [(6)=(7)] (8)	Cond. Diff. [(6)=(7)] (9)	Uncond. Diff. [(4)=(8)] (10)	Cond. Diff. [(4)=(8)] (11)
Male	.531	.536	.544	-.008***	-.002**	.531	.530	.000	-.001	-.009***	-.002
	[.174]	[.133]	[.122]	(.001)	(.001)	[.136]	[.123]	(.001)	(.001)	(.002)	(.002)
Minority	.216	.193	.185	.008***	-.002**	.213	.201	.011***	.002***	-.003	-.004***
	[.256]	[.235]	[.222]	(.002)	(.001)	[.255]	[.243]	(.002)	(.001)	(.003)	(.001)
English as the first language	.885	.890	.899	-.009***	-.004***	.880	.887	-.007***	.000	-.002	-.003***
	[.174]	[.167]	[.153]	(.002)	(.001)	[.177]	[.168]	(.002)	(.001)	(.002)	(.001)
Ever with SEN by age 11	.416	.422	.444	-.022***	-.014***	.412	.416	-.003***	-.011***	-.019***	-.002**
	[.199]	[.143]	[.158]	(.001)	(.001)	[.128]	[.129]	(.001)	(.001)	(.002)	(.001)
Ever eligible for FSM by age 11	.331	.378	.397	-.019***	-.006***	.366	.365	.001	-.002**	-.020***	-.002
	[.197]	[.188]	[.200]	(.002)	(.001)	[.183]	[.182]	(.002)	(.001)	(.003)	(.002)
Early years overall score	-.152	-.013	-.015	.003	-.004	-.008	-.008	-.000	-.001	.003	-.002
	[.686]	[.060]	[.049]	(.003)	(.003)	[.069]	[.042]	(.003)	(.002)	(.004)	(.004)
Age 7 overall score	-.210	-.132	-.168	.036***	.002	-.107	-.119	.011***	.011***	.024***	-.009***
	[.744]	[.307]	[.338]	(.003)	(.002)	[.295]	[.313]	(.003)	(.002)	(.004)	(.003)
Age 11 overall score	-.180	-.201	-.227	.026***	.008***	-.182	-.188	.006**	.011***	.020***	-.002
	[.590]	[.283]	[.283]	(.003)	(.002)	[.285]	[.270]	(.003)	(.002)	(.004)	(.003)
Offender	.042	.054	.091	-.036***	-.028***	.027	.036	-.009***	-.011***	-.027***	-.017***
	[.101]	[.103]	[.150]	(.001)	(.001)	[.054]	[.071]	(.001)	(.001)	(.001)	(.001)
N of students	154	169	179	-9.28***	-16.2***	166	190	-24.4***	-25.0***	15.1***	9.51***
	[75.6]	[73.1]	[92.8]	(.800)	(.669)	[66.5]	[83.3]	(.694)	(.633)	(1.06)	(.921)
Sample size	4,559	21,803	21,803	43,606	43,606	23,590	23,590	47,180	47,180	90,786	90,786

Notes: *** denotes significance at the 1

Table A7. Detailed Common Offense Descriptions that Lead to Caution Marginally

Rank	Count	Detailed offense description
Theft Offenses		
1	101,395	Burglary in a building other than a dwelling with intent to steal/inflict grievous bodily harm/commit damage – triable either way
2	46,714	Theft from a motor vehicle
3	31,128	Theft of pedal cycle
4	23,280	Theft in a dwelling other than from automatic machine or meter
5	12,962	Theft of a motor vehicle (excluding aggravated vehicle taking) – triable either way
6	594	Theft from a vehicle – other than a motor vehicle
Battery Offenses		
1	476,403	Common assault and battery
Drug-related Offenses		
1	4,638	Having possession of a controlled drug – class A (MDMA)
2	777	Production or being concerned in production of a controlled drug – class C (cannabis, including cannabis resin, cannabinol and cannabinol derivatives) (historic)
3	458	Supplying or offering to supply a controlled drug – class C (other)
Violent Offenses		
1	115,566	Causing harassment, alarm or distress
2	1,404	Making threats to kill
3	625	Sending letters etc. with intent to cause distress or anxiety
Fraud Offenses		
1	10,746	Failure to comply with requirements concerning approval and signing of abbreviated accounts
2	8,341	Travelling by railway without paying correct fare, failing to show ticket, failing to give name and address, etc.
3	1,240	Passing etc. counterfeit coin or note as genuine
4	1,008	Possession etc. of articles for use in frauds
5	476	Forgery or copying false instrument
Weapon Offenses		
1	56,984	Possession of offensive weapons without lawful authority or reasonable excuse
2	3,457	Possession of a firearm or imitation firearm, with intent to cause fear of violence (Group I)
3	1,716	Carrying a loaded or unloaded or imitation firearm or air weapon in public place
Sexual Offenses		
1	684	Sexual activity with a female child aged 13 to 15 – offender aged 18 or over – penetration
2	287	Exposing the person in any street, etc. or in view thereof, or in any place or public resort with intent to insult any female
Motoring Offenses		
1	23,994	Being carried knowing vehicle to have been taken or driven away
Other Offenses		
1	48,702	Being guilty, while drunk, of disorderly behaviour
2	12,358	Offence of harassment
3	4,330	Improper use of public electronic communications network
4	2,773	Contravention of a direction given by a constable under Section 30(4) [dispersal of groups and removal of persons under 16 to their place of residence]
5	1,583	Offences against private Acts relating to railways
6	1,056	Being found drunk in a highway or other public place whether a building or not, or a licensed premises
7	613	Railway byelaws as to the conduct of passengers
8	468	False alarms of fire (Fire Services Act 1947)
9	205	Offences against the State and Public Order: Other offences under which proceedings were taken
10	202	Possessing fireworks etc. in, or when entering, a designated sports ground

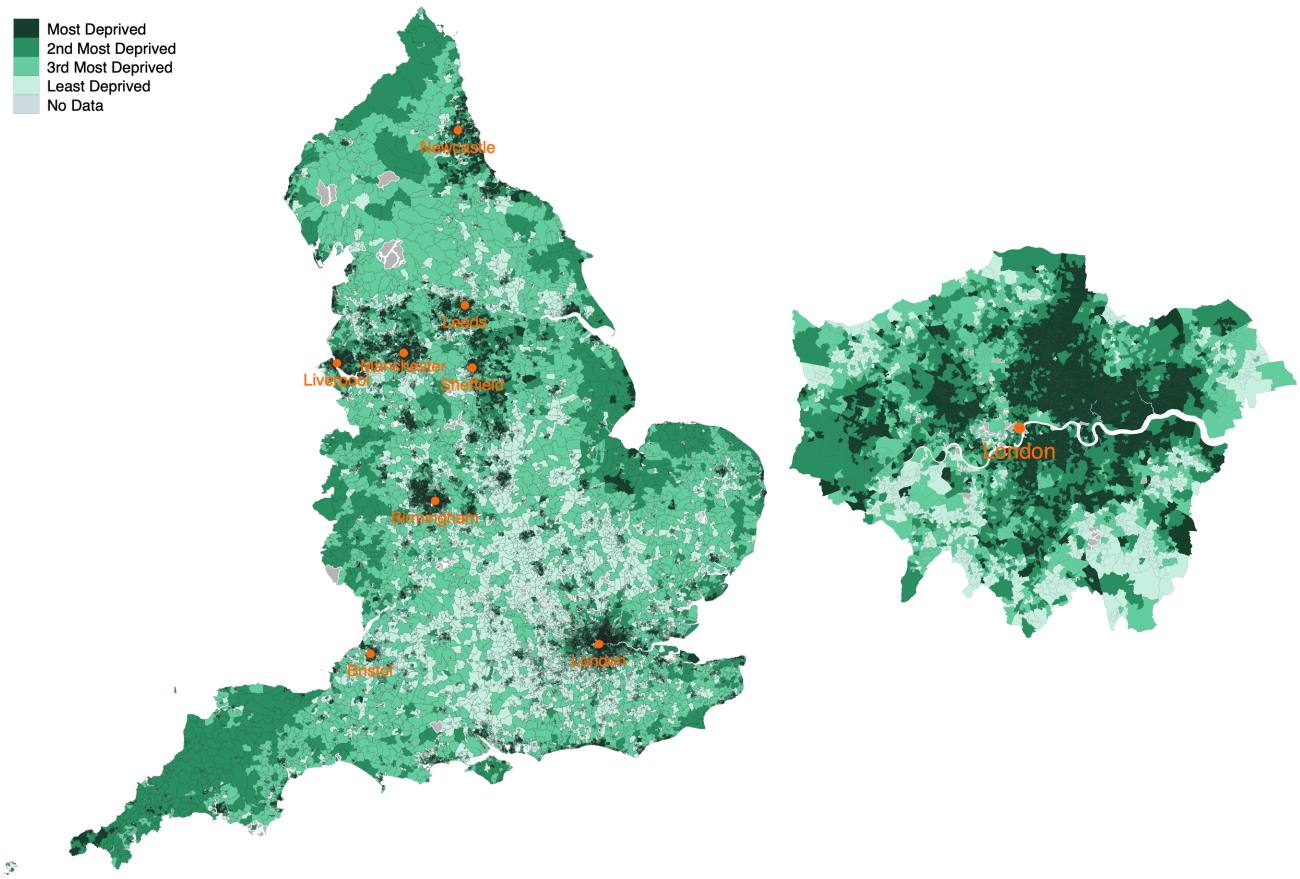


Figure A1. Deprivation Index across England and London

Note: The location of London in the right panel is that of the Prime Minister's Office, 10 Downing Street.

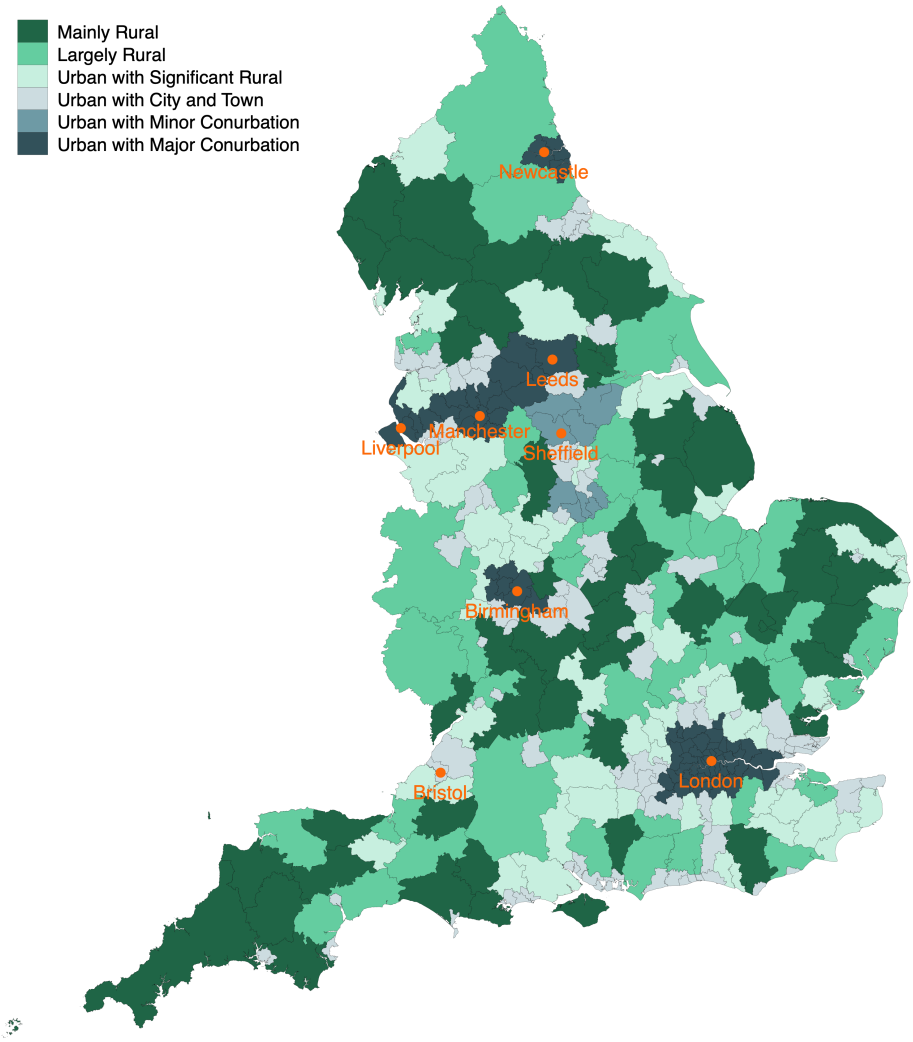


Figure A2. Rural and Urban Classification

B Details in Estimation

The estimation framework is as follows. There are $\mathcal{T}$ periods, indexed by $t = 1, 2, ..., \mathcal{T}$. Each period corresponds to a school census season, so the interval between periods is approximately four months. Let D_{it} denote the treatment indicator, where $D_{it} = 1$ if individual i has been exposed to an offender's transfer by time t. Because I exclude transfers occurring in the first term at age 11, $D_{i1} = 0$. I assume that treatment is irreversible: once a child is exposed to an offender's transfer, the treatment status cannot be reversed. Thus, $D_{i,t-1} = 1$ implies $D_{i,t} = 1$ a.s. for $t = 2, ..., \mathcal{T}$.

Let G_i denote the period in which individual i first becomes treated. If an individual is never treated, let $G_i = \infty$.[81] Define $G_{i,g}$ as an indicator taking the value one if individual i is first treated in period g, such that $G_{i,g} = \mathbb{1}\{G_i = g\}$. Define C_i as an indicator taking the value one if individual i is never treated, i.e., $C_i = \mathbb{1}\{G_i = \infty\} = 1 - D_{i,\mathcal{T}}$. Let $\bar{g} = \max_{i=1,...,n} G_i$ denote the latest treatment period observed in the data.

Let $p_{i,g,s}(X_i) = P(G_{i,g} = 1 | X_i, G_{i,g} + (1 - D_{i,s})(1 - G_{i,g}) = 1)$ denote the generalized propensity score, that is, the probability of first being treated in period g, conditional on pre-treatment covariates X_i and on either being treated in period g ($G_{i,g} = 1$) or belonging to the "not-yet-treated" group ($G_{i,g} = 0$ and $D_{i,s} = 0$). Also define $p_{i,g}(X_i) = p_{i,g,\mathcal{T}}(X_i) = P(G_{i,g} = 1 | X_i, G_{i,g} + C_i = 1)$, which is the probability of first being treated in period g, conditional on X_i and on either being treated in period g ($G_{i,g} = 1$ and $C_i = 0$) or belonging to the "never-treated" group ($G_{i,g} = 0$ and $C_i = 1$). Let $\mathcal{G} = \text{supp}(G_i) \backslash \{\bar{g}\} \subseteq \{2, 3, ...\mathcal{T}\}$, let $\mathcal{X} = \text{supp}(X_i) \subseteq \mathbb{R}^k$, and, for a generic $\delta \geq 0$, let $\mathcal{G}_\delta = \mathcal{G} \cap \{2 + \delta, 3 + \delta, ...\mathcal{T}\}$.

Finally, for $g = 2, ..., \mathcal{T}$, let $Y_{i,t}(g)$ denote the potential outcome that individual i would experience at time t if first treated in period g. The observed outcome $Y_{i,t}$ can then be expressed as

$$Y_{i,t} = Y_{i,t}(0) + \sum_{g=2}^{\mathcal{T}} (Y_{i,t}(g) - Y_{i,t}(0)) \times G_{i,g}. \tag{3}$$

Based on this expression, the average treatment effect on the treated (ATT) for the pair (g, t) is

$$ATT(g, t) = \mathbb{E}[Y_{i,t}(g) - Y_{i,t}(0) | G_{i,g} = 1], \tag{4}$$

which is the main building block of the framework.

In this paper, Y_{it} primarily takes three forms.

The first is an indicator equal to one if individual i has committed an offense *by* time t. This outcome allows me to examine whether previously nonoffending students begin offending following an offender's school transfer. As emphasized above, this transition is particularly important when

[81] In this paper, I omit the never-treated individuals, but I keep this notation for completeness.

considering potential consequences for human capital, relative to the literature studying peer effects among individuals who are already involved in crime, such as those in incarceration settings.

The second is an indicator equal to one if individual i commits an offense *at* time t. The third is the number of offenses committed by individual i *at* time t. I use these latter two outcomes to examine the effect of exposure on contemporaneous criminal activity.

Following the assumptions in Callaway and Sant'Anna (2021), I impose the following conditions. First, I assume random sampling, such that $\{Y_{i,1}, Y_{i,2}, ..., Y_{i,\mathcal{T}}, X_i, D_{i,1}, D_{i,2}, ..., D_{i,\mathcal{T}}\}_{i=1}^{n}$ is independently and identically distributed.

Second, under limited treatment anticipation, I assume that there is a known $\delta \geq 0$ such that

$$\mathbb{E}[Y_{i,t}(g) \mid X_i, G_{i,g} = 1] = \mathbb{E}[Y_{i,t}(0) \mid X_i, G_{i,g} = 1] \quad a.s.$$

for all $g \in \mathcal{G}, t \in \{1, ..., \mathcal{T}\}$ s.t. $t < g - \delta$. In the context of this paper, this assumption requires that children do not change their criminal behavior in anticipation of an offender's transfer before the relevant anticipation window. This assumption is plausible if children cannot predict when an offender will enter their school and grade, as discussed above.

Third, I impose conditional parallel trends using either (i) a "never-treated" group or (ii) a "not-yet-treated" group. Let δ be defined as above. Under (i), for each $g \in \mathcal{G}$ and $t \in \{2, ..., \mathcal{T}\}$ such that $t \geq g - \delta$,

$$\mathbb{E}[Y_{i,t}(0) - Y_{i,t-1}(0) \mid X_i, G_{i,g} = 1] = \mathbb{E}[Y_{i,t}(0) - Y_{i,t-1}(0) \mid X_i, C_i = 1] \; a.s.,$$

and under (ii), for each $g \in \mathcal{G}$ and $(s,t) \in \{2, ..., \mathcal{T}\} \times \{2, ..., \mathcal{T}\}$ such that $t \geq g - \delta$ and $t + \delta \leq s < \bar{g}$,

$$\mathbb{E}[Y_{i,t}(0) - Y_{i,t-1}(0) \mid X_i, G_{i,g} = 1] = \mathbb{E}[Y_{i,t}(0) - Y_{i,t-1}(0) \mid X_i, D_{i,s} = 0, G_{i,g} = 0] \; a.s.$$

These assumptions require that, conditional on X_i, untreated potential outcomes would evolve similarly across the treatment and relevant control groups in the absence of treatment. I consider both control-group definitions and show that the results are robust to the choice of control group.

Finally, I impose an overlap assumption. For each $t \in \{2, ..., \mathcal{T}\}, g \in \mathcal{G}$, there exists some $\varepsilon > 0$ such that $P(G_{i,g} = 1) > 0$ and $p_{i,g,t}(X_i) < 1 - \varepsilon \; a.s.$ This assumption requires a positive probability of first receiving treatment in each relevant treatment period while ruling out treatment with probability one conditional on the observed covariates. In the empirical setting, a school cannot reject a transfer solely because the student is an offender, nor can it ensure that it will receive an offender with certainty, making this assumption plausible.

Under these assumptions, the doubly robust estimand can be written as

$$ATT(g,t;\delta) = \mathbb{E}\left[\left(\frac{G_{i,g}}{\mathbb{E}[G_{i,g}]} - \frac{\frac{p_{i,g}(X_i)C_i}{1-p_{i,g}(X_i)}}{\mathbb{E}\left[\frac{p_{i,g}(X_i)C_i}{1-p_{i,g}(X_i)}\right]}\right)(Y_{i,t} - Y_{i,g-\delta-1} - m^{nev}_{i,g,t,\delta}(X_i))\right], \tag{5}$$

under the specification with the "never-treated" group, where $m^{nev}_{i,g,t,\delta}(X_i) = \mathbb{E}[Y_{i,t} - Y_{i,g-\delta-1}|X_i, C_i = 1]$, and

$$ATT(g,t;\delta) = \mathbb{E}\left[\left(\frac{G_{i,g}}{\mathbb{E}[G_{i,g}]} - \frac{\frac{p_{i,g,t+\delta}(X_i)(1-D_{i,t+\delta})(1-G_{i,g})}{1-p_{i,g,t+\delta}(X_i)}}{\mathbb{E}\left[\frac{p_{i,g,t+\delta}(X_i)(1-D_{i,t+\delta})(1-G_{i,g})}{1-p_{i,g,t+\delta}(X_i)}\right]}\right)(Y_{i,t} - Y_{i,g-\delta-1} - m^{ny}_{i,g,t,\delta}(X_i))\right], \tag{6}$$

under the specification with the "not-yet-treated" group, where $m^{ny}_{i,g,t,\delta}(X_i) = \mathbb{E}[Y_{i,t} - Y_{i,g-\delta-1}|X_i, D_{i,t+\delta} = 0, G_{i,g} = 0]$.

I then aggregate the group-time treatment effects by event time. The treatment effect e periods after exposure is

$$\theta_{es}(e) = \sum_{g\in\mathcal{G}} \mathbb{1}\{g + e \leq \mathcal{T}\}P(G = g|G + e \leq \mathcal{T})ATT(g, g + e), \tag{7}$$

where the primary parameter of interest in this paper is $\theta_{es}(0)$, which captures the treatment effect immediately following an offender's transfer.

C Robustness Checks

Table C1. Age of Not-Yet-Offenders' School Transfers

Age in Sept.	Not-yet-offenders' Transfers		
	Autumn (1)	Spring (2)	Summer (3)
A. Transfer			
11	-	1,430	1,997
12	4,687	1,861	1,948
13	5,301	1,696	1,254
14	2,717	1,065	617
15	751	383	60
B. First Contact to Others' Transfers (Treatment)			
11	-	173,126	180,795
12	210,195	121,780	105,445
13	119,196	67,767	62,753
14	58,013	34,375	22,418
15	16,761	3,380	517

Notes: Offenders are those who have committed a crime by the time of their school transfer, while non-offenders are those who have never committed a crime at all or those not yet by the time of their school transfer.

Table C2. Effects of Disadvantaged and Disruptive Students' Transfers

Characteristics of Transfer Students:	Any	FSM	SEN	FSM and SEN	Suspension	FSM, SEN, and Suspension
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Non-Offenders' Transfers						
Treatment Effects at $t = 0$	.125	.075***	.043***	.114***	.106***	.122***
	(.115)	(.008)	(.009)	(.010)	(.008)	(.012)
Pre-Treatment Mean	1.56	.632	.640	.744	1.22	1.39
Relative Effect Size	8.02%	11.9%	6.79%	15.3%	8.72%	8.74%
Sample Size (Panel Count)	9,455,532	14,409,248	14,286,249	13,819,441	12,551,511	10,406,529
Number of Individuals	642,916	979,095	970,886	939,161	852,880	707,334
Panel B: Offenders' Transfers						
Treatment Effects at $t = 0$	.193***	.208***	.205***	.218***	.192***	.224***
	(.009)	(.012)	(.012)	(.014)	(.013)	(.017)
Pre-Treatment Mean	1.50	1.52	1.52	1.53	1.50	1.52
Relative Effect Size	12.9%	13.7%	13.4%	14.2%	12.8%	14.8%
Sample Size (Panel Count)	15,458,144	12,327,449	11,676,655	9,720,359	10,750,223	7,022,390
Number of Individuals	1,050,567	837,843	793,664	660,754	730,468	477,164

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. Control variables include FSM and SEN status before secondary school, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, a linear month-of-birth term, and a linear age term. This table reports the treatment effects of exposure to the school transfer of an offender or a disadvantaged or disruptive student. The treatment group consists of students who are exposed to an individual with the characteristics indicated in each column transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. In Panel A, I exclude students who are exposed to an offender's transfer at the same time as the transfer of the specific type of individual considered in each column. Column (1) in Panel A adjusts the estimation using sampling weights based on treatment timing. In Columns (2) to (6) in Panel B, I exclude students who were exposed to the transfer of an offender without the specified characteristic before the transfer of the specific type of individual considered in each column. Data sources: Ministry of Justice and Department for Education.

Table C3. Crimes during School Terms

	During School Term*[1] (1)	During Holidays*[1] (2)	Weekdays (3)	Weekends (4)
Panel A: Effects on becoming an offender by each term				
Treatment Effects at $t = 0$	.136***	.090***	.150***	.073***
	(.008)	(.007)	(.009)	(.006)
Pre-Treatment Mean	.997	.665	1.14	.501
Relative Effect Size	13.6%	13.5%	13.1%	14.5%
Panel B: Effects on the number of crimes committed in each term				
Treatment Effects at $t = 0$	.050***	.016**	.046***	.020***
	(.010)	(.008)	(.011)	(.006)
Pre-Treatment Mean	.191	.121	.226	.086
Relative Effect Size	26.3%	13.1%	20.2%	23.6%
Sample Size (Panel Count)	15,458,144	15,458,144	15,458,144	15,458,144
Number of Individuals	1,050,567	1,050,567	1,050,567	1,050,567

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. School term dates vary slightly across local authorities in England. This table uses the school term dates for Brighton & Hove. Data sources: Ministry of Justice and Department for Education.

Table C4. Effects of Students' Transferring Out on Stayers

Students Transferring Out:	Offender (1)	Not-yet-offender (2)	Non-offender (3)
Panel A: Effects on Crime Onset			
Treatment Effects at $t = 0$	.137***	.118***	.105***
	(.010)	(.012)	(.027)
Pre-Treatment Mean	1.21	.763	.404
Relative Effect Size	11.3%	15.5%	25.9%
Panel B: Effects on Number of Crimes			
Treatment Effects at $t = 0$	.014	.095***	.014
	(.021)	(.027)	(.040)
Pre-Treatment Mean	.328	.222	.155
Relative Effect Size	4.32%	42.9%	9.04%
Sample Size (Panel Count)	12,072,353	15,788,824	14,858,403
Number of Individuals	819,109	1,070,911	1,006,868

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Data sources: Ministry of Justice and Department for Education.

Table C5. Transfer with a Residence Move

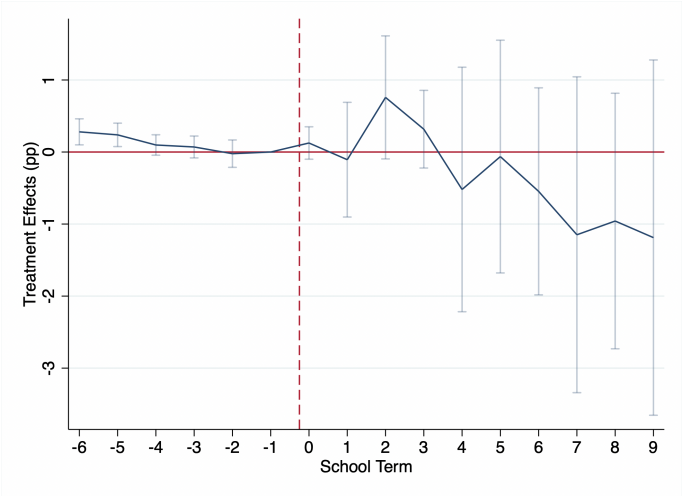
	No Residence Move	Residence Move				
		Any	0-25 miles	25-50 miles	50-100 miles	More than 100 miles
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment Effects at $t = 0$	.182***	.261***	.277***	.346***	.086	.276***
	(.012)	(.021)	(.028)	(.089)	(.055)	(.064)
Pre-Treatment Mean	1.50	1.53	1.53	1.53	1.63	1.58
Relative Effect Size	12.1%	17.0%	18.1%	22.6%	5.26%	17.4%
Sample Size (Panel Count)	10,837,428	4,939,848	3,130,345	731,958	568,499	597,536
Number of Individuals	736,373	335,947	212,850	49,722	38,635	40,797

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. Control variables include FSM and SEN status before secondary school, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, a linear month-of-birth term, and a linear age term. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Residential moves are identified based on changes in LSOA recorded in the NPD. Distance is measured between the population-weighted centroids of LSOAs, as provided by the Office for National Statistics. LSOA stands for Lower Layer Super Output Area, a geographic unit containing approximately 1,000 to 3,000 residents. Data sources: Ministry of Justice and Department for Education.

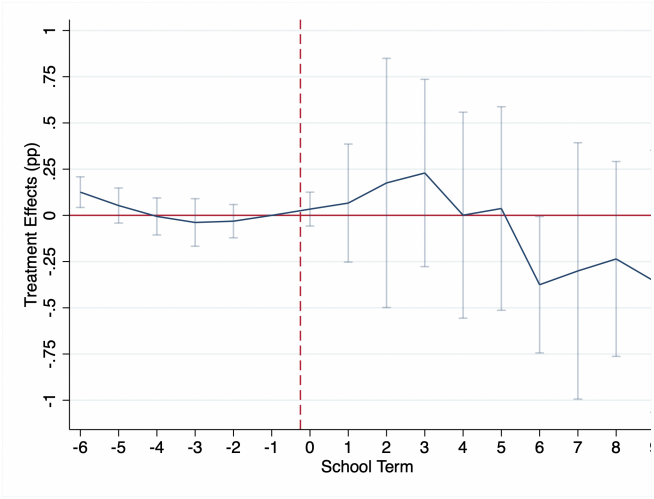
Table C6. Different Specifications

Effects on:	Crime Onset (1)	Any Crime (2)	Number of Crimes (3)
Poisson Regression	.130***	.021***	.030**
	(.010)	(.007)	(.013)
Logit Regression	14.55***	13.28***	-
	(1.38)	(2.41)	
Sample Size	15,262,170	15,262,170	15,262,170

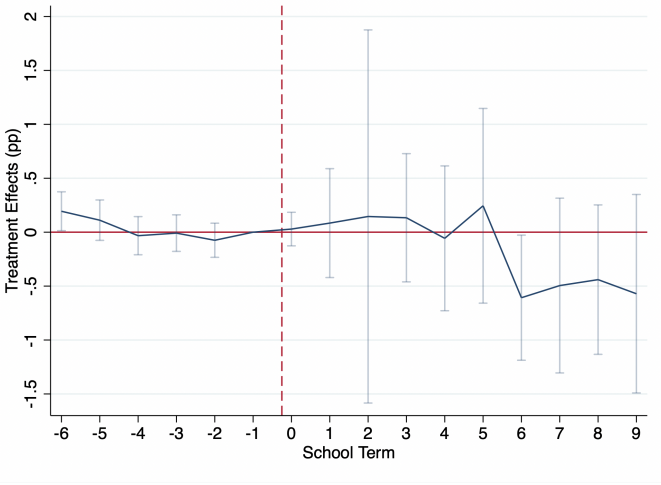
Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Wooldridge (2023, 2025), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Data sources: Ministry of Justice and Department for Education.



(a) Effects on Offending Initiation



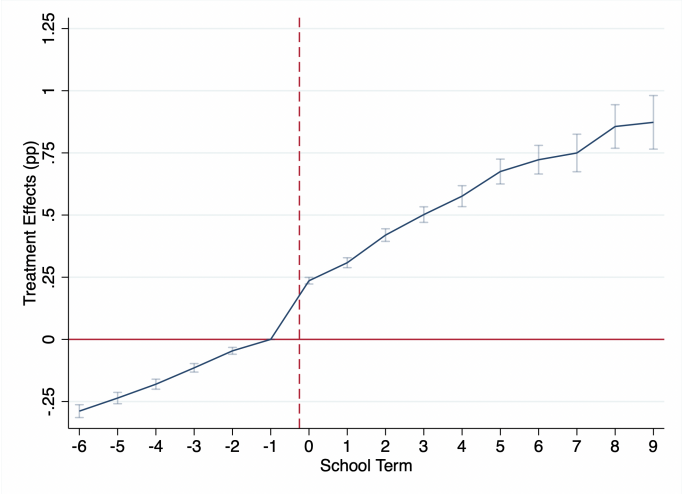
(b) Effects on Any Offending



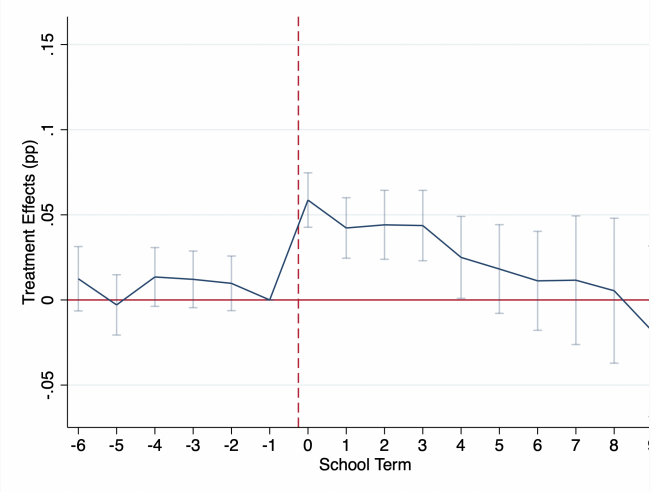
(c) Effects on Number of Crimes

Figure C1. Placebo Test: Receiving a Not-Yet Offender's Transfer

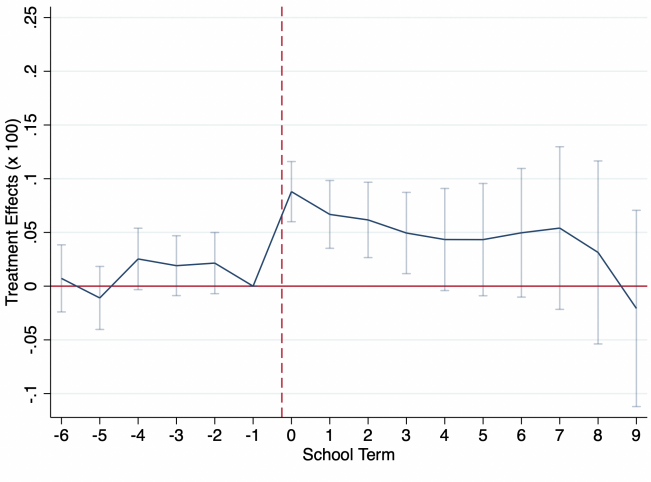
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021). I use the weight to adjust for differences in exposure timing across grades and school census seasons between offenders' and not-yet-offenders' transfers.



(a) Effects on Offending Initiation



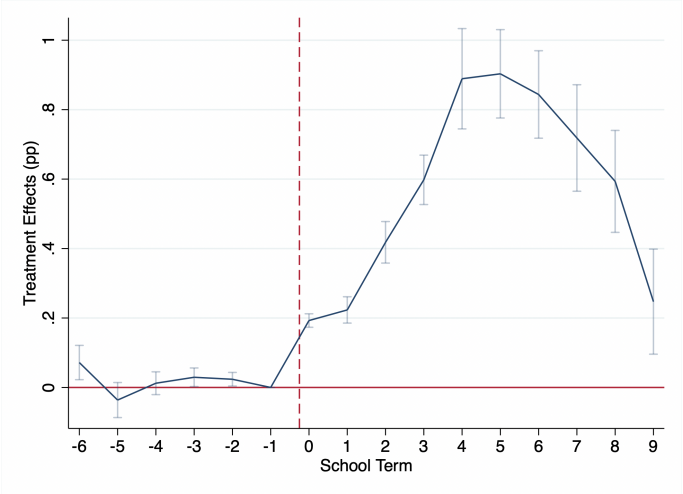
(b) Effects on Any Offending



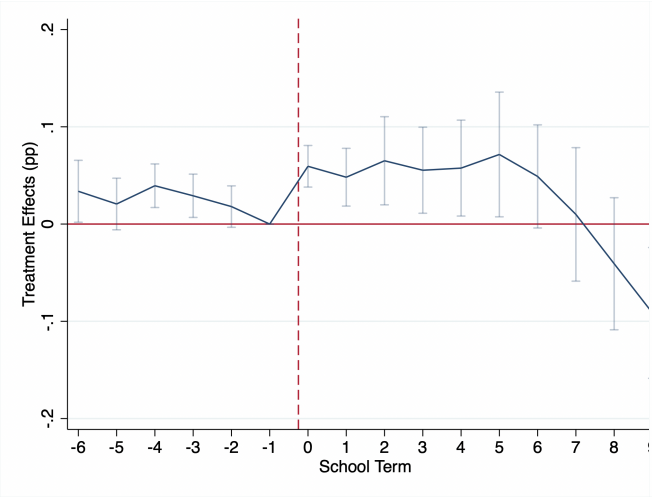
(c) Effects on Number of Crimes

Figure C2. Treatment Effects: Censored at the First Transfer

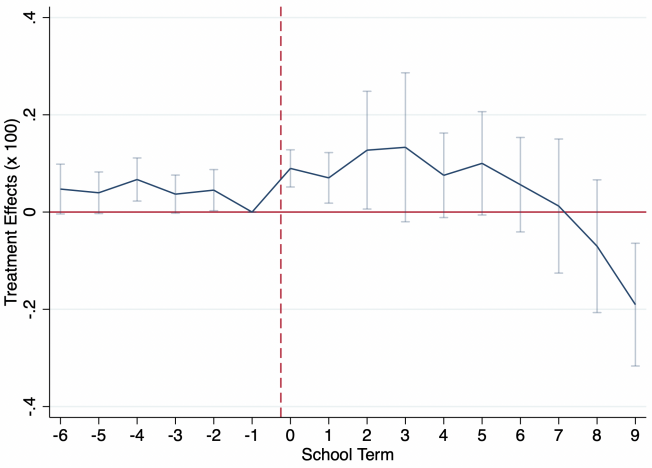
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021); Rambachan and Roth (2023).



(a) Effects on Offending Initiation



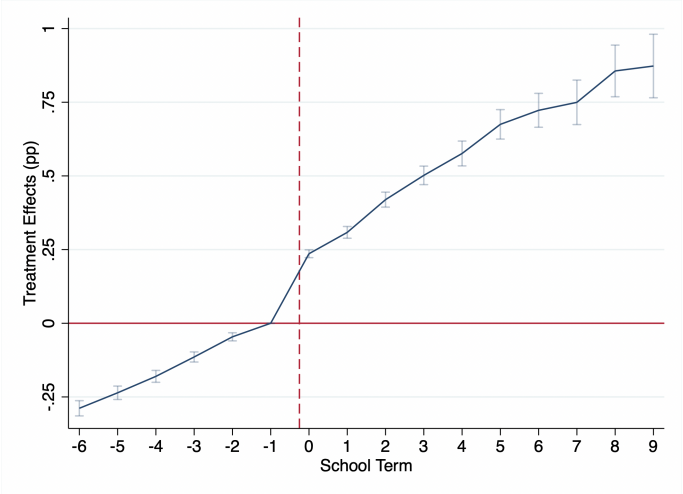
(b) Effects on Any Offending



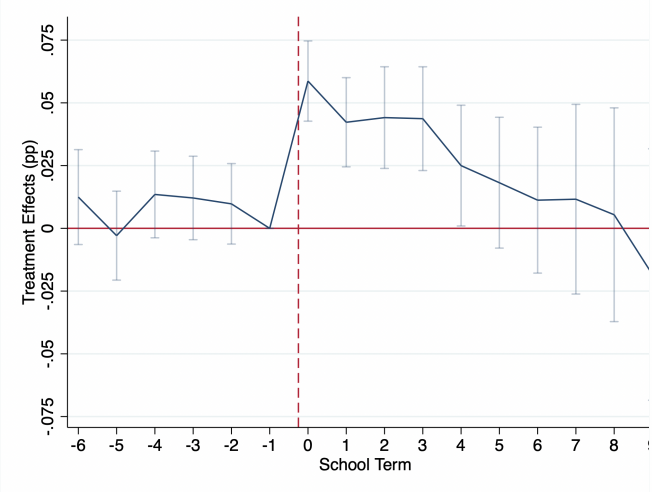
(c) Effects on Number of Crimes

Figure C3. Censored at the Second Treatment

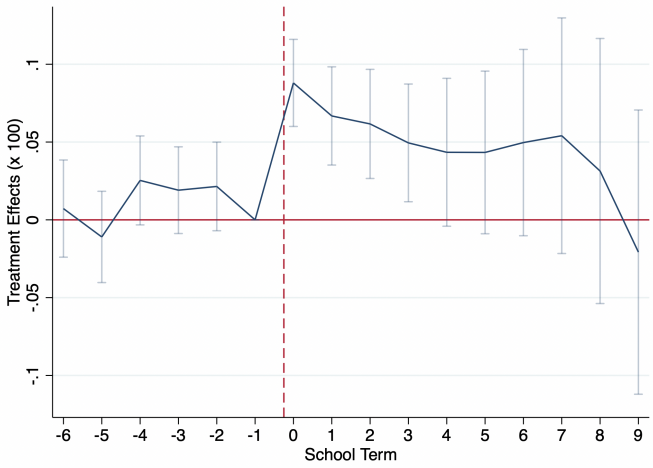
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021); Rambachan and Roth (2023).



(a) Effects on Offending Initiation



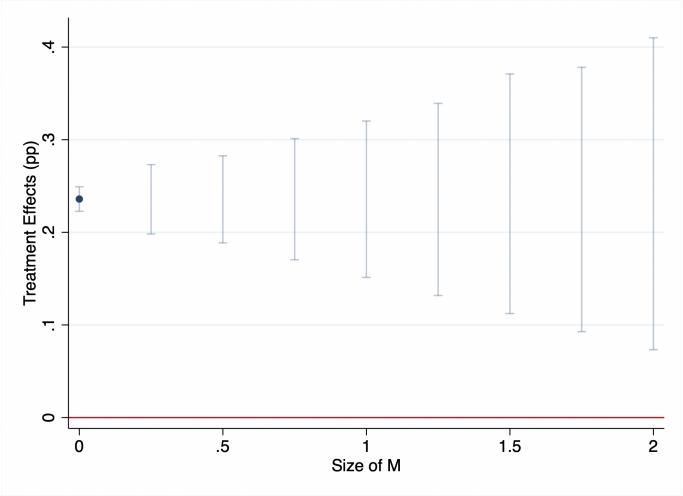
(b) Effects on Any Offending



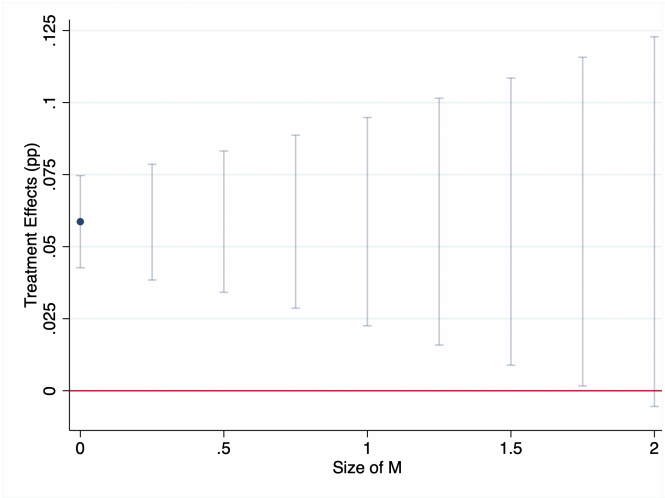
(c) Effects on Number of Crimes

Figure C4. Main Results without Control Variables

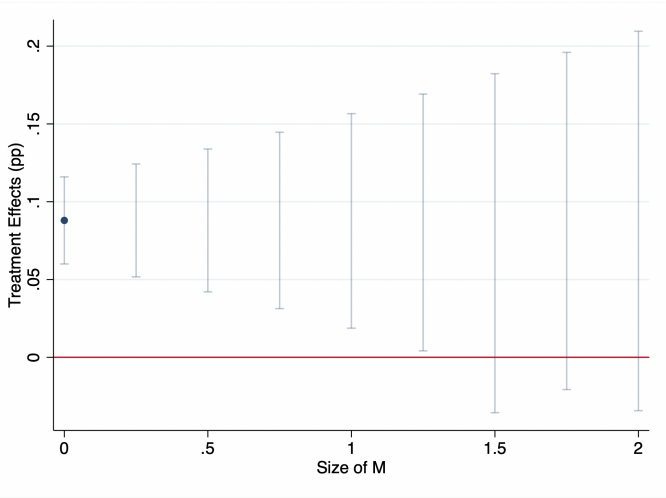
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).



(a) Offending Initiation



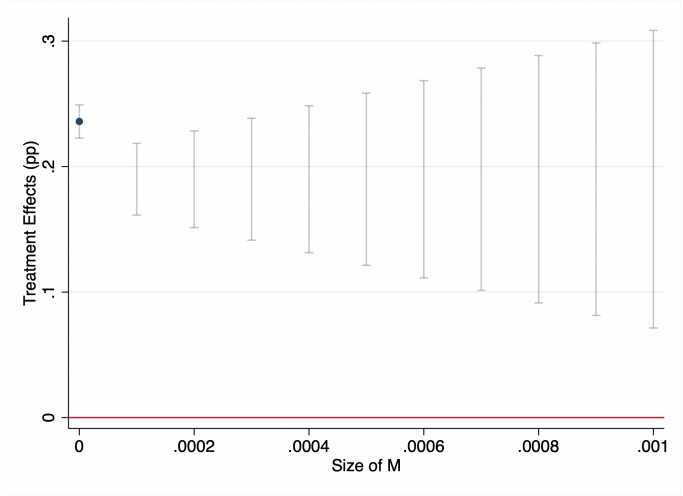
(b) Any Crime at Each Period



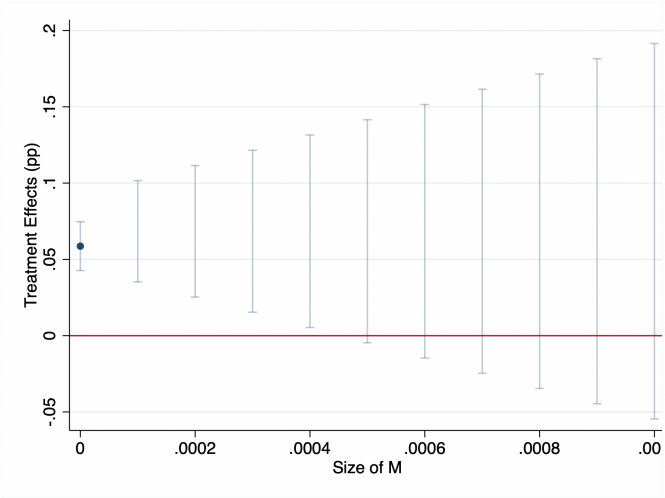
(c) N of Crimes at Each Period

Figure C5. Honest Difference-in-Differences without Control Variables, Relative Magnitude Restrictions

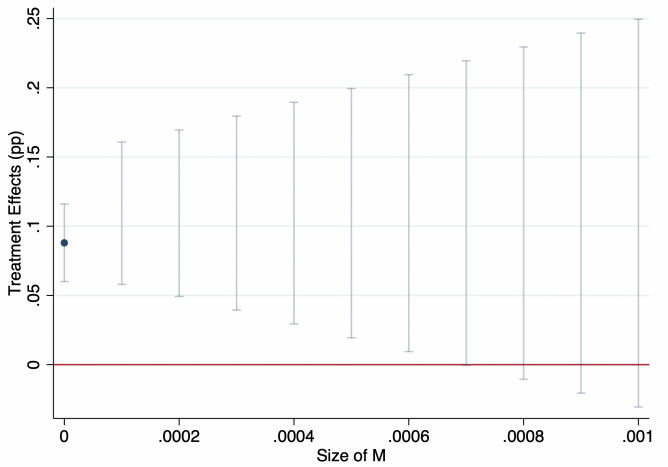
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021); Rambachan and Roth (2023).



(a) Offending Initiation



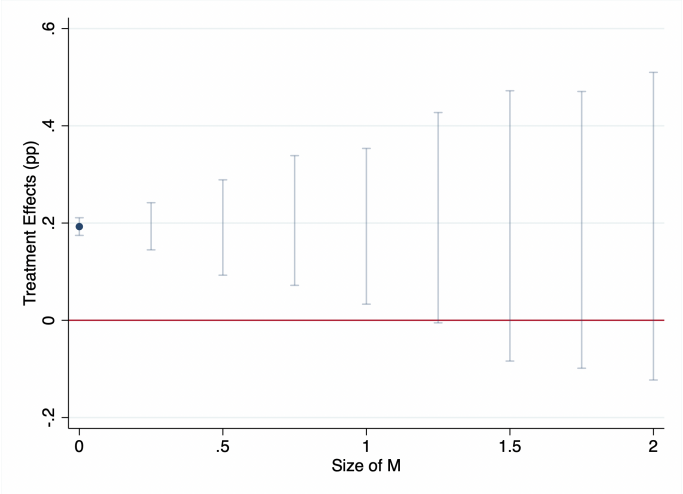
(b) Any Crime at Each Period



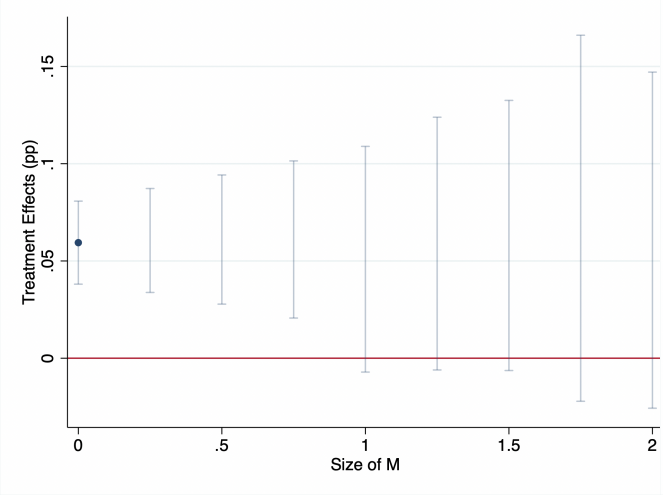
(c) N of Crimes at Each Period

Figure C6. Honest Difference-in-Differences without Control Variables, Second Difference Restrictions

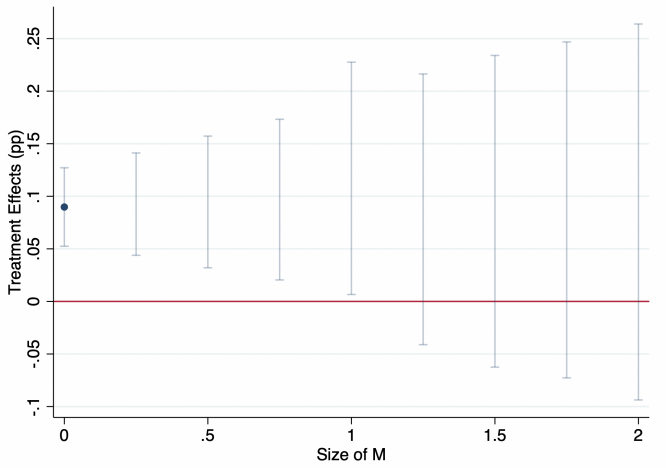
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021); Rambachan and Roth (2023).



(a) Offending Initiation



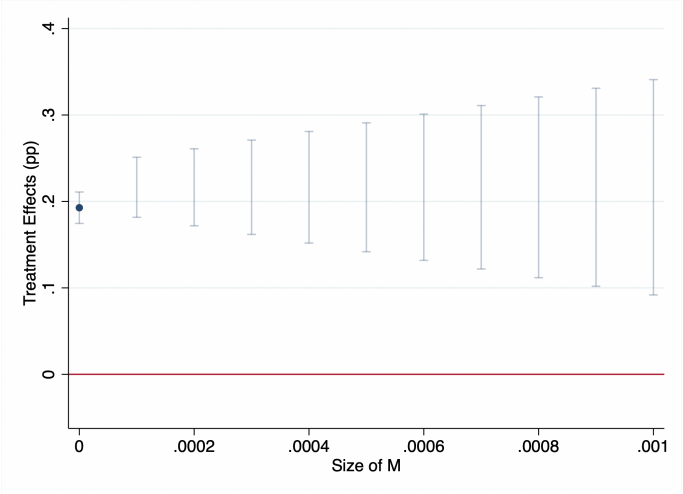
(b) Any Crime at Each Period



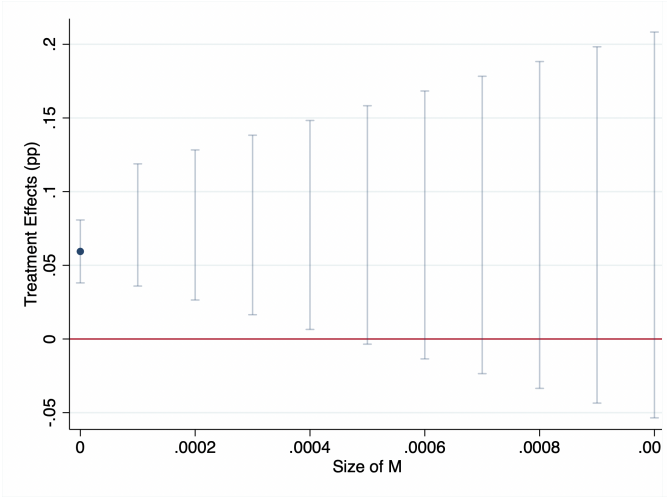
(c) N of Crimes at Each Period

Figure C7. Honest Difference-in-Differences, Relative Magnitude Restrictions

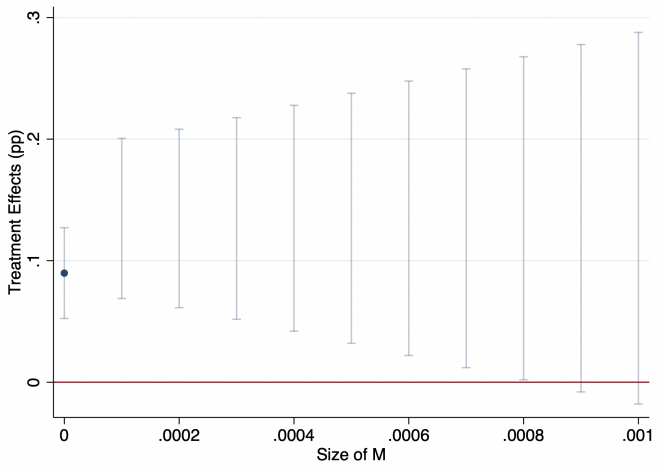
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).



(a) Offending Initiation



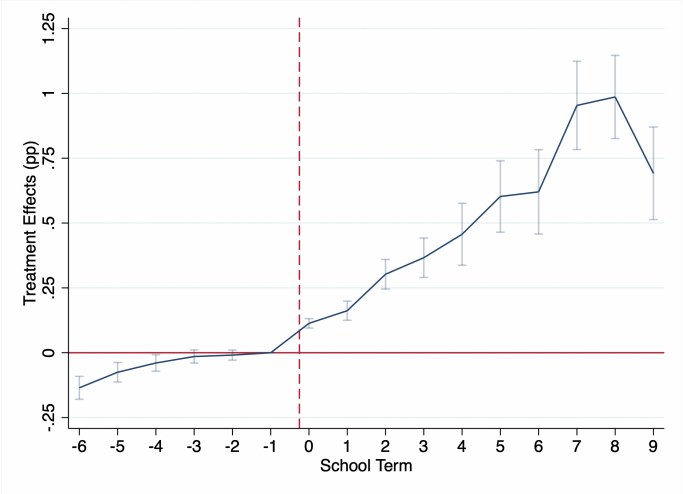
(b) Any Crime at Each Period



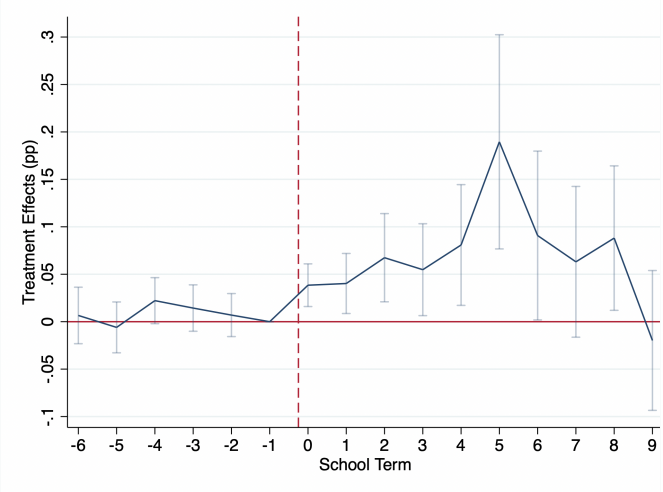
(c) N of Crimes at Each Period

Figure C8. Honest Difference-in-Differences, Second Difference Restrictions

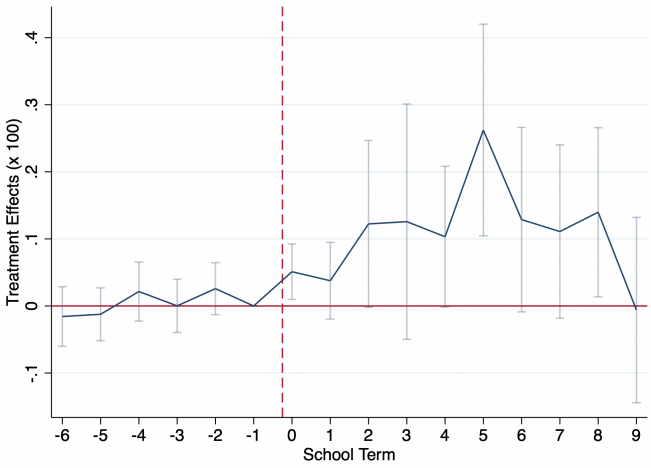
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).



(a) Offending Initiation



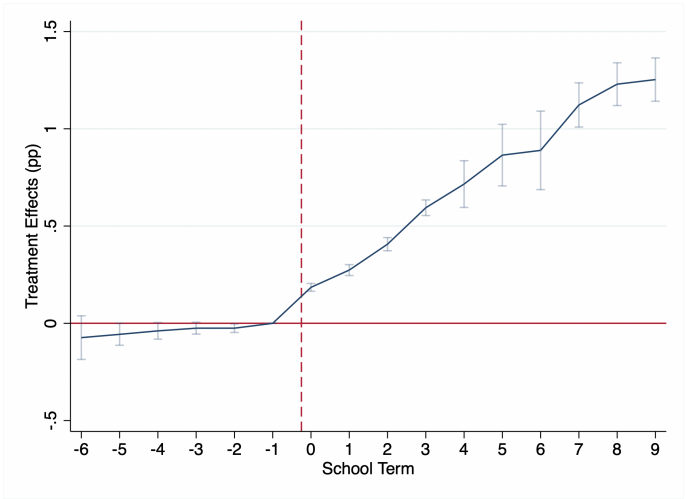
(b) Any Crime at Each Period



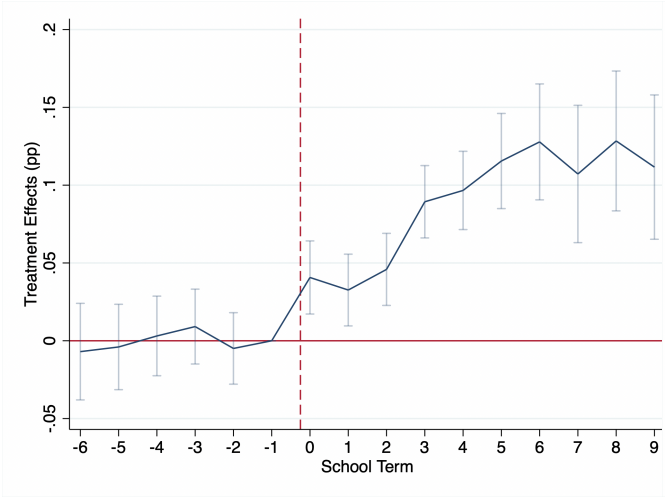
(c) N of Crimes at Each Period

Figure C9. Treatment Effects, Partial out with Demographics

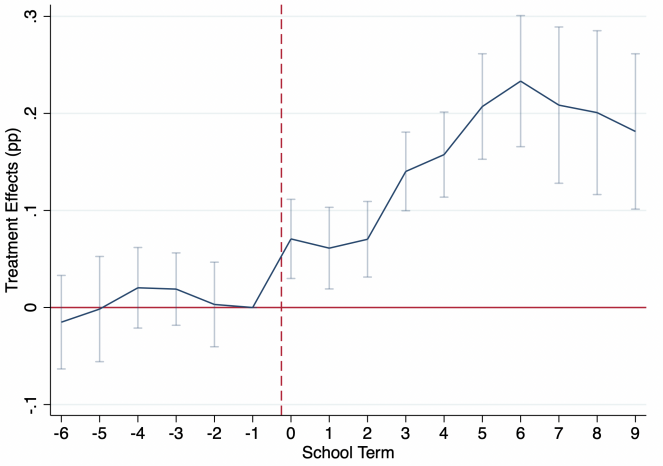
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).



(a) Offending Initiation



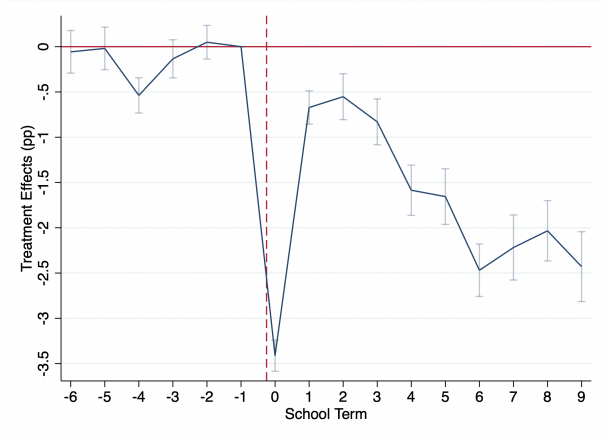
(b) Any Crime at Each Period



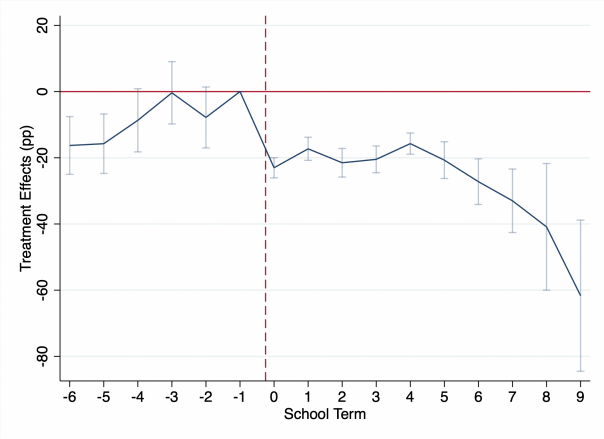
(c) N of Crimes at Each Period

Figure C10. Treatment Effects Including Never Treated Group

Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).



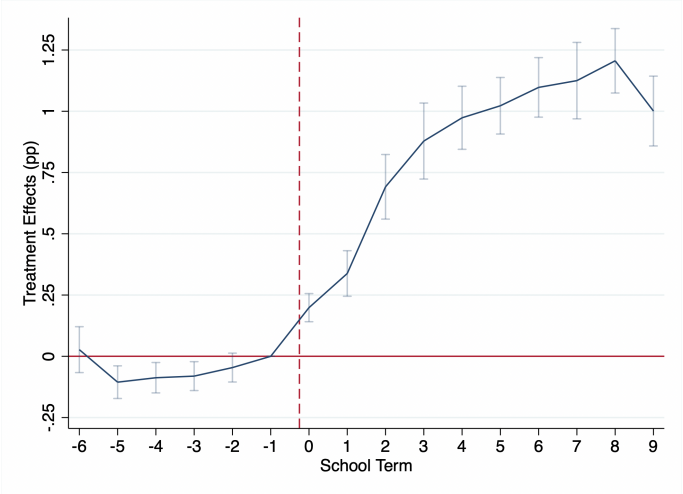
(a) Effects of Receiving an Offender Transfer



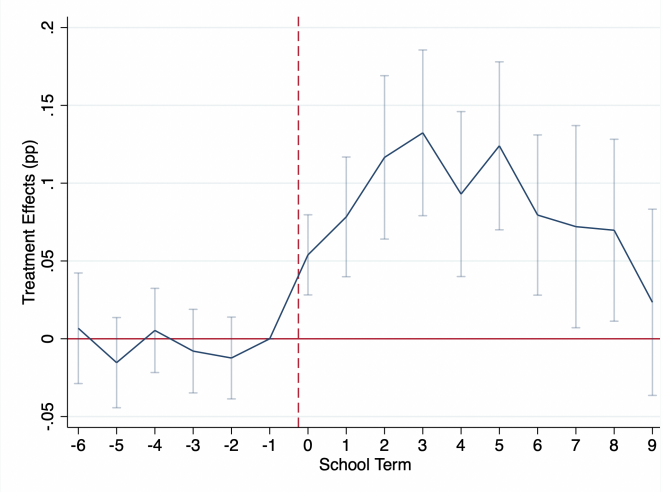
(b) Effects of Receiving a Non-offender Transfer

Figure C11. Effects on Transfer

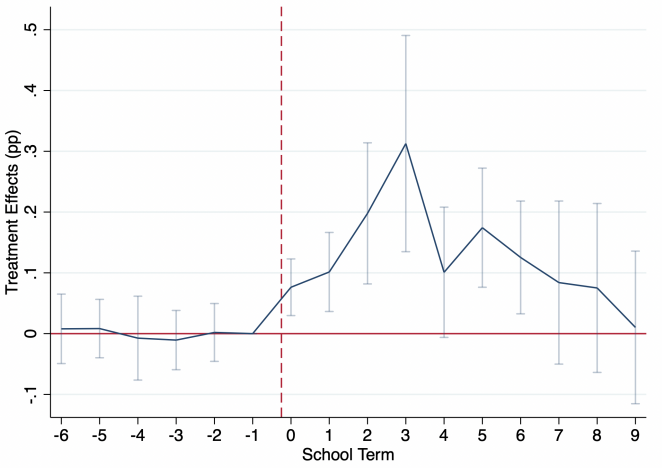
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).



(a) Offending Initiation



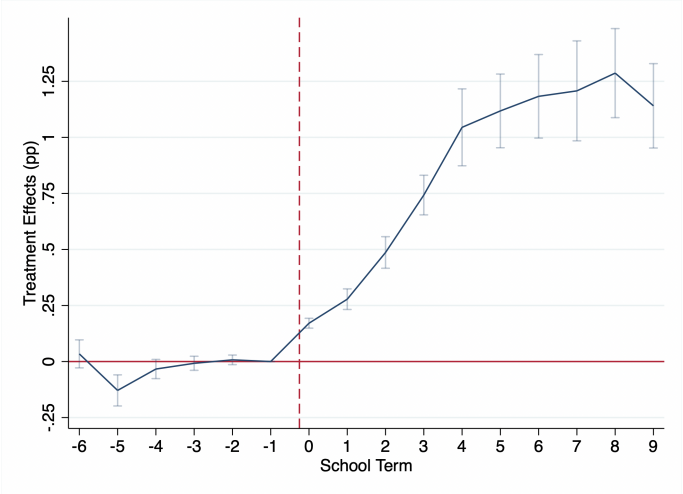
(b) Any Crime at Each Period



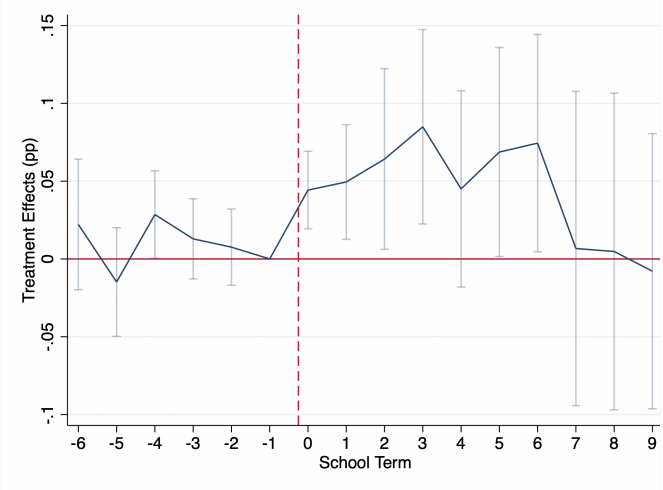
(c) N of Crimes at Each Period

Figure C12. Treatment Effects Focusing on Schools from which No Offender Left

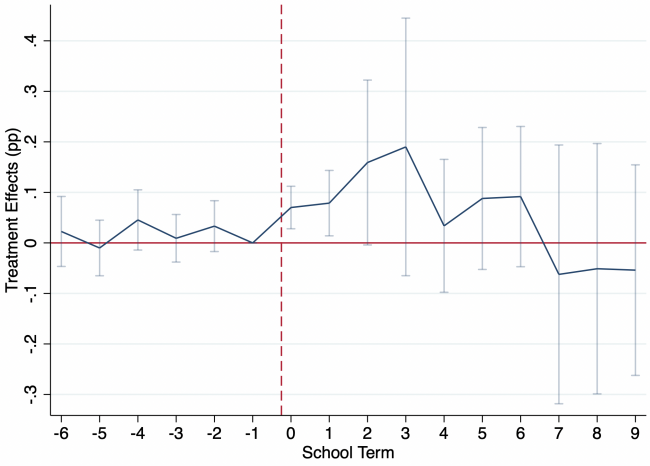
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).



(a) Offending Initiation



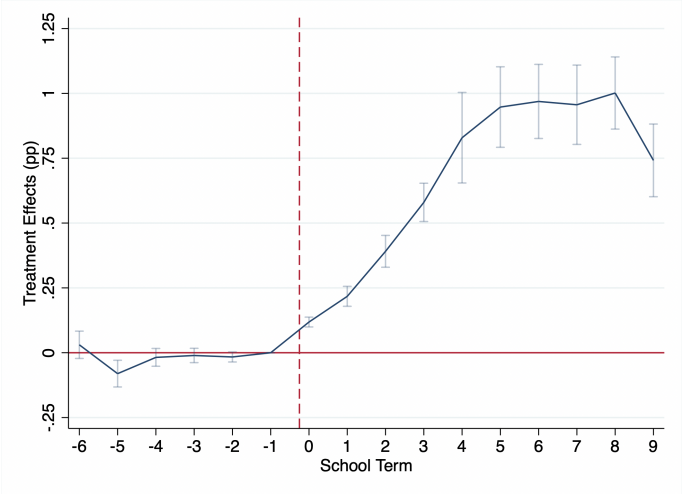
(b) Any Crime at Each Period



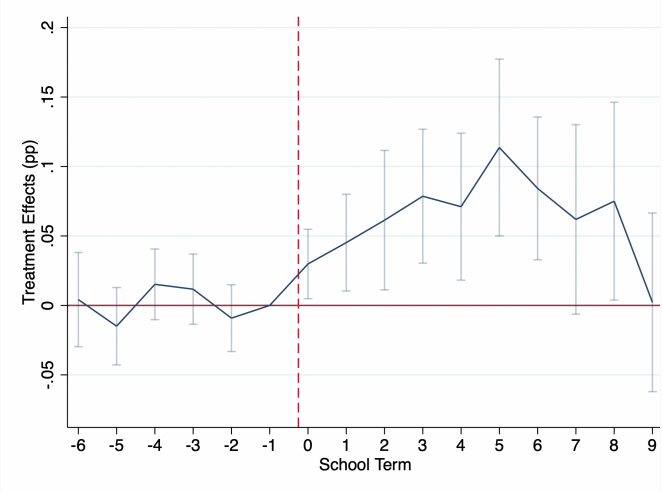
(c) N of Crimes at Each Period

Figure C13. Treatment Effects Excluding Those Who Move Back to the Previous School

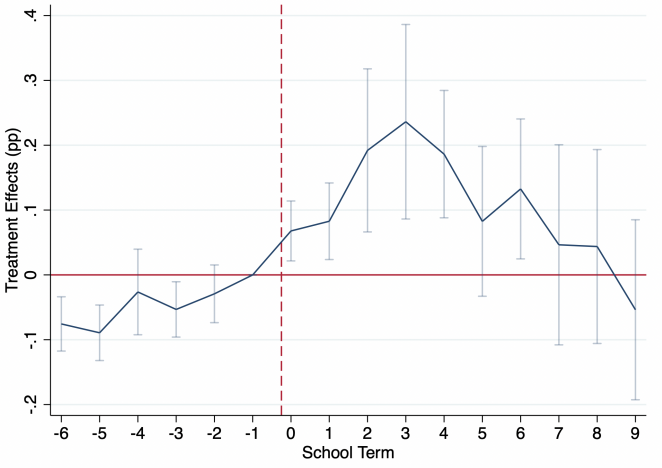
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).



(a) Offending Initiation



(b) Any Crime at Each Period



(c) N of Crimes at Each Period

Figure C14. Treatment Effects Excluding Those Who Have Been Exposed to a Mass Transfer

Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).

D Heterogeneity in Detail

Table D1. Treatment Intensity

Quartile in treatment intensity:	Lowest (1)	2nd Lowest (2)	3rd Lowest (3)	Highest (4)
Panel A: Mean of Intensity Measure				
(# Offenders Transfering)/(# Students)	.833	2.50	4.17	5.83
Panel B: Effects on Crime Onset				
Treatment Effects at $t = 0$	.092***	.093	.141***	.378***
	(.018)	(.154)	(.016)	(.029)
Pre-Treatment Mean	1.29	1.36	1.46	1.89
Relative Effect Size	7.09%	6.81%	9.66%	20.0%
Sample Size (Panel Count)	3,864,488	3,885,222	3,872,423	3,836,011
Number of Individuals	262,642	262,641	262,642	262,642

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. Control variables include FSM and SEN status before secondary school, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, a linear month-of-birth term, and a linear age term. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Treatment intensity is measured as the ratio of the number of offenders transferring into a school-grade to the number of students in that school-grade. Data sources: Ministry of Justice and Department for Education.

Table D2. Heterogeneity in Demographic Characteristics

Heterogeneity in Gender

Population	All (1)	Boys (2)	Girls (3)
Panel A: Effects of Any Offenders' Transfers			
Treatment Effects at $t = 0$	.193***	.249***	.126***
	(.009)	(.016)	(.011)
Pre-Treatment Mean	1.50	1.98	.979
Ratio of the Spontaneous Effects to the Pre-Treatment Mean	12.9%	12.6%	12.9%
Sample Size (Panel Count)	15,458,144	8,013,566	7,444,578
Number of Individuals	1,050,567	544,283	506,284
Panel B: Effects of Transfers of Offenders of the Same Gender			
Treatment Effects at $t = 0$	.217***	.261***	.136***
	(.014)	(.019)	(.022)
Pre-Treatment Mean	1.63	1.93	1.04
Ratio of the Spontaneous Effects to the Pre-Treatment Mean	13.3%	13.5%	13.1%
Sample Size (Panel Count)	8,448,209	5,534,952	2,913,257
Number of Individuals	573,876	375,792	198,084
Panel C: Effects of Transfers of Offenders of the Opposite Gender			
Treatment Effects at $t = 0$	.138**	-.136	.119***
	(.058)	(.108)	(.014)
Pre-Treatment Mean	1.35	2.10	.941
Ratio of the Spontaneous Effects to the Pre-Treatment Mean	10.2%	–6.50%	12.7%
Sample Size (Panel Count)	7,009,935	2,478,614	4,531,321
Number of Individuals	476,691	168,491	308,200

Heterogeneity in First Language

Population	Native English Speakers (1)	Non-native English Speakers (2)
Panel A: Effects of Transfers of Offenders with the Same First Language Status		
Treatment Effects at $t = 0$	.503***	.478***
	(.013)	(.108)
Pre-Treatment Mean	1.34	.540
Ratio of the Spontaneous Effects to the Pre-Treatment Mean	37.5%	88.6%
Sample Size (Panel Count)	14,601,647	416,685
Number of Individuals	834,068	21,763
Panel B: Effects of Transfers of Offenders with a Different First Language Status		
Treatment Effects at $t = 0$	-.304	.007
	(.226)	(.028)
Pre-Treatment Mean	2.01	.883
Ratio of the Spontaneous Effects to the Pre-Treatment Mean	–15.1%	.804%
Sample Size (Panel Count)	948,034	1,315,063
Number of Individuals	44,262	76,774

Heterogeneity in Ethnicity

	White British (1)	Non White British (2)
Panel A: Effects of Transfers of Offenders of the Same Ethnicity		
Treatment Effects at $t = 0$	-.290**	.048
	(.129)	(.137)
Pre-Treatment Mean	1.54	1.11
Ratio of the Spontaneous Effects to the Pre-Treatment Mean	–18.8%	4.35%
Sample Size (Panel Count)	1,856,895	1,346,979
Number of Individuals	126,035	91,035
Panel B: Effects of Transfers of Offenders of a different Ethnicity		
Treatment Effects at $t = 0$	.255***	.176***
	(.037)	(.013)
Pre-Treatment Mean	1.27	1.59
Ratio of the Spontaneous Effects to the Pre-Treatment Mean	20.1%	11.1%
Sample Size (Panel Count)	1,574,950	10,895,631
Number of Individuals	107,049	741,223

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. I exclude observations that were exposed to another type of offender prior to treatment. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Data sources: Ministry of Justice and Department for Education.

Table D3. Offender's Experience

Quartile in offender density:	One Crime (1)	Two to Five Crimes (2)	More than Five Crimes (3)
Panel A: Effects on Crime Onset			
Treatment Effects at $t = 0$	.198***	.275***	.539***
	(.012)	(.024)	(.165)
Pre-Treatment Mean	1.46	1.65	2.10
Relative Effect Size	13.5%	16.7%	25.7%
Panel B: Effects on Number of Crimes			
Treatment Effects at $t = 0$	.085***	.154***	.474*
	(.023)	(.041)	(.259)
Pre-Treatment Mean	.365	.385	.552
Relative Effect Size	23.4%	39.9%	86.0%
Sample Size (Panel Count)	11,380,413	4,191,421	341,441
Number of Individuals	773,254	285,159	23,257

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Data sources: Ministry of Justice and Department for Education.

Table D4. Treatment Age Heterogeneity

Treatment Age:	Age 11 (1)	Age 12 (2)	Age 13 (3)	Age 14 (4)
Treatment Effects at $t = 0$	.084***	.061***	.065***	.172***
	(.018)	(.022)	(.025)	(.064)
Pre-Treatment Mean	.834	1.31	1.69	1.99
Relative Effect Size	10.1%	4.67%	3.83%	8.63%
Sample Size (Panel Count)	3,127,728	3,081,488	3,051,872	3,085,214
Number of Individuals	91,301	276,133	341,758	262,657

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. Control variables include FSM and SEN status before secondary school, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, a linear month-of-birth term, and a linear age term. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade at the specified age, while the control group consists of students who will be treated later at the same age. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. Data sources: Ministry of Justice and Department for Education.

Table D5. Heterogeneity in Early Years Educational Achievement

Educational Achievement:	High	Low
Treatment Effects at $t = 0$	(1)	(2)
Overall Test Score	.115**	.129*
	(.049)	(.072)
Pre-Treatment Mean	.675	1.40
Relative Effect Size	17.1%	9.23%
Personal, social and emotional development	.054	.191***
	(.068)	(.069)
Pre-Treatment Mean	.703	1.38
Relative Effect Size	7.66%	13.9%
Communication, language and literacy	.117**	.138*
	(.048)	(.074)
Pre-Treatment Mean	.692	1.39
Relative Effect Size	16.9%	9.93%
Mathematical development	.088	.155**
	(.055)	(.072)
Pre-Treatment Mean	.764	1.32
Relative Effect Size	11.5%	11.7%
Knowledge and understanding of the world	.168***	.079
	(.058)	(.063)
Pre-Treatment Mean	.889	1.18
Relative Effect Size	18.9%	6.72%
Physical development	.112*	.151**
	(.066)	(.061)
Pre-Treatment Mean	.914	1.16
Relative Effect Size	12.2%	13.0%
Creative development	.102*	.158**
	(.059)	(.069)
Pre-Treatment Mean	.863	1.21
Relative Effect Size	11.8%	13.1%
Sample Size (Panel Count)	273,240	272,122
Number of Individuals	18,525	18,523

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. Control variables include FSM and SEN status before secondary school, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, a linear month-of-birth term, and a linear age term. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. All test scores are standardized within each cohort. Early-years test scores are available only for a quasi-random 10% of the sample comprising children born on the 5th, 15th, or 25th of each month (Department for Education, 2010). Data sources: Ministry of Justice and Department for Education.

Table D6. Heterogeneity in Individual Characteristics

	With FSM and SEN (1)	With FSM Only (2)	With SEN Only (3)	Neither FSM nor SEN (4)
Treatment Effects at $t = 0$	.570***	.233***	.116***	.072***
	(.041)	(.029)	(.019)	(.008)
Pre-Treatment Mean	4.17	2.04	1.37	.578
Relative Effect Size	13.7%	11.4%	8.47%	12.4%
Sample Size (Panel Count)	2,357,011	2,199,327	3,277,133	7,624,673
Number of Individuals	161,381	150,051	222,544	516,591

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. Control variables include FSM and SEN status before secondary school, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, a linear month-of-birth term, and a linear age term. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. The variable for SEN is a dummy variable that takes the value one if there is any explicit record on SEN provision before the secondary school, and zero if otherwise. The variable for FSM is also a dummy variable that takes the value one if there is any record as eligible for FSM before the secondary school and zero if otherwise. Data sources: Ministry of Justice and Department for Education.

Table D7. School Types

School Types:	Academy Converter (1)	Academy Sponsor Led (2)	Community School (3)	Foundation School (4)	Voluntary Aided School (5)	Voluntary Controlled School (6)
Panel A: Effects on Crime Onset						
Treatment Effects at $t = 0$	-.027	.104	.139***	-.023	.649*	.467**
	(.035)	(.066)	(.014)	(.301)	(.386)	(.229)
Pre-Treatment Mean	1.12	2.14	1.50	1.61	1.15	1.37
Relative Effect Size	–2.39%	4.86%	9.21%	–1.40%	56.5%	34.1%
Panel B: Effects on Number of Crimes						
Treatment Effects at $t = 0$	-.025	-.122	.049*	-.136	.782***	.247*
	(.088)	(.095)	(.029)	(.303)	(.230)	(.147)
Pre-Treatment Mean	.334	.521	.353	.373	.341	.314
Relative Effect Size	–7.40%	–23.4%	13.9%	–36.3%	229%	78.4%
Sample Size (Panel Count)	2,000,104	989,257	5,710,743	3,747,099	1,272,572	285,724
Number of Individuals	134,819	68,313	386,644	252,803	85,623	19,495

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. School type is measured at the time of treatment. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Data sources: Ministry of Justice and Department for Education.

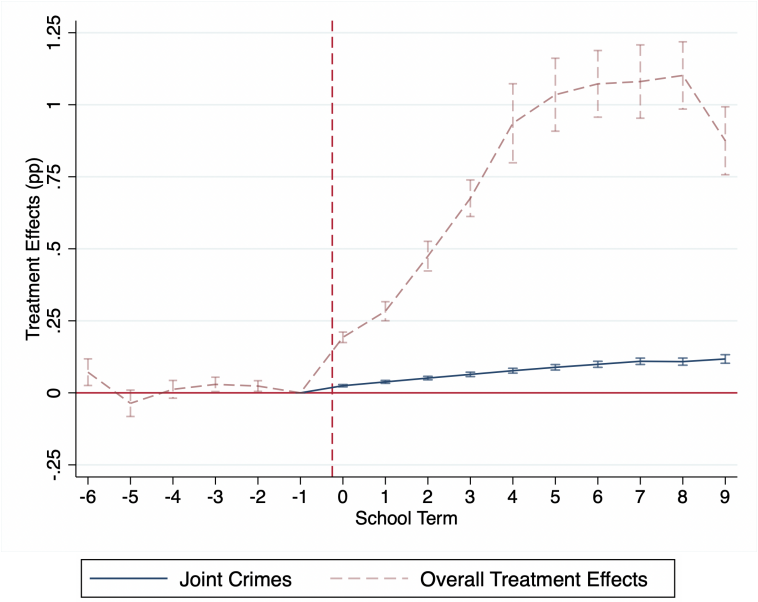
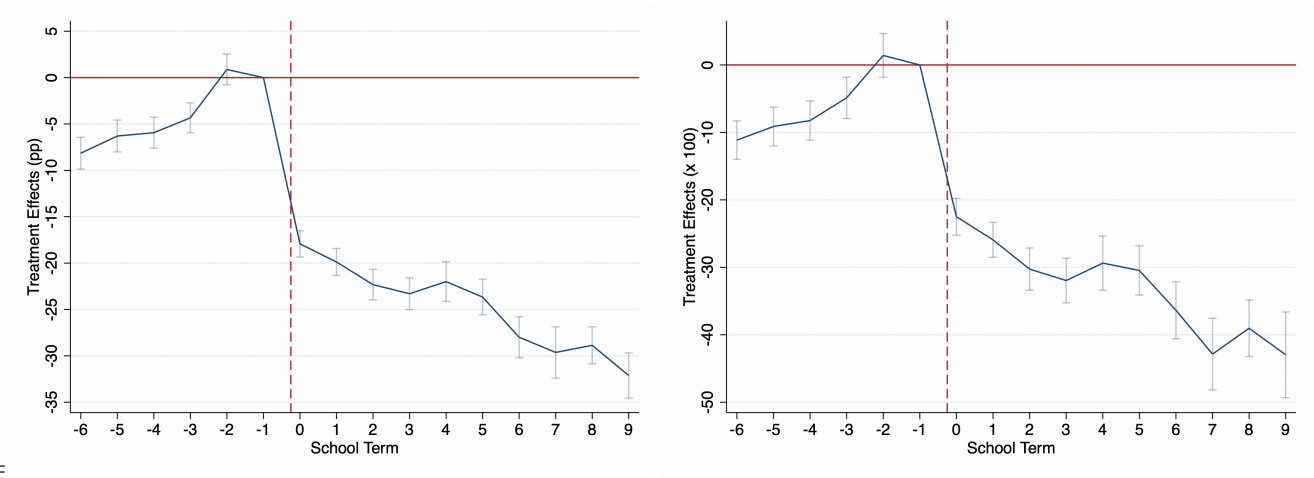


Figure D1. Any Offending with Transfer Offenders

Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).



(a) Any Crime at Each Period

(b) N of Crimes at Each Period

Figure D2. Treatment Effects of Moving on Movers: Offenders at the Time of Transfer

Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).

E Simulation Details

In this section, I describe the matching problem mathematically. The setting is a special case of the generalized theory discussed in Bogomolnaia and Moulin (2001).

Suppose that there is a finite set of transfer students $s \in S$ and a finite set of school grades $g \in G$.[82] Denote $N = |S|$. I assume a null school grade $\emptyset \in G$. In this exercise, I consider swapping students transfer *after* they are initially assigned to a school, so the outside option by assigned to $\emptyset$ represents being unmatched with an alternative destination school and ending up with the original destination school.

Each student s has a strict preference $\succ_s$ over G.[83] For school grade $g, g' \in G$, $g \succ_s g'$ means that student s strictly prefers g to s'. Assume $\succ_s$ satisfies completeness and transitivity. For any $s \in S$ and $g, g' \in G$, denote $g \succsim_s g'$ if $g \succ_s g'$ or $g = g'$. I assume that school grades do not have any preference.[84] Denote $\succ = (\succ_1, \succ_2, \cdots, \succ_N)$ as a preference profile.

Define the probability that student $s \in S$ is assigned to $g \in G$ as P_{sg}, where $P_{sg} \in [0, 1]$ and $\sum_{g \in G} P_{sg} = 1$ for all $s \in S$, and a row vector of these probabilities for s as $P_s = (P_{sg})_{g \in G}$. P is a matrix that has P_s in each row, as $P = (P_s)_{s \in S}$.

I consider a random priority mechanism (RP mechanism, also known as a random serial dictatorship) as an assignment mechanism. The procedure is as follows:

Step 1. Assign a random number to each student $s \in S$.

Step 2. From the top student on the priority list based on their random numbers, assign the most preferred school grade remaining in the pool to the student.

Step 3. Repeat Step 2 until all students are assigned to a school grade (including $\emptyset$).

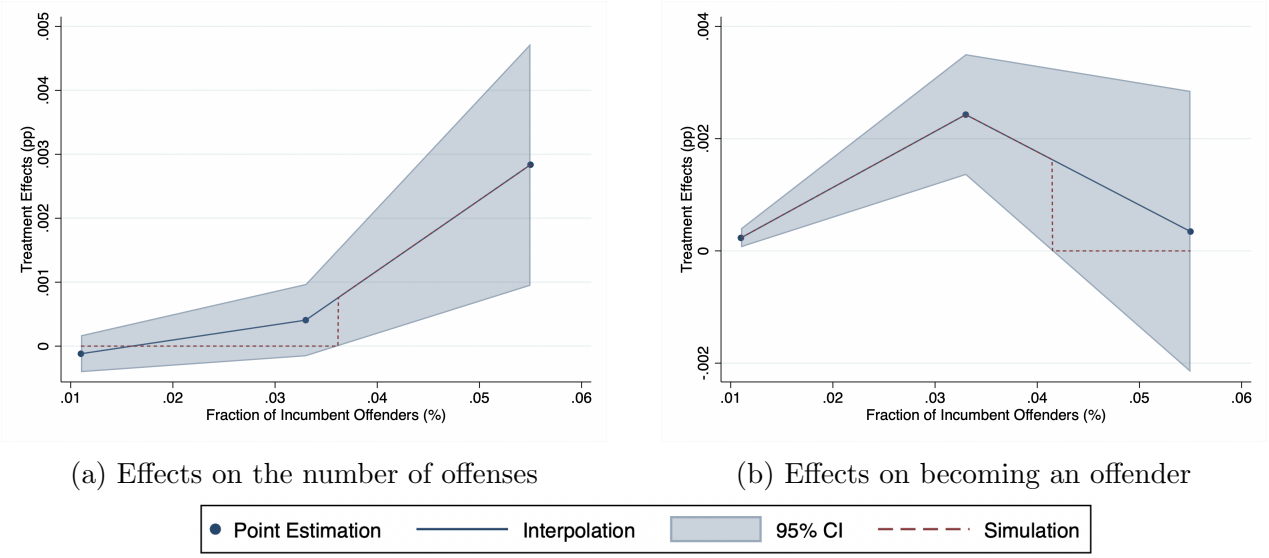
This RP mechanism has several desirable properties. First, it is strategy-proof. Each student chooses the best possible choice at each round, and the random number assignment is random, so there is no incentive to manipulate. It also satisfies a notion of fairness, equal treatment of equals, which is characterized as assigning equal probability to those who have the same preference, i.e,. for $s, s' \in S$ and for any $g \in G$, $P_{sg} = P_{s'g}$. Further, it is ex-post efficient, meaning that once the allocation has been made, there is no way to benefit any student without sacrificing others by swapping schools.

[82] I can discuss the theory at the school level by introducing a grade-level capacity constraint as a type-specific quota as in Abdulkadiroğlu and Sönmez (2003). But for simplicity of notation, I consider this at a school-grade level.

[83] In the empirical exercise, I ensure preferences are strict by ordering their preferred school by (1) distance from the original destination, (2) the size of the peer effects, and (3) random number.

[84] This is equivalent to assuming that all school grades $g \in G$ have a weak preference $\succsim_g$, but they are indifferent across all students. I.e., for any $s, s' \in S$, $s \sim_g s'$.

This allocation does not satisfy ordinary efficiency, meaning that if agents have Von Neumann-Morgenstern preferences, there is a Pareto-efficient allocation. However, there is no allocation mechanism that have all the above desirable properties in addition to the Pareto efficiency (Bogomolnaia and Moulin, 2001). There is another allocation mechanism that satisfies ex-ante efficiency and envy-free, which means that the probability assignment to each school is always the most preferred compared to others. However, it does not satisfy strategy-proofness and hence there is a room for manipulation. Therefore I use this RP mechanism.

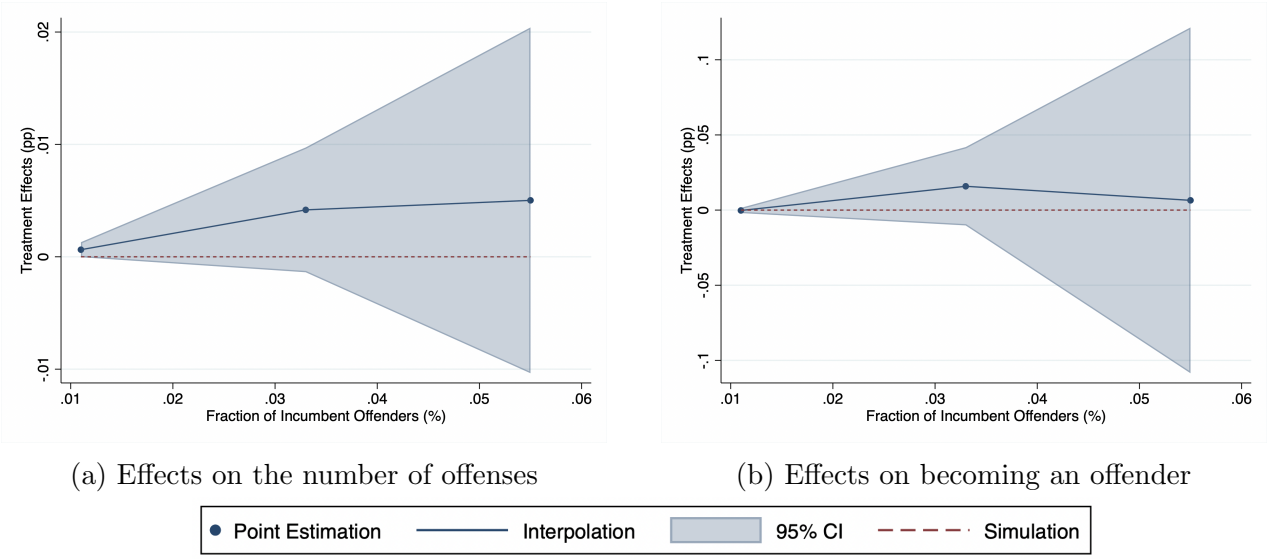


(a) Effects on the number of offenses

(b) Effects on becoming an offender

Figure E1. Interpolation of the Treatment Effects

Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades, corresponding to Table 8. The estimation is based on Callaway and Sant'Anna (2021). I interpolate the treatment effects linearly.



(a) Effects on the number of offenses

(b) Effects on becoming an offender

Figure E2. Placebo: Interpolation of the Treatment Effects

Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades, corresponding to Table 8. The estimation is based on Callaway and Sant'Anna (2021). I interpolate the treatment effects linearly.

F Other Results

Table F1. Local Deprivation: Other Crime Types

Quartile in Deprivation Index:	Least Deprived (1)	2nd Least Deprived (2)	3rd Least Deprived (3)	Most Deprived (4)
Battery	.024***	.040***	.068***	.068***
	(.008)	(.008)	(.013)	(.010)
Pre-Treatment Mean	.172	.233	.357	.460
Relative Effect Size	14.2%	17.3%	19.1%	14.9%
Drug	.013***	.014***	.023***	.019***
	(.005)	(.005)	(.006)	(.005)
Pre-Treatment Mean	.024	.032	.043	.040
Relative Effect Size	53.4%	42.0%	53.1%	48.8%
Violence	.006	.018***	.033***	.062***
	(.006)	(.006)	(.010)	(.009)
Pre-Treatment Mean	.083	.128	.215	.316
Relative Effect Size	7.56%	14.0%	15.2%	19.7%
Weapon	.006*	.004	.011**	.009*
	(.003)	(.002)	(.005)	(.005)
Pre-Treatment Mean	.025	.029	.048	.077
Relative Effect Size	22.6%	13.5%	22.7%	12.1%
Arson/Damage	.030***	.025***	.062***	.085***
	(.008)	(.009)	(.013)	(.010)
Pre-Treatment Mean	.185	.295	.412	.604
Relative Effect Size	16.2%	8.42%	15.1%	14.0%
Sexual	-.000	.006**	.011***	.007**
	(.002)	(.003)	(.004)	(.004)
Pre-Treatment Mean	.032	.023	.040	.053
Relative Effect Size	-.783%	27.6%	28.3%	13.5%
Motoring	.000	.003	.006*	.010***
	(.003)	(.003)	(.004)	(.004)
Pre-Treatment Mean	.007	.010	.016	.035
Relative Effect Size	4.45%	30.1%	40.9%	29.9%
Others	.004	.008**	.017***	.012**
	(.003)	(.004)	(.004)	(.006)
Pre-Treatment Mean	.023	.034	.042	.073
Relative Effect Size	19.0%	23.3%	41.1%	16.6%
Sample Size (Panel Count)	3,299,643	3,503,179	3,970,134	4,685,188
Number of Individuals	223,594	237,727	270,064	319,182

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Deprivation is measured using the 2007 Income Deprivation Affecting Children Index (IDACI), defined as the proportion of children aged 0 to 15 living in income-deprived families at the LSOA level. LSOA stands for Lower Layer Super Output Area, a geographic unit containing approximately 1,000 to 3,000 residents. Data sources: Ministry of Justice and Department for Education.

Table F2. Local Deprivation: Measure in 2004

Quartile in Deprivation Index:	Least Deprived (1)	2nd Least Deprived (2)	3rd Least Deprived (3)	Most Deprived (4)
All Crimes	.078***	.112***	.202***	.317***
	(.017)	(.016)	(.023)	(.021)
Pre-Treatment Mean	.801	1.09	1.60	2.21
Relative Effect Size	9.80%	10.3%	12.6%	14.3%
Theft	.023**	.037***	.086***	.135***
	(.010)	(.010)	(.016)	(.014)
Pre-Treatment Mean	.342	.453	.655	.931
Relative Effect Size	6.60%	8.18%	13.1%	14.5%
Battery	.019**	.038***	.070***	.073***
	(.008)	(.008)	(.013)	(.010)
Pre-Treatment Mean	.176	.232	.355	.461
Relative Effect Size	10.9%	16.4%	19.8%	15.7%
Drug	.013***	.010**	.024***	.020***
	(.004)	(.005)	(.007)	(.006)
Pre-Treatment Mean	.023	.031	.044	.040
Relative Effect Size	57.6%	32.8%	55.4%	49.6%
Violence	.012**	.016***	.033***	.059***
	(.006)	(.006)	(.010)	(.009)
Pre-Treatment Mean	.083	.126	.213	.320
Relative Effect Size	14.6%	12.3%	15.7%	18.6%
Fraud	-.001	.002	.004	.009**
	(.002)	(.003)	(.006)	(.004)
Pre-Treatment Mean	.012	.024	.033	.046
Relative Effect Size	–4.87%	8.01%	11.3%	20.1%
Weapon	.007**	.003	.009*	.011**
	(.003)	(.003)	(.006)	(.005)
Pre-Treatment Mean	.025	.028	.050	.077
Relative Effect Size	28.7%	12.3%	18.3%	14.1%
Arson/Damage	.025***	.030***	.062***	.083***
	(.009)	(.008)	(.013)	(.010)
Pre-Treatment Mean	.184	.299	.408	.605
Relative Effect Size	13.4%	10.2%	15.2%	13.7%
Sexual	-.001	.007**	.011***	.007**
	(.002)	(.003)	(.004)	(.003)
Pre-Treatment Mean	.032	.023	.041	.052
Relative Effect Size	–2.72%	29.9%	26.6%	13.4%
Motoring	.001	.003	.007*	.010**
	(.003)	(.002)	(.004)	(.004)
Pre-Treatment Mean	.007	.009	.016	.034
Relative Effect Size	8.58%	34.9%	40.5%	27.8%
Others	.005	.009**	.016***	.011*
	(.003)	(.004)	(.005)	(.006)
Pre-Treatment Mean	.023	.032	.044	.072
Relative Effect Size	20.9%	27.0%	35.9%	15.8%
Sample Size (Panel Count)	3,269,879	3,558,504	3,971,545	4,658,216
Number of Individuals	221,566	241,552	270,143	317,306

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Deprivation is measured using the 2004 Income Deprivation Affecting Children Index (IDACI), defined as the proportion of children aged 0 to 15 living in income-deprived families at the LSOA level. LSOA stands for Lower Layer Super Output Area, a geographic unit containing approximately 1,000 to 3,000 residents. Data sources: Ministry of Justice and Department for Education.

Table F3. Treatment Effects on Each Crime Type

Effects on:	Violence with Injury (1)	Violence without Injury (2)	Rape (3)	Other Sexual Offence (4)	Robbery (5)	Domestic Burglary (6)	Theft of Vehicle (7)	Theft from Vehicle (8)	Theft from Person (9)	Criminal Damage (Arson) (10)	Criminal Damage (Other) (11)	Fraud (12)	Commercial Theft (13)
Time −6	.009	.005	-.000	-.006**	.004	.007**	.003**	.003*	-.004	.002	.009	.001	.001*
	(.009)	(.005)	(-.000)	(-.006)	(.004)	(.007)	(.003)	(.003)	(-.004)	(.002)	(.009)	(.001)	(.001)
Time −5	.005	.005	.000	-.003*	.001	.005*	.003***	.001	.007	.000	-.003	.002	.001*
	(.005)	(.005)	(.000)	(-.003)	(.001)	(.005)	(.003)	(.001)	(.007)	(.000)	(-.003)	(.002)	(.001)
Time −4	.002	.011***	-.000	-.002	-.000	.003	.002*	.000	.012*	-.001	.000	.003	.001*
	(.002)	(.011)	(-.000)	(-.002)	(-.000)	(.003)	(.002)	(.000)	(.012)	(-.001)	(.000)	(.003)	(.001)
Time −3	.008	.009***	.000	-.004***	-.003	.001	-.000	-.001	.006	-.002**	.005	.001	.001
	(.008)	(.009)	(.000)	(-.004)	(-.003)	(.001)	(-.000)	(-.001)	(.006)	(-.002)	(.005)	(.001)	(.001)
Time −2	-.010*	.009***	.000	.001	-.002	.002	.001	-.001	.010	.000	.006	.003	.000
	(-.010)	(.009)	(.000)	(.001)	(-.002)	(.002)	(.001)	(-.001)	(.010)	(.000)	(.006)	(.003)	(.000)
Time −1	0	0	0	0	0	0	0	0	0	0	0	0	0
Time 0	.014**	.010***	.000	.000	-.004**	.003	-.001	-.003	.016**	-.000	.010**	-.002	.001*
	(.014)	(.010)	(.000)	(.000)	(-.004)	(.003)	(-.001)	(-.003)	(.016)	(-.000)	(.010)	(-.002)	(.001)
Time 1	.019**	.011**	-.000	-.003	-.000	.004	.000	-.004	.012	-.002	.003	.001	.001**
	(.019)	(.011)	(-.000)	(-.003)	(-.000)	(.004)	(.000)	(-.004)	(.012)	(-.002)	(.003)	(.001)	(.001)
Time 2	.011	.020*	.000	.001	.002	.014*	.005**	-.003	.021*	-.002*	-.005	-.000	-.001
	(.011)	(.020)	(.000)	(.001)	(.002)	(.014)	(.005)	(-.003)	(.021)	(-.002)	(-.005)	(-.000)	(-.001)
Time 3	.027*	.013***	.000	-.002	-.008	.012***	.004	-.006***	.025	-.001	.032	-.003	-.000
	(.027)	(.013)	(.000)	(-.002)	(-.008)	(.012)	(.004)	(-.006)	(.025)	(-.001)	(.032)	(-.003)	(-.000)
Time 4	.010	.019*	.002*	.003	-.007	.005	.001	-.006*	.008	-.003	.007	.001	-.000
	(.010)	(.019)	(.002)	(.003)	(-.007)	(.005)	(.001)	(-.006)	(.008)	(-.003)	(.007)	(.001)	(-.000)
Time 5	-.009	.009	.002***	.015	.011	.001	.000	-.004	.037**	-.001	.008	.011	-.000**
	(-.009)	(.009)	(.002)	(.015)	(.011)	(.001)	(.000)	(-.004)	(.037)	(-.001)	(.008)	(.011)	(-.000)
Time 6	.001	.009	.001	.000	-.003	.009*	.003	-.001	.006	-.002	.032**	.001	.001
	(.001)	(.009)	(.001)	(.000)	(-.003)	(.009)	(.003)	(-.001)	(.006)	(-.002)	(.032)	(.001)	(.001)
Time 7	-.016	.013	.001*	.012*	.002	.014***	.009***	.000	-.004	.001	.010	-.006	.002
	(-.016)	(.013)	(.001)	(.012)	(.002)	(.014)	(.009)	(.000)	(-.004)	(.001)	(.010)	(-.006)	(.002)
Time 8	-.025	.010	.003*	.007	-.001	.007	.010*	-.002	-.008	.000	.017	.001	.001
	(-.025)	(.010)	(.003)	(.007)	(-.001)	(.007)	(.010)	(-.002)	(-.008)	(.000)	(.017)	(.001)	(.001)
Time 9	-.027*	.016**	.001	-.003	.005	.009	-.001	-.000	-.031*	-.006***	-.008	.000	.001
	(-.027)	(.016)	(.001)	(-.003)	(.005)	(.009)	(-.001)	(-.000)	(-.031)	(-.006)	(-.008)	(.000)	(.001)
Sample Size (Panel Count)	15,458,144	15,458,144	15,458,144	15,458,144	15,458,144	15,458,144	15,458,144	15,458,144	15,458,144	15,458,144	15,458,144	15,458,144	15,458,144
Number of Individuals	1,050,567	1,050,567	1,050,567	1,050,567	1,050,567	1,050,567	1,050,567	1,050,567	1,050,567	1,050,567	1,050,567	1,050,567	1,050,567

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. Control variables include FSM and SEN status before secondary school, an indicator for whether English is the first language, ethnicity fixed effects, cohort fixed effects, a linear month-of-birth term, and a linear age term. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Data sources: Ministry of Justice and Department for Education.

Table F4. Rural and Urban Areas

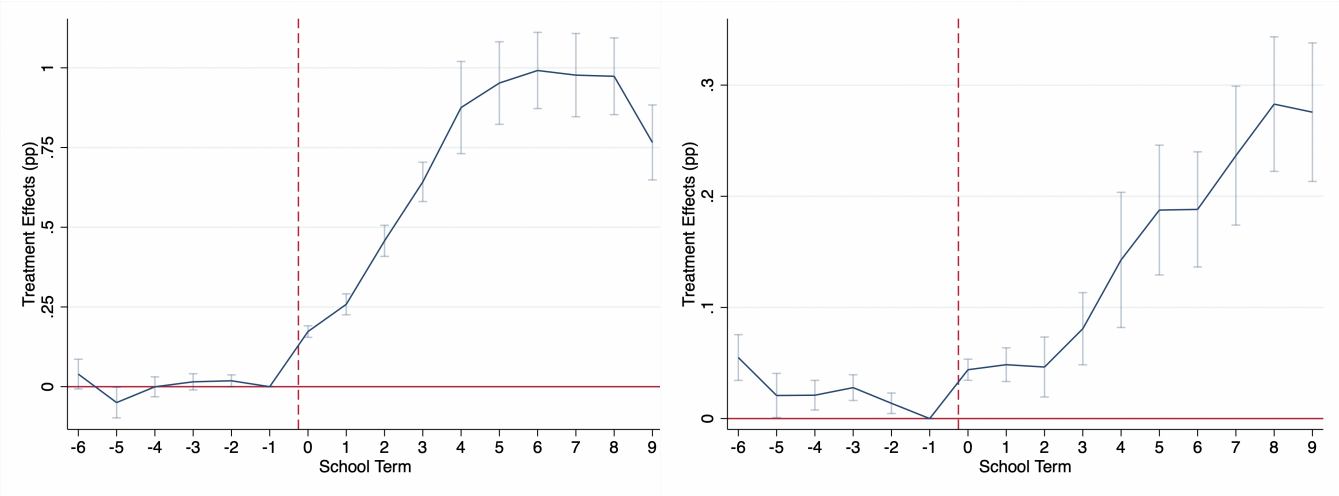
	Rural Areas			Urban Areas		
	Mainly Rural	Largely Rural	Urban with Significant Rural	Urban with City and Town	Urban with Minor Conurbation	Urban with Major Conurbation
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment Effects at $t = 0$	.147**	.176***	.227**	.215***	.267**	.200***
	(.064)	(.032)	(.105)	(.019)	(.118)	(.020)
Pre-Treatment Mean	1.39	1.51	1.54	1.60	1.78	1.38
Relative Effects Size	10.6%	11.7%	14.7%	13.5%	15.0%	14.5%
Sample Size (Panel Count)	1,450,874	1,991,579	2,000,476	4,549,275	732,882	4,733,058
Number of Individuals	99,039	136,555	136,240	309,107	49,785	319,841

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard errors reported in parentheses are clustered at the treatment level. This table reports the treatment effects of exposure to an offender's school transfer. The treatment group consists of students who are exposed to an offender transferring into their school-grade, while the control group consists of students who will be treated later. Never-treated students are excluded from the estimation. The definitions of urban and rural areas are based on the 2011 Local Authority Classification. The estimation follows Callaway and Sant'Anna (2021), and the table reports the contemporaneous effects at $t = 0$. The relative effect size is the ratio of the contemporaneous effect to the pre-treatment mean. All coefficients are multiplied by 100. Data sources: Ministry of Justice and Department for Education.

Table F5. School Characteristics: Offender-Dense Schools

				Differences between Lowest and Middle Terciles			Differences between Middle and Highest Terciles			Differences between Lowest and Highest Terciles		
	Lowest Dense	Middle Dense	Highest Dense	Uncond. Diff. [(1) = (2)]	Cond. Diff. [(1) = (2)]	Cond. Diff. [(1) = (2)]	Uncond. Diff. [(2) = (3)]	Cond. Diff. [(2) = (3)]	Cond. Diff. [(2) = (3)]	Uncond. Diff. [(1) = (3)]	Cond. Diff. [(1) = (3)]	Cond. Diff. [(1) = (3)]
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Male	.506	.523	.525	.017***	-.000***	-.000	.002	-.029	-.071	-.026***	-.005***	-.000***
				(.001)	(.000)	(.000)	(.001)	(.000)	(.000)	(.001)	(.000)	(.000)
Minority	.197	.201	.158	.005***	-.004***	.002**	-.043***	-.029***	.002	-.039***	-.033***	.001
				(.001)	(.001)	(.001)	(.001)	(.001)	(.001)	(.001)	(.001)	(.001)
English as the first language	.897	.892	.915	-.005***	-.000***	.000	.022***	-.000***	.000	.017***	-.000***	-.000
				(.001)	(.000)	(.000)	(.001)	(.000)	(.000)	(.001)	(.000)	(.000)
Ever with SEN by age 11	.330	.362	.402	.032***	-.000***	.000***	.040***	-.000***	-.000***	.072***	-.000***	-.000***
				(.001)	(.000)	(.000)	(.001)	(.000)	(.000)	(.001)	(.000)	(.000)
Ever eligible for FSM by age 11	.244	.292	.351	.048***	.000***	.000***	.059***	-.000***	-.000***	.107***	.000***	-.000***
				(.001)	(.000)	(.000)	(.001)	(.000)	(.000)	(.001)	(.000)	(.000)
Age 5 overall score	.043	-.029	-.118	-.071***	-.026***	-.005	-.090***	-.026**	-.018	-.161***	-.043***	.009
	[.956]	[.966]	[.985]	(.011)	(.010)	(.023)	(.015)	(.013)	(.032)	(.015)	(.013)	(.043)
Age 7 overall score	.061	-.018	-.121	-.078***	-.040***	-.004	-.104***	-.069***	-.012***	-.182***	-.108***	-.014***
	[.935]	[.957]	[.976]	(.002)	(.002)	(.003)	(.002)	(.002)	(.003)	(.002)	(.002)	(.004)
Age 11 overall score	.049	-.041	-.169	-.090***	-.061***	.001	-.128***	-.097***	-.012***	-.218***	-.152***	-.007
	[.957]	[.980]	[.993]	(.003)	(.002)	(.004)	(.003)	(.002)	(.004)	(.003)	(.003)	(.006)
Cummulative N of offenses | any	2.01	2.00	2.34	-.013	-.038	.011***	.342***	.241***	.022***	.329***	.217***	.026***
	[2.16]	[2.31]	[2.84]	(.063)	(.063)	(.001)	(.057)	(.058)	(.001)	(.059)	(.066)	(.002)
Sample size	344,858	344,858	344,858	689,716	689,716	689,716	689,716	689,716	689,716	689,716	689,716	689,716

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard deviations are reported in brackets, and heteroskedasticity-robust standard errors clustered at the secondary-school level are reported in parentheses. I use the earliest available secondary-school record. The sample in Columns (1)–(4) consists of observations in the NPD with the full set of control variables. The sample size corresponds to the sum of the 1995/96–1998/99 cohorts reported in Column (2) of Table A1 and Column (1) of Table A2. Conditional differences control for gender, minority status, an indicator for English as the first language, and SEN and FSM status at age 11, unless the variable is itself the outcome. All test scores are standardized within each cohort, except for the indicator variables for the age 16 tests. Early-years test scores are available only for a quasi-random 10% of the sample comprising children born on the 5th, 15th, or 25th of each month (Department for Education, 2010). The age 7 overall score is calculated as a weighted average of teachers' assessments of Reading, Writing, Speaking and Listening, Math, and Overall Science. Within English, Reading, Writing, and Speaking and Listening are weighted 1:1:2 so that the four underlying abilities contribute equally. English, Math, and Science are then weighted 1:1:1 in constructing the overall score. The age 11 overall score is calculated as a weighted average of English, Math, and Science, with equal weight assigned to each subject. Some observations are missing information for certain variables. The reported sample size is the number of observations with demographic information. Data sources: Ministry of Justice and Department for Education.

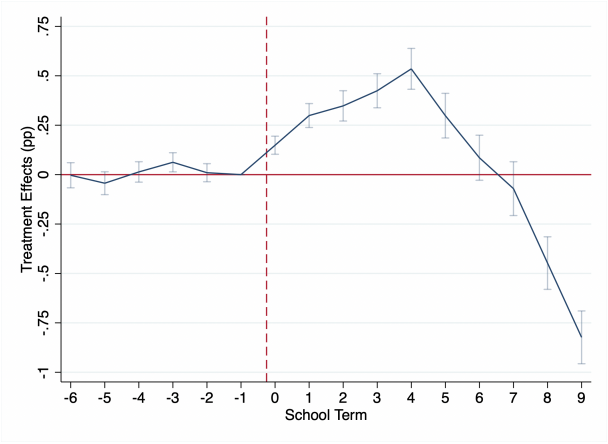


(a) Effects on Caution Experience

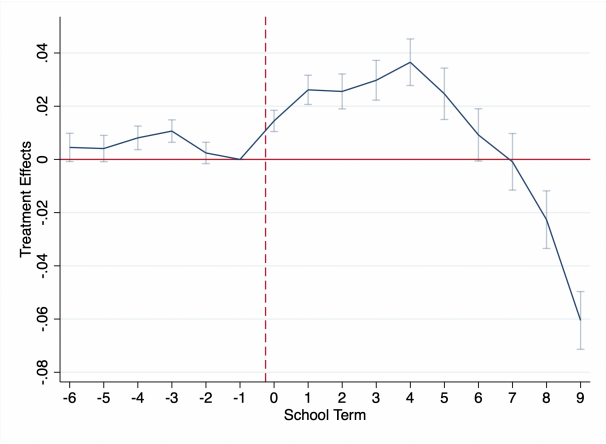
(b) Effects on Conviction Experience

Figure F1. Treatment Effects on Cautions and Convictions Experience

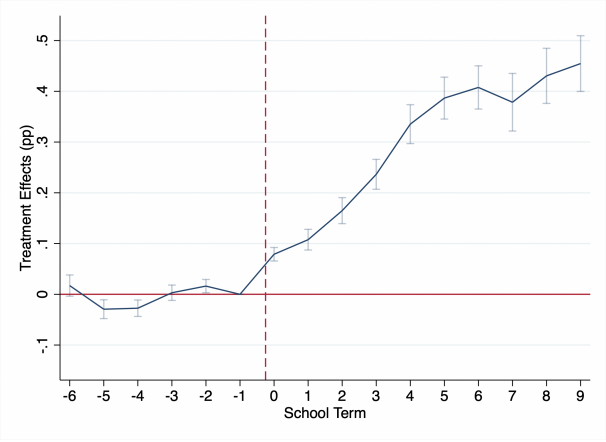
Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).



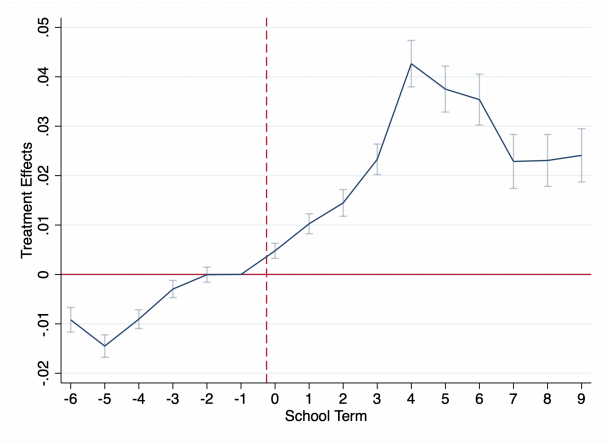
(a) Prob. of Any Fixed Exclusion



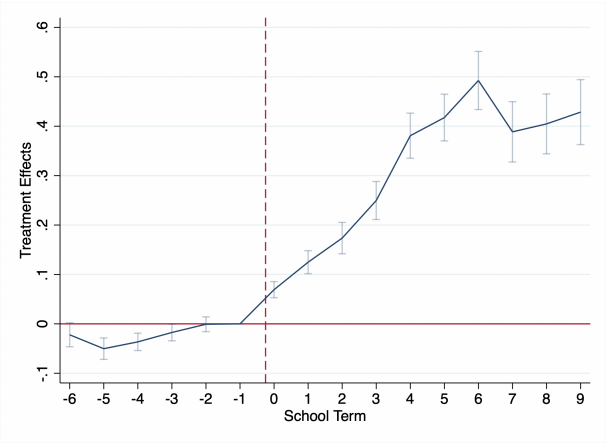
(b) N of Fixed Exclusion sessions



(c) Prob. of Unauthorized Absence



(d) Prob. of Any Fixed Exclusion or Unauthorized Absence



(e) N of Fixed Exclusion sessions or Unauthorized Absence

Figure F2. Effects on Problematic Behaviors

Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).

G Results for 1991–94 cohorts

Table G1. Sample Selection

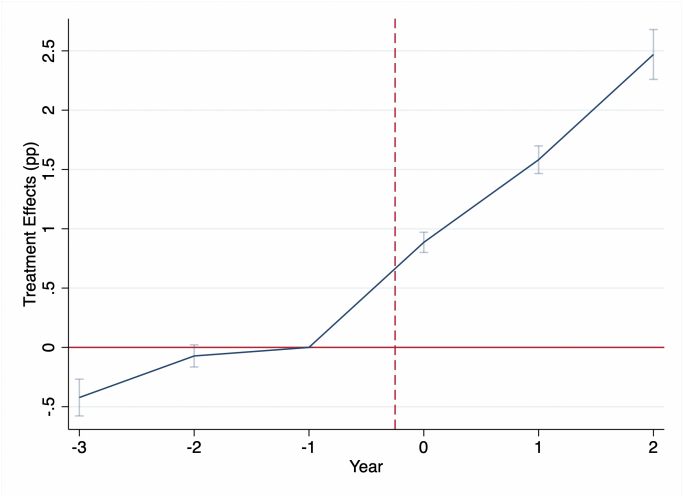
	Among NPD sample with controls Complete Records during Secondary School				Among Those in Column (1) Experienced a School-grade less than 20 Students				Among Those in Column (5) Those Treated at Age 13/14 or not			
	Complete	Not Complete	Uncond. Diff. [(1) = (2)]	Cond. Diff. [(1) = (2)]	Never Experienced	Experienced	Uncond. Diff. [(5) = (6)]	Cond. Diff. [(5) = (6)]	Not Treated at Age 13/14	Treated at Age 13/14	Uncond. Diff. [(9) = (10)]	Cond. Diff. [(9) = (10)]
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
A. Demographics												
Male	.498	.564	-.066***	-.023***	.498	.802	-.304***	-.175***	.491	.506	-.015***	-.004***
			(.001)	(.001)			(.009)	(.010)			(.001)	(.001)
Minority	.161	.312	-.151***	-.043***	.161	.143	.018**	.014**	.173	.146	.026***	.000
			(.001)	(.000)			(.008)	(.007)			(.001)	(.000)
English as the first language	.923	.799	.125***	.040***	.923	.952	-.028***	-.010**	.918	.930	-.012***	-.001***
			(.001)	(.000)			(.005)	(.004)			(.000)	(.000)
Ever with SEN by age 11	.236	.457	-.221***	-.169***	.235	.784	-.549***	-.408***	.225	.247	-.022***	-.000
			(.001)	(.001)			(.009)	(.010)			(.001)	(.001)
Ever eligible for FSM by age 11	.179	.305	-.126***	-.057***	.179	.437	-.258***	-.171***	.171	.189	-.018***	.000
			(.001)	(.001)			(.011)	(.012)			(.001)	(.001)
Offender by age 16	.080	.198	-.118***	-.104***	.079	.444	-.365***	-.343***	.074	.085	-.011***	.001**
			(.001)	(.001)			(.011)	(.012)			(.000)	(.000)
Any school transfer by age 16	.175	.229	-.054***	-.026***	.174	.830	-.655***	-.654***	.170	.179	-.009***	.018***
			(.001)	(.001)			(.008)	(.009)			(.001)	(.001)
N of school transfers by age 16 | any transfer	1.10	1.16	-.055***	-.034***	1.10	1.24	-.137***	-.099***	1.11	1.09	.026***	.046***
	[.325]	[.406]	(.001)	(.001)	[.323]	[.500]	(.012)	(.013)	[.348]	[.291]	(.001)	(.001)
B. Education												
Age 11 score (English, math, science)	.071	-.528	.598***	.288***	.072	-.912	.984***	.263***	.130	-.002	.131***	.008***
	[.960]	[1.13]	(.002)	(.002)	[.959]	[1.11]	(.025)	(.023)	[.962]	[.951]	(.001)	(.001)
Age 11 score: English	.082	-.549	.631***	.295***	.083	-.904	.988***	.228***	.141	.010	.131***	.007***
	[.953]	[1.13]	(.002)	(.002)	[.952]	[1.07]	(.024)	(.022)	[.955]	[.943]	(.001)	(.001)
Age 11 score: math	.086	-.537	.622***	.300***	.087	-.755	.842***	.223***	.140	.019	.121***	.008***
	[.958]	[1.09]	(.002)	(.002)	[.957]	[1.08]	(.024)	(.024)	[.959]	[.950]	(.001)	(.001)
Age 11 score: science	.106	-.605	.711***	.369***	.106	-.719	.825***	.255***	.151	.050	.101***	.007***
	[.927]	[1.17]	(.002)	(.002)	[.926]	[1.15]	(.026)	(.025)	[.923]	[.927]	(.001)	(.001)
Age 14 score (English, math, science)	.063	-.609	.672***	.465***	.064	-.760	.824***	.334***	.137	-.027	.164***	.009***
	[.951]	[1.23]	(.003)	(.002)	[.950]	[1.15]	(.026)	(.026)	[.966]	[.922]	(.001)	(.001)
Age 14 score: English	.077	-.604	.681***	.443***	.078	–1.06	1.14***	.468***	.150	-.013	.163***	.007***
	[.961]	[1.10]	(.002)	(.002)	[.960]	[.978]	(.022)	(.020)	[.972]	[.937]	(.001)	(.001)
Age 14 score: math	.082	-.485	.567***	.357***	.082	-.150	.232***	.016	.142	.007	.135***	.013***
	[.949]	[1.15]	(.002)	(.002)	[.949]	[1.19]	(.027)	(.029)	[.962]	[.926]	(.001)	(.001)
Age 14 score: science	.089	-.531	.620***	.453***	.089	-.304	.393***	.244***	.117	.055	.062***	.000
	[.940]	[1.17]	(.002)	(.002)	[.939]	[1.13]	(.025)	(.027)	[.931]	[.949]	(.001)	(.002)
Age 16 score: taken	1.00	.857	.142***	.150***	1.00	.998	.002	.002	1.00	1.00	.000	-.000
			(.000)	(.001)			(.001)	(.001)			(.000)	(.000)
Age 16 score: 5 A* – G	.935	.585	.349***	.262***	.935	.335	.600***	.552***	.940	.929	.011***	-.002***
			(.001)	(.001)			(.011)	(.011)			(.000)	(.000)
Age 16 score: 5 A* – C	.620	.261	.359***	.216***	.621	.112	.509***	.249***	.649	.586	.063***	.003***
			(.001)	(.001)			(.007)	(.008)			(.001)	(.001)
Age 16 score: point score | taken	.171	-.779	.950***	.734***	.173	–1.37	1.55***	1.13***	.214	.120	.095***	.002
	[.880]	[1.14]	(.002)	(.002)	[.878]	[.956]	(.022)	(.023)	[.874]	[.881]	(.001)	(.001)
Sample size	1,949,897	549,972	2,499,869	2,499,869	1,947,914	1,983	1,949,897	1,949,897	1,085,361	862,553	1,947,914	1,947,914

Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard deviations are reported in brackets, and heteroskedasticity-robust standard errors clustered at the secondary-school level are reported in parentheses. I use the earliest available secondary-school record. The sample in Columns (1)–(4) consists of observations in the NPD with the full set of control variables. The sample size corresponds to the sum of the 1990/91–1993/94 cohorts reported in Column (2) of Table A1 and Column (1) of Table A2. Conditional differences control for gender, minority status, an indicator for English as the first language, and SEN and FSM status at age 11, unless the variable is itself the outcome. All test scores are standardized within each cohort, except for the indicator variables for the age 16 tests. The age 11 and 14 overall scores are calculated as a weighted average of English, Math, and Science, with equal weight assigned to each subject. Some observations are missing information for certain variables. The reported sample size is the number of observations with demographic information. Data sources: Ministry of Justice and Department for Education.

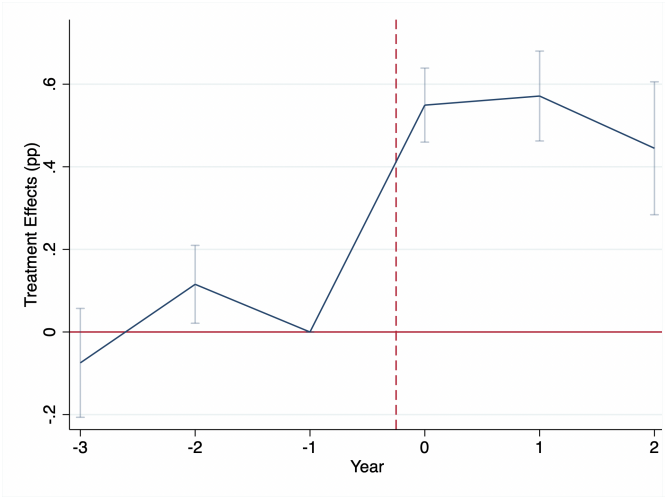
Table G2. Summary Statistics and Transfer Experience

	Treatment Status				Transfer by Age 16			
	Control	Treated	Uncond. Diff. [(1) = (2)]	Cond. Diff. [(1) = (2)]	Non-movers	Movers	Uncond. Diff. [(5) = (6)]	Cond. Diff. [(5) = (6)]
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Demographics								
Male	.503	.509	-.006***	-.004***	.508	.496	.012***	.037***
			(.001)	(.001)			(.001)	(.002)
Minority	.154	.139	.015***	.001	.146	.147	-.001	.000
			(.001)	(.001)			(.001)	(.001)
English as the first language	.926	.934	-.008***	.001**	.929	.933	-.003***	-.006***
			(.001)	(.000)			(.001)	(.001)
Ever with SEN by age 11	.243	.252	-.009***	.004***	.241	.276	-.034***	-.048***
			(.001)	(.001)			(.001)	(.002)
Ever eligible for FSM by age 11	.186	.193	-.007***	.004***	.180	.231	-.051***	-.091***
			(.001)	(.001)			(.001)	(.002)
Offender by age 16	.084	.087	-.002***	.002**	.078	.121	-.043***	-.078***
			(.001)	(.001)			(.001)	(.002)
Any school transfer by age 16	.224	.138	.087***	.038***	0	1	-	-
			(.001)	(.001)				
N of school transfers by age 16 | any transfer	1.10	1.06	.042***	.057***	-	1.09	-	-
	[.315]	[.248]	(.001)	(.002)		[.291]		
B. Education								
Age 11 score (English, math, science)	.008	-.011	.019***	-.004***	.029	-.140	.169***	.097***
	[.950]	[.951]	(.002)	(.002)	[.945]	[.965]	(.003)	(.004)
Age 11 score: English	.023	-.001	.025***	-.003	.037	-.112	.149***	.081***
	[.942]	[.944]	(.002)	(.002)	[.940]	[.950]	(.003)	(.004)
Age 11 score: math	.025	.014	.011***	-.005***	.049	-.118	.167***	.093***
	[.950]	[.950]	(.002)	(.002)	[.943]	[.968]	(.003)	(.004)
Age 11 score: science	.059	.041	.018***	-.003	.074	-.059	.133***	.085***
	[.926]	[.929]	(.002)	(.002)	[.921]	[.951]	(.003)	(.004)
Age 14 score (English, math, science)	-.006	-.047	.041***	-.005**	-.004	-.131	.127***	.102***
	[.921]	[.923]	(.002)	(.002)	[.917]	[.940]	(.003)	(.004)
Age 14 score: English	.001	-.026	.027***	-.014***	.015	-.142	.157***	.136***
	[.935]	[.938]	(.002)	(.002)	[.934]	[.938]	(.003)	(.004)
Age 14 score: math	.027	-.011	.038***	-.000	.020	-.050	.069***	.063***
	[.930]	[.921]	(.002)	(.003)	[.921]	[.947]	(.003)	(.004)
Age 14 score: science	.070	.041	.029***	.007**	.063	.015	.048***	.017***
	[.946]	[.951]	(.002)	(.003)	[.943]	[.971]	(.003)	(.005)
Age 16 score: taken	1.00	1.00	.000	.000	1.00	.999	.001***	.002***
			(.000)	(.000)			(.000)	(.000)
Age 16 score: 5 A* – G	.933	.926	.007***	-.003***	.938	.887	.051***	.084***
			(.001)	(.001)			(.001)	(.002)
Age 16 score: 5 A* – C	.595	.577	.018***	-.003**	.604	.501	.103***	.121***
			(.001)	(.001)			(.001)	(.002)
Age 16 score: point score | taken	.130	.110	.020***	-.002	.156	-.045	.201***	.301***
	[.877]	[.884]	(.002)	(.002)	[.864]	[.934]	(.003)	(.004)
Sample size	412,447	450,106	862,553	862,553	707,893	154,660	862,553	862,553

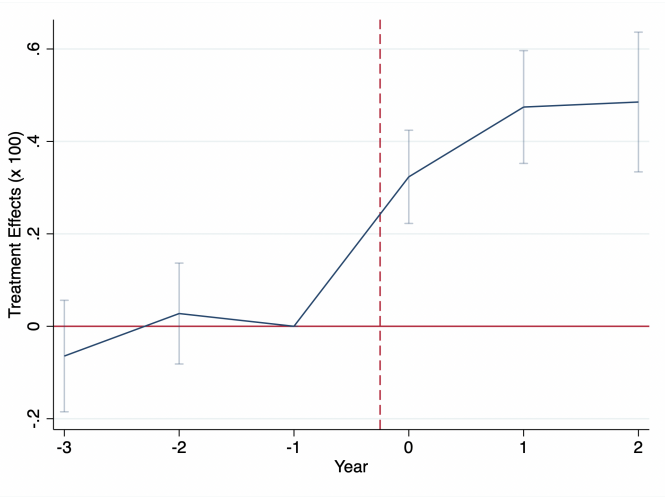
Notes: *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level. Standard deviations are reported in brackets, and heteroskedasticity-robust standard errors clustered at the secondary-school level are reported in parentheses. I use the earliest available secondary-school record. Conditional differences control for gender, minority status, an indicator for whether English is the first language, and SEN and FSM status at age 11, unless the variable is itself the outcome. All test scores are standardized within each cohort, except for the indicator variables for the age 16 tests. The age 11 and age 14 overall scores are calculated as weighted averages of English, Math, and Science, with equal weight assigned to each subject. Some observations are missing information for certain variables. The reported sample size is the number of observations with demographic information. Data sources: Ministry of Justice and Department for Education.



(a) Offending Initiation



(b) Any Crime at Each Period



(c) N of Crimes at Each Period

Figure G1. Treatment Effects for 1991–94 cohorts

Note: This figure reports the treatment effects of exposure to offenders transferring into children's schools and grades. The treatment effect is standardized at $t = -1$. The treatment group is those who are exposed to an offender transferring into their school-grade at $t = 0$, while the control group is those who will be treated later. I exclude the never-treated group from the estimation. The estimation is based on Callaway and Sant'Anna (2021).